

Mémoire

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Chapter 1

Moduli Stacks of G -bundles

1.1 Stacks of principal G -bundle (Bun_G) on a curve

1.1.1 First properties of Bun_G

Let X be a scheme over S and particularly choose G to be the general linear group GL_r over X , namely G is the sheaf $\underline{\text{Isom}}_{\mathcal{O}_X}(\mathcal{O}_X^r, \mathcal{O}_X^r)$. Then there is a dictionary to identify vector bundles (of rank r) and GL_r -torsors as following,

For $\mathcal{E} \in \text{Obj}(\text{Coh}(X))$ a locally free sheaf of rank r , define a sheaf $T(\mathcal{E}) := \underline{\text{Isom}}_{\mathcal{O}_X}(\mathcal{O}_X^r, \mathcal{E})$. It can be verified that the sheaf $T(\mathcal{E})$, under the GL_r -action given by $(g, \phi) \mapsto \phi \circ g^{-1}$ (locally for each open set), is a GL_r -torsor (for Zariski topology, thus fppf-torsor). Conversely, if $\mathcal{T}$ is some GL_r -torsor for fppf topology, then it can be glued into a vector bundle of rank r in Zariski topology from the fact that

Lemma 1.1.1. (i) Assume M is any A -module, then M is of finite presentation and flat if and only if M is finite and locally free.
(ii) Assume $A \rightarrow B$ is faithfully flat ring morphism, then M is finite presentation and flat over if and only if $M \otimes_A B$ is.

For the language of stacks and the related definition, see the Appendix A.1 for a throughout introduction.

Definition 1.1.2. Fix a rank r , let $\text{Bun}_{r,X} \rightarrow \text{Sch}/S$ fibered in groupoid with $\text{Bun}_{r,X}(U)$ contains isomorphism classes of rank r vector bundles on $X \times_S U$, for $U \in \text{Obj}(\text{Sch}/S)$.

By previous dictionary, we know $\text{Bun}_{r,X}$ is an S -prestack, and as GL_r is affine over X , we know the fppf descents are effective, hence $\text{Bun}_{r,X}$ is an S -stack.

For the algebraicity of $\text{Bun}_{r,X}$, let us first check the representability of the diagonal, this uses the criterion as in Lemma A.1.22. For $U \in \text{Obj}(\text{Sch}/S)$, $\mathcal{E}, \mathcal{E}'$ be two locally free sheaves over X_U of same rank r , then we know all of $\underline{\text{Hom}}(\mathcal{E}, \mathcal{E})$, $\underline{\text{Hom}}(\mathcal{E}', \mathcal{E}')$, $\underline{\text{Hom}}(\mathcal{E}, \mathcal{E}')$, $\underline{\text{Hom}}(\mathcal{E}', \mathcal{E})$ are representable by scheme, then consider the map $\underline{\text{Hom}}(\mathcal{E}, \mathcal{E}') \times \underline{\text{Hom}}(\mathcal{E}', \mathcal{E}) \rightarrow \underline{\text{Hom}}(\mathcal{E}, \mathcal{E}) \times \underline{\text{Hom}}(\mathcal{E}', \mathcal{E}')$ given by $(f, g) \mapsto (g \circ f, f \circ g)$. The preimage of $(\text{Id}_{\mathcal{E}}, \text{Id}_{\mathcal{E}'})$ can be identified with $\underline{\text{Isom}}(\mathcal{E}, \mathcal{E}')$, hence the diagonal is in fact representable by a scheme.

In the rest of this section, without specification, we take $S = \text{Spec}(\mathbf{k})$ for $\mathbf{k}$ a field and take X to be a smooth, geometrically connected projective curve over S and we want to seek for $\text{Bun}_{r,X}$ a presentation.

We give $\text{Bun}_{r,X}$ an atlas given by the union of substacks $\mathcal{U}_{d,N}^r$, where for $S' \rightarrow S$, the objects in $\text{Obj}(\mathcal{U}_{d,N}^r)_{S'}$ are those sheaves in $\text{Obj}(\text{Bun}_{r,X})_{S'}$ so that (we denote by $p_{S'} : X_{S'} \rightarrow S'$ the projection):

- (i) $R^1(p_{S'})_* \mathcal{E}(d) = 0$.
- (ii) $p_{S'}^* \circ (p_{S'})_* \mathcal{E}(d) \rightarrow \mathcal{E}(d)$ is surjective.
- (iii) $(p_{S'})_* \mathcal{E}(d)$ is of rank N .

It is clear that $\mathcal{U}_{d,N}^r$ is an S -stack, and by Serre's vanishing theorem, we know there is a cover $\text{Bun}_{r,X} = \bigcup_{d,N} \mathcal{U}_{d,N}^r$. Further, the natural inclusion $\mathcal{U}_{d,N}^r \rightarrow \text{Bun}_{r,X}$ is representable by open immersion. It follows from the fact that both (i) and (ii) are open conditions, since they are satisfied on the open of $X_{S'}$ obtained from cutting the support of $R^1(p_{S'})_* \mathcal{E}(d)$ (resp. $\text{Coker}(p_{S'}^* \circ (p_{S'})_* \mathcal{E}(d))$).

$(p_{S'})_* \mathcal{E}(d) \rightarrow \mathcal{E}(d))$. For condition (iii), notice that the rank of push-ward sheaf $(p_S)_* \mathcal{E}(d) = dn \cdot \deg(\mathcal{O}(1)) + \deg(\mathcal{E}) + (1 - g_X)r$. Moreover, the condition (iii) enables us to attach to each $\mathcal{U}_{d,N}^r$ a GL_r -torsor, by joining local trivialization. Namely, set $\tilde{\mathcal{U}}_{d,N}^r$ to have objects $\mathrm{Obj}(\tilde{\mathcal{U}}_{d,N}^r(S')) := \{(\mathcal{E}, \phi) \mid \mathcal{E} \in \mathrm{Obj}(\mathcal{U}_{d,N}^r(S')), \phi : \mathcal{O}_{S'}^N \xrightarrow{\sim} (p_{S'})_* \mathcal{E}(d)\}$. Next, we see that $\tilde{\mathcal{U}}_{d,N}^r$ is included in the Hilbert scheme $\mathrm{Quot}_{\mathcal{O}_{X/\mathbf{k}}^N/X/\mathbf{k}}$. This inclusion, is again an open immersion. Hence we constructed a presentation $\coprod_{d,N} \tilde{\mathcal{U}}_{d,N}^r \rightarrow \mathrm{Bun}_{r,X}$. This concludes to:

Proposition 1.1.3. *For $X \rightarrow S$ projective scheme, the S -stack $\mathrm{Bun}_{r,X} = \mathrm{Bun}_{\mathrm{GL}_r}(X)$ is an algebraic stack.*

For general affine algebraic group G over X , we use the geometric G -bundles to substitute the notion of locally free sheaves. Notice for the vector bundle case, $\mathcal{E} \rightarrow \mathrm{Spec}(\mathrm{Sym}(\mathcal{E}^\vee))$ gives a correspondence from locally free sheaves and geometric vector bundles. In general case, we associate for any $U \in \mathrm{Obj}(\mathrm{Sch}/S)$ with a X_U -scheme P and group action $G_U \times_U P \rightarrow P$. Moreover, we require an fppf covering $\{U_i \rightarrow U\}$ so that $U_i \times_U P \simeq U_i \times_S G$ (an fppf-trivialization), call such P a principal G -bundle, clearly P is a G -torsor. Since G is affine, all G -torsors come from principal G -bundles.

Now for any G affine algebraic group, define for $X/\mathbf{k}$ the fibered category $\mathrm{Bun}_G(X) \rightarrow \mathrm{Sch}/\mathbf{k}$ with $\mathrm{Bun}_G(X)(U)$ containing isomorphism classes of (geometric) G -bundles on X_U , $U \in \mathrm{Obj}(\mathrm{Sch}/\mathbf{k})$.

Proposition 1.1.4. *$\mathrm{Bun}_G(X)$ is an algebraic stack, locally of finite presentation.*

Proof. It is clear that Bun_G is a $\mathbf{k}$ -stack from the affineness. Choose $G \rightarrow \mathrm{GL}_n$ to be a closed embedding, then we have a map $\mathrm{Bun}_G \rightarrow \mathrm{Bun}_n$ by $E \mapsto E \times^G \mathrm{GL}_n$. By [Stacks, Tag 05UM], we reduce to prove the map is representable. For $U \in \mathrm{Obj}(\mathrm{Sch}/\mathbf{k})$ and a section $\eta : U \rightarrow \mathrm{Bun}_{\mathrm{GL}_n}$, corresponding to a geometric vector bundle F on X_U , the fiber product $\mathcal{Y} := \mathrm{Bun}_G \times_{\mathrm{Bun}_{\mathrm{GL}_n}} U$ is the data of $\{(E, \alpha) \mid E \in \mathrm{Obj}(\mathrm{Bun}_G(X)(U')); \alpha : E \times^G \mathrm{GL}_n \xrightarrow{\sim} F_{U'}\}$, where $U' \in \mathrm{Obj}(\mathrm{Sch}/U)$. For two pairs $(E_1, \alpha_1), (E_2, \alpha_2)$, we have $\alpha_2^{-1} \circ \alpha_1$ is an isomorphism, and since G faithfully acting on GL_n , we can determine the isomorphism $\phi : E_1 \rightarrow E_2$. From the affineness of G , we know the fiber product $\mathcal{Y}$ is an U -space. And the set of pairs can be functorially identified with those sections $\mathrm{Hom}(X_{U'}, (F \times^{\mathrm{GL}_n} \mathrm{GL}_n / G)_{U'})$, this is deduced from diagrams

$$\begin{array}{ccc} \sigma^*(F_{U'}) & \xrightarrow{\sigma'} & F_{U'} \\ \downarrow & & \downarrow \\ X_{U'} & \xrightarrow{\sigma} & (F \times^{\mathrm{GL}_n} \mathrm{GL}_n / G)_{U'} \end{array} \quad \begin{array}{ccc} \sigma^*(F_{U'}) \times^G \mathrm{GL}_n & & \\ \downarrow & \searrow \alpha & \\ \sigma^*(F_{U'}) & \xrightarrow{\sigma'} & F_{U'} \end{array}$$

where the vertical arrows in first diagram are as G -torsors, and α in the second one is an isomorphism. Thus we are reduced to show the representability of the functor $\mathrm{Sch}/U \rightarrow \mathrm{SET}, U' \mapsto \mathrm{Hom}(X_{U'}, (F \times^{\mathrm{GL}_n} \mathrm{GL}_n / G)_{U'})$. It is equivalent to show that $F/G := F \times^{\mathrm{GL}_n} \mathrm{GL}_n / G \rightarrow X_U$ is quasi-projective. It is known for $G \hookrightarrow \mathrm{GL}_n$ closed embedding, there exists a representation V of GL_n and a line $\ell \in V$ so that $G \simeq \mathrm{Stab}_{\mathrm{GL}_n}(\ell)$, and thus we have an embedding $\mathrm{GL}_n / G \subseteq \mathbb{P}(V^\vee)$, thus $F/G \subseteq \mathbb{P}(F \times^{\mathrm{GL}_n} V^\vee)$. $\square$

From above argument, we know that Bun_G is quasi-compact and locally of finite type (thus locally of finite presentation by definition), due to the existence of presentation with those properties. We have also

Proposition 1.1.5. *The algebraic stack $\mathrm{Bun}_G(X)$ is smooth for affine smooth algebraic group G .*

Proof. We use an infinitesimal criterion. Recall for $f : X \rightarrow S$ a morphism of schemes, we say it is formally smooth if for every

solid commutative diagram: $\begin{array}{ccc} T_0 & \xrightarrow{\quad} & X \\ i \downarrow & \nearrow & \downarrow f \\ T & \xrightarrow{\quad} & S \end{array}$ where T is affine scheme and T_0 is the quotient by a square zero ideal, there exists

a lifting $T \rightarrow X$ such that the diagram with dotted arrow commutes. It is known for f a morphism locally of finite presentation, formally smoothness is equivalent to smoothness. It can be generalized to stacks:

Lemma 1.1.6. [LM00, Proposition 4.15] Let $\mathcal{X} \rightarrow S$ be an algebraic stack locally of finite presentation, then $\mathcal{X} \rightarrow S$ is smooth if

and only if for any solid commutative 2-square:

$$\begin{array}{ccc} T_0 & \longrightarrow & \mathcal{X} \\ i \downarrow & \nearrow & \downarrow f \\ T & \longrightarrow & S \end{array}, \text{ with } T_0, T \text{ as the above case of schemes, there is a dotted lifting}$$

$T \rightarrow \mathcal{X}$ that makes the 2-diagram commute.

We prove first the smoothness for $G = \mathrm{GL}_n$. Choose a section $\eta_0 : T_0 = \mathrm{Spec}(A/I) \rightarrow \mathrm{Bun}_{n,X}$, it corresponds to $\mathcal{E}_0$ a vector bundle on X_{T_0} . Then we can find for X_{T_0} an affine cover $X_{T_0} = \cup_i U_{0,i}$ so that $U_{0,i} = \mathrm{Spec}(A_i/IA_i)$ and the thickening $U_i = \mathrm{Spec} A_i$ is a covering of X_T , also $\mathcal{E}_0(U_{0,i}) =: M_{0,i}$ is free $A_{0,i} := A_i/IA_i$ module. Then locally each $M_{0,i}$ can be lift to M_i that is free on U_i . The remaining is the lift glue data, we need to consider the cocycle condition, The gluing map $\phi_{0,i,j} : M_{0,i} \otimes A_{0,i,j} \rightarrow M_{0,j} \otimes A_{0,i,j}$ can be lift to $\phi_{i,j} : M_i \otimes A_{i,j} \rightarrow M_j \otimes A_{i,j}$, while the cocycle $\delta_{i,j,k} := \phi_{j,k} \circ \phi_{i,j} \circ \phi_{i,k}^{-1} - \mathrm{id}$ locates in the second order cohomology group $H^2(X_{T_0}, \mathrm{End}(\mathcal{E}_0) \otimes I)$, which vanishes due to the dimension assumption on X .

For general G , consider $\eta : T_0 \rightarrow \mathrm{Bun}_G(X)$ is a G -bundle P_0 on X_{T_0} . Again, choose covering $U_{0,i}$ and U_i for X_{T_0} and X_T respectively, we can define sheaves of groups (on X_{T_0}) $\mathcal{G}_0 : U_{0,i} \mapsto \mathrm{Hom}(U_{0,i}, G)$ and $\mathcal{G} : U_{0,i} \mapsto \mathrm{Hom}(U_i, G)$. Then $P_0 \rightarrow X_{T_0}$ is a $\mathcal{G}$ -torsor and to lift P_0 is to give a $\mathcal{G}$ -torsor on X_T . Use the infinitesimal criterion on G , since $G \rightarrow S$ is smooth, we can always lift $U_{0,i} \rightarrow G$ to $U_i \rightarrow G$. That is to say $\mathcal{G} \rightarrow \mathcal{G}_0$ is surjective. Fit this sheaf projection to $0 \rightarrow \mathcal{A} \rightarrow \mathcal{G} \rightarrow \mathcal{G}_0 \rightarrow 0$, then the cocycle difference for local $\mathcal{G}$ -torsors is reduce to compute the cohomology $H^2(X_{T_0}, P \times^{\mathcal{G}_0} \mathcal{A})$ (for a description, see [Wan11]), while the sheaf in this cohomology can be identified with $\mathfrak{g} \otimes \mathcal{I}$, $\mathcal{I}$ is the ideal sheaf corresponding to $X_{T_0} \rightarrow X_T$. Again, it vanishes due to the coherence of the sheaf and the dimension of X . $\square$

Remark. For properties proved in this section, the condition that X is a curve over $\mathbf{k}$ can be loosen. For example, in the proof of algebraicity, we can generalize to the case when $X \rightarrow S$ projective, locally of finite presentation. For smoothness, we can see that dimension condition is necessary through the proof, and in fact it is enough to add the assumption that $X \rightarrow S$ is of relative dimension 1. The idea behind the proof of general cases is however similar, we do not expand here, in [Wan11], there are detailed description.

Proposition 1.1.7. The algebraic stack $\mathrm{Bun}_{n,X}$ has dimension $n^2(g_X) - 1$. Generally, assume G is a connected semisimple group, then $\mathrm{Bun}_G(X)$ has dimension $\dim G(g_X - 1)$.

Proof. The proof uses deformation theory, we give an explanation to it rather a complete formulation. For some algebraic variety Y over $\mathbf{k}$, its Zariski tangent space at $y : \mathrm{Spec}(\mathbf{k}'/\epsilon) \rightarrow Y$ can be identified with the fiber of $Y(\mathbf{k}'[\epsilon]/(\epsilon^2)) \rightarrow Y(\mathbf{k}')$ over y .

Now for a (smooth) algebraic stack $\mathcal{Y}$ over $\mathbf{k}$, the fiber of $\xi : \mathrm{Spec}(\mathbf{k}') \rightarrow \mathcal{Y}$ can be regarded as groupoid of the fiber of a point $\mathcal{Y}(\mathbf{k}'[\epsilon]/(\epsilon^2)) \rightarrow \mathcal{Y}$ over ξ , denoted by $T_{\mathcal{Y},\xi}$, it is a $\mathbf{k}'$ -linear subcategory of $\mathcal{Y}(\mathbf{k}'[\epsilon]/(\epsilon^2))$. The dimension of ξ in X , can be taken as the $\mathbf{k}'$ dimension of the isomorphism classes in $T_{\mathcal{Y},\xi}$ minus the $\mathbf{k}'$ dimension of automorphism group at some point in $T_{\mathcal{Y},\xi}$.

In the case when $\mathcal{Y} = \mathrm{Bun}_G(X)$, a point $\eta : \mathrm{Spec}(\mathbf{k}') \rightarrow \mathrm{Bun}_G(X)$ is a G -bundle P on $X_{\mathbf{k}'}$. The dimension of η is

$$H^1(X_{\mathbf{k}'}, (\mathfrak{g}_P)_{\mathbf{k}'}) - H^0(X_{\mathbf{k}'}, (\mathfrak{g}_P)_{\mathbf{k}'}) = -\chi((\mathfrak{g}_P)_{\mathbf{k}'}), \quad \mathfrak{g}_P := P \times^G \mathfrak{g}.$$

By Riemann-Roch, $\dim_{\eta}(\mathrm{Bun}_G) = \dim G(g_X - 1) - \deg(\mathfrak{g}_P)$.

When $G = \mathrm{GL}_n$, the dimension is $-\chi(\mathrm{End}(\mathcal{E})) = n^2(g_X - 1)$, where $\mathcal{E}$ is the rank n sheaf corresponding to η .

In general, the action of G on $\bigwedge^n \mathfrak{g}$ defines a morphism $G \rightarrow \mathbb{G}_m$. If G is semisimple, by the triviality of the center, we know $\bigwedge^n \mathfrak{g}_P$ does not depend on the twist of G -bundle P , thus we can just take the trivial one. and then $\deg(\mathfrak{g}) = 0$ due to the isomorphism $\mathfrak{g} \rightarrow \mathfrak{g}^*$. $\square$

Irreducibility of $\mathrm{Bun}_{r,d}$, to be added.

1.1.2 The Harder-Narasimhan filtrations

The Harder-Narasimhan filtrations was firstly introduced in to study the cohomology of moduli space of vector bundles, it turns out that it can give a stratification of Bun_G for general reductive groups. From now on, we always take X to be a smooth, geometrically

connected projective curve of genus g_X .

We start with the case of vector bundles:

Definition 1.1.8. Let $\mathcal{E}$ be a vector bundle, a subsheaf $\mathcal{E}'$ of $\mathcal{E}$ is a **subbundle** if the quotient $\mathcal{E}/\mathcal{E}'$ is locally free.

For $\mathcal{E}$ vector bundle, its slope is defined to be $\mu(\mathcal{E}) := \frac{\deg(\mathcal{E})}{\text{rank}(\mathcal{E})}$, this definition can be extended to coherent sheaves, where the slope of torsion sheaves is counted as ∞ . The vector bundle $\mathcal{E}$ is called **semi-stable** (resp. **stable**) if for any (non-zero) subbundle $\mathcal{E}' \subseteq \mathcal{E}$, the slopes satisfy $\mu(\mathcal{E}') \leq \mu(\mathcal{E})$ (resp. $\mu(\mathcal{E}') < \mu(\mathcal{E})$).

For a short exact sequence of vector bundles $0 \rightarrow \mathcal{E} \rightarrow \mathcal{E}' \rightarrow \mathcal{E}'' \rightarrow 0$, we know $\mu(\mathcal{E}') \in [\min\{\mu(\mathcal{E}), \mu(\mathcal{E}'')\}, \max\{\mu(\mathcal{E}), \mu(\mathcal{E}'')\}]$ from the additivity of rank and degree on $\text{Coh}(X)$. Thus for semi-stable $\mathcal{E}$, its quotient vector bundles have slope $\geq \mu(\mathcal{E})$. We then have the following immediate lemma is useful, and will be repeatedly use in the sequel

Lemma 1.1.9. Let $\mathcal{E}, \mathcal{E}'$ be two semi-stable vector bundles, then

- (i) Assume if $\mu(\mathcal{E}) > \mu(\mathcal{E}')$, then $\text{Hom}(\mathcal{E}, \mathcal{E}') = 0$.
- (ii) Assume if $\mu(\mathcal{E}) + 2g_X - 2 < \mu(\mathcal{E}')$, then $\text{Ext}^1(\mathcal{E}, \mathcal{E}') = 0$.

Proof. For (i), if there is a non-zero morphism $f : \mathcal{E} \rightarrow \mathcal{E}'$, then we have $\mu(\mathcal{E}) \leq \mu(\text{Im}(f)) \leq \mu(\mathcal{E}')$, a contradiction.

For (ii), we use the Serre duality, then $\text{Ext}^1(\mathcal{E}, \mathcal{E}') = H^1(X, \mathcal{E}^\vee \otimes \mathcal{E}) \simeq H^0(X, \text{Hom}(\mathcal{E}', \Omega_X \otimes \mathcal{E}))$, we conclude with (i), as the degree of $\Omega_X \otimes \mathcal{E} = (2g_X - 2)\text{rank}(\mathcal{E}) + \deg(\mathcal{E})$, hence $\mu(\Omega_X \otimes \mathcal{E}) = 2g_X - 2 + \mu(\mathcal{E}) < \mu(\mathcal{E}')$ by assumption. $\square$

Theorem 1.1.10. For each vector bundle $\mathcal{E}$, there is a unique filtration $0 \subsetneq \mathcal{E}_0 \subsetneq \dots \subsetneq \mathcal{E}_k$ so that $\mathcal{E}_i/\mathcal{E}_{i-1}$ is semistable. Moreover, the slopes $\mu_i := \mu(\mathcal{E}_i/\mathcal{E}_{i-1})$ satisfy $\mu_1 > \dots > \mu_k$.

Proof. Fix a vector bundle $\mathcal{E}$, we first notice that its subbundles have degrees bounded. Due to Riemann-Roch, we have $\deg(\mathcal{E}') + \text{rk}(\mathcal{E}')(1 - g_X) = \chi(\mathcal{E}') \leq h^0(\mathcal{E}') \leq h^0(\mathcal{E})$, for any subbundle $\mathcal{E}' \subseteq \mathcal{E}$. If $\mathcal{E}$ itself is semistable, then we are done.

Assume $\mathcal{E}$ is not semistable, then by above argument, we can deduce that the set $\{\mu(\mathcal{E}')\}_{\mathcal{E}' \subseteq \mathcal{E}}$ have a maximal element, denote by μ_1 . We first claim there is a subbundle maximal in the sense of inclusion that has slope μ_1 . In fact, if there are subbundles $\mathcal{E}', \mathcal{E}''$ attach the maximal slope, then they are semistable by definition. $\mathcal{E}' \oplus \mathcal{E}''$ is also semistable due to the following lemma:

Lemma 1.1.11. Let $\text{VB}(X)_\mu$ be the full subcategory of $\text{Coh}(X)$ containing semistable vector bundles of same slope μ , then it is an Abelian category and is stable by extension.

Proof of Lemma. For any inclusion of sheaves $\mathcal{E}' \hookrightarrow \mathcal{E}$ with both $\mathcal{E}, \mathcal{E}'$ vector bundles, define the saturation $(\mathcal{E}')^\#$ in $\mathcal{E}$ to be $\text{Ker}(\mathcal{E} \rightarrow (\mathcal{E}/\mathcal{E}')^{\text{cotor}})$, where by $\mathcal{F}^{\text{cotor}}$ we mean the locally free part of coherent $\mathcal{F}$, then $(\mathcal{E}')^\# \subseteq \mathcal{E}$ either a subbundle or equal. For any $f : \mathcal{E} \rightarrow \mathcal{E}'$ in VB_μ , from semistability of $\mathcal{E}, \mathcal{E}'$, we have $\mu = \mu(\mathcal{E}) \leq \mu(\text{Im}(f)) \leq \mu((\text{Im}(f))^\#) \leq \mu(\mathcal{E}') = \mu$. $\square$

Using the above lemma, we consider $f : \mathcal{E}' \oplus \mathcal{E}'' \rightarrow \mathcal{E}$, then $\mu((\text{Im}(f))^\#) \geq \mu(\mathcal{E}' \oplus \mathcal{E}'') = \mu_1$. If $(\text{Im}(f))^\# = \mathcal{E}$, we are done, otherwise we get a subbundle of slope μ_1 with higher rank, the claim is thus proved.

Choose $\mathcal{E}_1$ maximal subbundle with maximal slope μ_1 , we then proceed to $\mathcal{E}/\mathcal{E}_1$ and repeat the above. This process ends in finite step as for the finiteness of rank. We get a filtration $0 \subsetneq \mathcal{E}_0 \subsetneq \dots \subsetneq \mathcal{E}_k$ as wanted. The uniqueness is easy. $\square$

As a consequence of the theorem, we know for each vector bundle $\mathcal{E}$, the rank and degree of $\mathcal{E}_i$ in its Harder-Narasimhan filtration are determined, call the sequence of rank-degree data of the filtration the **Harder-Narasimhan(HN) type** of $\mathcal{E}$. Drop the vertices on the plane to have coordinate $(\text{rk}(\mathcal{E}_i), \deg(\mathcal{E}_i))$, and lining up two neighbour vertices, we get a polygon in $\mathbb{Q}^2$, called the **Harder-Narasimhan(HN) Polygon**, which only depends on the HN-type. For a fixed class (r, d) , there is a partial order on the set of HN-types with total rank r and total degree d , we say HN-types $\mathcal{P}' \geq \mathcal{P}$ if the HN-polygon of $\mathcal{P}$ lies beneath the one of $\mathcal{P}'$.

With the help of Harder-Narasimhan filtration, we could endow a stratification on the stack $\text{Bun}_{r,d}(X)$. Fix a HN-type for (r, d) , let $\text{Bun}_{r,d}^{\leq \mathcal{P}}$ to be the stack with the data $(S, \mathcal{E})$, where $\mathcal{E} \in \text{Bun}_{r,d}(X)(S)$, so that for each point $s \in S$, $\mathcal{E}_s$ has HN-type $\leq \mathcal{P}$.

Theorem 1.1.12. $\text{Bun}_{r,d}^{\leq \mathcal{P}} \subseteq \text{Bun}_{r,d}(X)$ is an open substack.

Proof. For each presentation $T \rightarrow \text{Bun}_{r,d}$, we want to show the fiber product $T^{\leq \mathcal{P}} := T \times_{\text{Bun}_{r,d}} \text{Bun}_{r,d}^{\leq \mathcal{P}}$ is open in T . To do this, let us consider the HN-filtration for families:

Proposition 1.1.13. *Let $T \rightarrow \operatorname{Spec} \mathbf{k}$ be a scheme, We fix $\mathcal{E}$ locally free sheaf on X_T and a HN-type $\mathcal{P}$ of $(\operatorname{rk}(\mathcal{E}), \deg(\mathcal{E})) =: (r, d)$. Set $T^{\leq \mathcal{P}}$ to contain the points $t \in T$ such that the HN-type of $\mathcal{E}_t \leq \mathcal{P}$. Then $T^{\leq \mathcal{P}} \subseteq T$ is open.*

Proof of Proposition 1.1.13. The first step is to consider whether the semistable part is open, we have

Lemma 1.1.14. *The set of points with semistable HN-type (r, d) is open.*

Proof. When $T := \operatorname{Spec}(\mathbf{k}') \xrightarrow{p} \operatorname{Spec}(\mathbf{k})$ is a point, with $\mathbf{k}'/\mathbf{k}$ is a finite extension, then $p^*(\mathcal{E}_\bullet)$ is a HN-filtration of $p^*(\mathcal{E})$ if and only if $\mathcal{E}_\bullet$ is a HN-filtration of $\mathcal{E}$, where $0 \subsetneq \mathcal{E}_1 \subsetneq \dots \subsetneq \mathcal{E}_k = \mathcal{E}$ is a strictly increasing filtration. It suffices to show $p^*\mathcal{E}$ semistable if and only if $\mathcal{E}$ semistable, the only if part is clear as $\mu(p^*\mathcal{E}') = \deg p \cdot \mu(\mathcal{E}')$, $\forall \mathcal{E}' \subseteq \mathcal{E}$, for the converse side, see [Le 97, Theorem 10.1.3]. For general T , as we saw in the proof of 1.1.10, any subbundle $\mathcal{V}' \subseteq \mathcal{V}$ has degree bounded by $h^0(\mathcal{V})$. Set $d_m := \sup_{t \in T} h^0(X_t, \mathcal{E}_t)$, then the vector bundle $\mathcal{E}_t$ is semistable with slope $\mu = \frac{d}{r}$ if and only if it has no subbundle $\mathcal{E}'_t$ with slope $\mu < \mu(\mathcal{E}'_t) < (g_X - 1) + d_m$, if and only if it has no quotient bundle $\mathcal{E}''_t$ with slope $\mu > \mu(\mathcal{E}''_t) > \mu''$, where μ'' is a certain lower bound, can be take as $\mu - (d_m + (r - 1)(g_X - 1))$ when $g_X \geq 1$ or $\mu - (d_m + 1)$ when $g_X = 0$.

Thus $\mathcal{E}_t$ is semistable if and only if it has no quotient bundle with certain finitely many Hilbert polynomials $P_1, \dots, P_k$. Then we can consider $\operatorname{Quot}_{\mathcal{E}/X_T/T}^{P_i} \longrightarrow T$, it has closed image, denoted by T_i . And the complement of $T \setminus \bigcup_{i=1}^k T_i$ is open in T to be the semistable locus with local slope exactly μ . $\square$

Now to prove $T^{\leq \mathcal{P}}$ is open, we use the fact that a constructible set is open if and only if it is stable under generalization. Thus we remains to show the set is constructible and stable under generalization. The first half is due to a stratification $T = \bigcup_i T_i$ such that the HN-type on each stratum is constant. The second half it due to an observation that for a DVR which is not a field, the HN-polygon at generic point is beneath the one at special point. Hence the Proposition is proved, and thus the theorem. $\square$

$\square$

We can generalize the Harder-Narasimhan filtration to reductive algebraic groups. Notice in previous (usual) HN-filtration for vector bundles, the data of ranks in the HN-type indicates the group of automorphisms of the filtration, namely a (standard) parabolic subgroup of GL_n . For general reductive G , we associate G -bundles with proper parabolic reduction.

Assume G is a reductive group over a perfect field $\mathbf{k}$, we choose $P_0 \subseteq G$ a minimal parabolic subgroup of G with Levi subgroup M_0 . By standard parabolic groups we mean those parabolic subgroups containing P_0 .

For P a standard parabolic, write $P = M_P N_P$ its Levi decomposition with M_P Levi subgroup and N_P the unipotent radical. Set also Z_P the center of the Levi M_P , define $A_P := \operatorname{Hom}(\mathbb{G}_m, Z_P) \otimes \mathbb{G}_m$ to be the maximal split torus in the center Z_P , and $A'_P := \operatorname{Hom}(\operatorname{Hom}(M_{P,\text{ab}}, \mathbb{G}_m), \mathbb{G}_m)$ to be the maximal split quotient torus in $M_{P,\text{ab}}$. Then there is an isogeny to be the composition

$$A_P \hookrightarrow Z_P \hookrightarrow M_P \twoheadrightarrow M_{P,\text{ab}} \twoheadrightarrow A'_P$$

which induces an inclusion of cocharacter groups $X_*(A_P) := \operatorname{Hom}(\mathbb{G}_m, A_P) \simeq \operatorname{Hom}(\mathbb{G}_m, Z_P) \hookrightarrow X_*(A'_P) = \operatorname{Hom}(\mathbb{G}_m, A'_P)$, such that $\mathbb{R} \otimes X_*(A_P) \simeq \mathbb{R} \otimes X_*(A'_P)$. Denote this $\mathbb{R}$ -linear spanning by $\mathfrak{a}_P$. Also denote $X^*(A_P)$ (resp. $X^*(A'_P)$) the dual of cocharacter groups, with a pairing $\langle \cdot, \cdot \rangle$ their $\mathbb{R}$ -linear expansion is $\mathfrak{a}_P^*$.

Definition 1.1.15. Let P be a standard parabolic and $\mathcal{E}_P$ be a principal P -bundle on X . Define the **slope** $\mu(\mathcal{E}_P)$ to be an element in $X_*(A'_P)$ such that

$$\langle \xi, \mu(\mathcal{E}_P) \rangle = \mu(\mathcal{E}_{P,\xi}) \in \mathbb{Z}$$

where $\mathcal{E}_{P,\xi}$ is a line bundle $\mathcal{E} \times^P \mathbb{G}_m$ on X with respect to the P -action on $\mathbb{G}_m$ through the composition $P \twoheadrightarrow A'_P \xrightarrow{\xi} \mathbb{G}_m$.

For two standard parabolics P and Q , if $P \subseteq Q$, then $A_Q \hookrightarrow A_P \hookrightarrow A'_P \twoheadrightarrow A'_Q$ an isogeny, which induces the embedding $\mathfrak{a}_Q \hookrightarrow \mathfrak{a}_P$ and the projection $\mathfrak{a}_P \twoheadrightarrow \mathfrak{a}_Q$. Thus we can decompose $\mathfrak{a}_P$ so that $\mathfrak{a}_Q$ is as a direct summand, write $\mathfrak{a}_P = \mathfrak{a}_P^Q \oplus \mathfrak{a}_Q$, also the corresponding dual space $\mathfrak{a}_P^* = (\mathfrak{a}_P^Q)^* \oplus \mathfrak{a}_Q^*$.

Let Δ_{P_0} be the set of simple roots, it is a basis of $(\mathfrak{a}_{P_0}^G)^*$. For P containing P_0 , let Δ_P be the restriction of simple roots to $\mathfrak{a}_P$, and it is a basis of $(\mathfrak{a}_P^G)^*$. Generally, for $P \subseteq Q$ two standard parabolics, let $\Delta_P^Q := \Delta_{P \cap M_Q}$, as a basis of $(\mathfrak{a}_P^Q)^*$.

Definition 1.1.16. For a principal P -bundle $\mathcal{E}_P$ on X with P standard parabolic, it is called **semi-stable** if for each standard $Q \subseteq P$ with $|\Delta_P^Q| = 1$ and each Q -bundle $\mathcal{E}_Q$ with $\mathcal{E}_Q \times^Q P \simeq \mathcal{E}_P$, the slope $\mu(\mathcal{E}_Q)$ satisfy

$$\langle \check{\alpha}_Q^P, \mu(\mathcal{E}_Q) \rangle \leq 0$$

with $\check{\alpha}_Q^P$ the unique simple root in Δ_Q^P . **(Do we need k to be alg. closed?)**

For reductive G , let $\text{Bun}_G^{\text{ss}}(X)$ (resp. $\text{Bun}_{G, \mu'_G}^{\text{ss}}$) be the stack of semistable G -bundles (resp. semistable of slope $\mu'_G \in X_*(A'_G)$) on X . For $P \subseteq G$ standard, as M_P is reductive, define the semistable locus of P -bundles to be

$$\text{Bun}_P(X) \times_{\text{Bun}_{M_P}(X)} \text{Bun}_{M_P}^{\text{ss}}(X) =: \text{Bun}_P^{\text{ss}}(X)$$

Proposition 1.1.17 (Existence of Harder-Narasimhan filtration). *Let $\mathfrak{P}$ be the set of pairs (P, μ'_P) where P is a standard parabolic and $\mu'_P \in X_*(A'_P)$. Then for each G -bundle $\mathcal{E}$ of slope μ'_G , there exists a unique $(P, \mu'_P) \in \mathfrak{P}$ and a reduction of $\mathcal{E}_P$ of $\mathcal{E}$ such that μ'_P restricts to μ'_G , where the restriction corresponds to $\mathfrak{a}_P = \mathfrak{a}_P^G \oplus \mathfrak{a}_G$. We call (P, μ'_P) above the **Harder-Narasimhan type** of $\mathcal{E}$.*

Similar to Theorem 1.1.12, we have:

Proposition 1.1.18. *Let $P \subseteq G$ a standard parabolic as above, then $\text{Bun}_P^{\text{ss}}(X) \subseteq \text{Bun}_P(X)$ is an open substack. Moreover, $\text{Bun}_P^{\text{ss}}(X) = \bigcup_{\mu'_P \in X_*(A'_P)} \text{Bun}_{P, \mu'_P}^{\text{ss}}(X)$.*

The vector bundle stack wrt HN-filtration for G , to be added

1.2 Geometry on Bun_G

1.2.1 Volume of Bun_G and Tamagawa numbers

Assume $\mathcal{C}$ a groupoid, if it further satisfies for any object $C \in \mathcal{C}$, the set of automorphisms $\text{Aut}_{\mathcal{C}}(C)$ has finite cardinal, then we can define the Volume of $\mathcal{C}$ to be $\int_{\mathcal{C}} d\mu = \sum_{C \in \text{Obj}(\mathcal{C})/\sim} \frac{1}{|\text{Aut}_{\mathcal{C}}(C)|} =: \text{Vol}(\mathcal{C})$.

For example, if we consider a groupoid SET^{fin} of finite sets, with bijective morphism (thus between sets with same cardinal), then $\text{Vol}(\text{SET}^{\text{fin}}) = \sum_n \frac{1}{|\mathfrak{S}_n|} = \sum_n \frac{1}{n!} = e$.

Assume X is a smooth, geometrically connected projective curve over $\mathbb{F}_q$, in this section (and the following ones), we want to count the volume of stack of principal bundles on X , namely the mass of groupoid of fiber $\text{Vol}(\text{Bun}_G(X)(\mathbb{F}_q))$. We first recall Weil's dictionary:

Let K_X be the function field of X , for $x \in X$ closed point, denote $\mathcal{O}_x = \widehat{\mathcal{O}_{X,x}} \simeq \kappa(x)[[t]]$, where $\kappa(x)$ is the residue at x . Denote $K_x := \text{Frac}(\mathcal{O}_x)$, and set the Idele group of X to be $\mathbb{A}_{0,X} = \prod_{x \in X} \mathcal{O}_x^\times$. For $S \subseteq |X|$ a finite subset of closed points on X , set $\mathbb{A}_X^S := \prod_{x \in S} K_x \times \prod_{x \notin S} \mathcal{O}_x^\times$. The Adele ring of X is $\mathbb{A}_X := \varinjlim_{S \subseteq X} \mathbb{A}_X^S$, which is locally compact commutative ring.

Proposition 1.2.1 (Weil's uniformization theorem). *There is a bijection*

$$\text{Bun}_{n,X}(\mathbb{F}_q) \simeq \text{GL}_n(K_X) \backslash \text{GL}_n(\mathbb{A}_X) / \text{GL}_n(\mathbb{A}_{0,X}) = \text{GL}_n(K_X) \backslash \text{GL}_n(\mathbb{A}_X) / \prod_{x \in X} \text{GL}_n(\mathcal{O}_x)$$

Proof. First notice for locally free sheaves of rank n , they are generically free, with automorphism group $\text{GL}_n(K_X)$ acting on the section at generic point η . On a closed point $x \in X$, for a fixed vector bundle $\mathcal{E}$, notice there is a bijection between modifications of $\mathcal{E}$ at x and $\mathcal{O}_x$ -lattices in $\mathcal{E}_x \otimes_{\mathcal{O}_{X,x}} K_x \simeq K_x^n$, while the later one is identified with $\text{GL}_n(K_x) / \text{GL}_n(\mathcal{O}_x)$, $n = \text{rk}(\mathcal{E})$. $\square$

The general uniformization for reductive groups is subtler.

Theorem 1.2.2. [GL19, Proposition 1.3.2.10] *Let X be a curve as before, and $G \rightarrow X$ is smooth affine group scheme with connected fibers, and assume the generic fiber of G is semisimple and simply connected, then*

$$\text{Bun}_G(X)(\mathbb{F}_q) \simeq G(K_X) \backslash G(\mathbb{A}_X) / \prod_{x \in X} G(\mathcal{O}_x)$$

A sketch of proof. The proof use a sequence of results. First we show under our assumption, any G -bundle is generically trivial.

Lemma 1.2.3 (Harder). *Let X be a curve as before, G a simply connected and semisimple algebraic group over K_X and P_0 a G -bundle over K_X , then P_0 is a trivial G -bundle if and only if for any $x \in X$ closed point, $\text{Spec}(K_x) \times_{\text{Spec}(K_X)} P$ is trivial on $\text{Spec}(K_x)$.*

Lemma 1.2.4 (Lang). *For any connected algebraic group G over finite field $\mathbf{k}$, any G -bundle on $\text{Spec}(\mathbf{k})$ is trivial.*

We can thus consider P a G -bundle over X , its restriction onto $\text{Spec}(\kappa(x)) = \text{Spec}(\mathbb{F}_{q^{\deg_x}})$, $\forall x \in X$ is thus trivial by Lang's lemma. By Hensel's lemma, the trivialization of P at some x can be extended to the infinitesimal neighbour, i.e a trivialization over $\text{Spec}(\mathcal{O}_x)$. And the restriction to $\text{Spec}(K_x)$ is thus trivial. By Harder's Lemma, we know P is generically trivial. We can thus find P an open set U so that P is trivial on U .

Lemma 1.2.5 (Beauville-Laszlo). *Let X be a Dedekind scheme and G a flat affine group scheme over X , S finite set of closed point on X , then we have a fiber product*

$$\begin{array}{ccc} \text{Tors}_G(X) & \longrightarrow & \text{Tors}_G(X - S) \\ \downarrow & & \downarrow \\ \prod_{x \in S} \text{Tors}_G(\text{Spec}(\mathcal{O}_x)) & \longrightarrow & \prod_{x \in S} \text{Tors}_G(\text{Spec}(K_x)) \end{array}$$

to avoid ambiguity we use $\text{Tors}_G(Y)$ to denote the G -torsors on Y , by substituting the notation Bun_G .

The lemma tells a gluing procedure to construct G -bundles, for an element $g = (g_x)_{x \in X} \in G(\mathbb{A}_X)$ with $g_x \in G(K_x)$, $x \in S$. we can associate P_g a G -bundle on X with trivialization on $X - S$ and trivializations on $\text{Spec}(\mathcal{O}_x)$ for each $x \in S$. The argument in the first step shows that any G -bundle arises in this way as we can modify P on $S = X - U$. $\square$

Remark. A proof of uniformization theorem using loop group argument can be found in [Sor00, Section 5].

Under the uniformization, we have immediately Weil's conjecture (function field version) of Tamagawa number in the form of volume of stacks. For an introduction to Tamagawa measure of function field, see [GL19]

Corollary 1.2.6. *Let X be a smooth, geometrically connected projective curve over $\mathbb{F}_q$ and $G \rightarrow X$ with semisimple and simply connected generic fiber. Let μ_{Tam} denote the Tamagawa measure on $G(\mathbb{A}_X)$, then the Tamagawa number of G , i.e the measure of $G(K_X) \backslash G(\mathbb{A}_X)$ is $\text{Vol}(\text{Bun}_G(X)) \cdot \mu_{\text{Tam}}(G(\mathbb{A}_0))$.*

Proof. We identify $\text{Bun}_G(X)(\mathbb{F}_q)$ with $G(K_X) \backslash G(\mathbb{A}_X) / G(\mathbb{A}_0)$. For an element $\bar{g}$ in the double quotient, denote $\mathbb{O}_{\bar{g}}$ its inverse image in $G(K_X) \backslash G(\mathbb{A}_X)$ $\square$

Tamagawa, Atiyah-Bott, to be added.

1.2.2 Trace formula

1.2.3 Compactification by Laumon-Gaitsgory

to be added

1.3 Geometric Eisenstein series

Chapter 2

Hall Algebras

2.1 Ringel-Hall algebras, definition and first properties

An ultimate reference for this section is [Sch06].

Let $\mathcal{A}$ be an abelian category with finite global dimension, require also $|\text{Ext}^i(M, N)| < \infty, \forall M, N \in \mathcal{A}, \forall i \geq 0$, where by Ext^i we mean the Yoneda i -extension. Such a category is called a **finitary category**. set the **Euler form** to be $\langle M, N \rangle := \prod_{i=0}^{\infty} \left(\#(\text{Ext}^i(M, N))^{\frac{(-1)^i}{2}} \right), \forall M, N \in \mathcal{A}$. Notice it only depends on the classes of object in $K(\mathcal{A})$. And a symmetric form $(\cdot, \cdot) := K(\mathcal{A}) \times K(\mathcal{A}) \rightarrow \mathbb{C}$ called **symmetric Euler form** is set to be $(M, N) := \langle M, N \rangle \cdot \langle N, M \rangle, \forall M, N \in \mathcal{A}$. If $\mathcal{A}$ is $\mathbf{k}$ -linear with $\mathbf{k}$ being finite cardinal q , we can use $\dim \text{Ext}^i$ for a substitute $|\text{Ext}^i|$, in such case, define also the Euler form as $\langle M, N \rangle := \sum_i (-1)^i \dim \text{Ext}^i(M, N)$.

Definition 2.1.1. Assume that $\mathcal{A}$ is finitary, define the **Hall algebra** $\mathbf{H}_{\mathcal{A}}$ of $\mathcal{A}$ to have the underlying vector space

$$\text{Fun}_0 \left(\bigoplus_{M \in \text{Obj}(\mathcal{A})/\sim} [M], \mathbb{C} \right) = \bigoplus_{\alpha \in K(\mathcal{A})} \mathbf{H}_{\mathcal{A}}[\alpha], \quad \text{the function space on the isomorphism classes of } \mathcal{A} \text{ with finite support.}$$

the multiplication on $\mathbf{H}_{\mathcal{A}}$ is defined as

$$f * g(M) = \sum_{N \subseteq M} \langle M/N, N \rangle f(M/N) g(N), \quad \forall f, g \in \mathbf{H}_{\mathcal{A}}$$

since $f \in \mathbf{H}_{\mathcal{A}}$ is spanned by the characteristic function $\mathbf{1}_{[M]}$ of isomorphism class of $M \in \mathcal{A}$, we can write the multiplication as

$$\mathbf{1}_{[M]} * \mathbf{1}_{[N]} = \langle M, N \rangle \sum_R \frac{1}{a_M a_N} \mathbf{P}_{M,N}^R \mathbf{1}_{[R]}, \quad a_M = |\text{Aut}_{\mathcal{A}}(M)| \text{ and,}$$

$\mathbf{P}_{M,N}^R$ is the number of short exact sequence of the form $0 \rightarrow N \rightarrow R \rightarrow M \rightarrow 0$

the equivalence of two formulas is due the fact that $\left| \frac{1}{a_M a_N} \mathbf{P}_{M,N}^R \right| = |\{L \subseteq R | L \simeq N, R/L \simeq M\}|$.

The multiplication formula is well-defined on $\mathbf{H}_{\mathcal{A}}$ as $\mathcal{A}$ is finitary, and we have the unit of $(\mathbf{H}_{\mathcal{A}}, *)$ is given by the zero object. A simple computation shows

Lemma 2.1.2. [Sch06, Proposition 1.1] $(\mathbf{H}_{\mathcal{A}}, *)$ is an associative algebra.

However, even with finitary condition, an object $M \in \mathcal{A}$ could occur as the extension of $(N, M/N)$ for infinitely many such pairs. The multiplication in the Hall algebra encodes the extension in $\mathcal{A}$, to reverse the process, i.e. to obtain a co-multiplication, we extends $\mathbf{H}_{\mathcal{A}} \otimes \mathbf{H}_{\mathcal{A}}$ to be the whole function space $\mathbf{H}_{\mathcal{A}} \hat{\otimes} \mathbf{H}_{\mathcal{A}} := \text{Fun}(K(\mathcal{A}) \otimes K(\mathcal{A}), \mathbb{C}) = \prod_{\alpha, \beta \in K(\mathcal{A})} \mathbf{H}_{\mathcal{A}}[\alpha] \hat{\otimes} \mathbf{H}_{\mathcal{A}}[\beta]$.

Define the co-multiplication Δ on $\mathbf{H}_{\mathcal{A}}$ to be:

$$\Delta(f)(M, N) := \frac{\langle M, N \rangle^{-1}}{|\mathrm{Ext}^1(M, N)|} \sum_{\xi \in \mathrm{Ext}^1(M, N)} f(X_{\xi}), \quad \text{where } \xi = 0 \longrightarrow N \longrightarrow X_{\xi} \longrightarrow M \longrightarrow 0$$

Lemma 2.1.3. *Expressing in characteristic functions, the co-multiplication becomes [Sch06, Proposition 1.5]*

$$\Delta(\mathbf{1}_{[R]}) = \sum_{M, N} \langle M, N \rangle \frac{1}{a_R} \mathbf{P}_{M, N}^R \mathbf{1}_{[M]} \otimes \mathbf{1}_{[N]}$$

Proof. Examine on each object $M \otimes N \in \mathbf{H}_{\mathcal{A}} \otimes \mathbf{H}_{\mathcal{A}}$, it suffices to prove $\frac{|\{\xi \in \mathrm{Ext}^1(M, N) | X_{\xi} \simeq R\}|}{|\mathrm{Ext}^1(M, N)|} = \langle M, N \rangle^2 \frac{\mathbf{P}_{M, N}^R}{a_R}$ and thus suffices to prove $|\{\xi \in \mathrm{Ext}^1(M, N) | X_{\xi} \simeq R\}| = \mathbf{P}_{M, N}^R \frac{|\mathrm{Hom}(M, N)|}{|\mathrm{Aut}(R)|}$. The group acts on the underlying set of $\mathbf{P}_{M, N}^R$ by $\phi \cdot (a, b) = (\phi \circ a, b \circ \phi^{-1})$, where we use (a, b) to denote $N \xrightarrow{a} R \xrightarrow{b} M$. The stabilizer of any (a, b) in the set does NOT depend on the point itself, it is isomorphic to $\mathrm{Hom}(M, N)$. The Lemma follows from the definition of Yoneda 1-extension. $\square$

it turns out that $\mathbf{H}_{\mathcal{A}}$ is an Hopf algebra, to see this, we set the (twisted) multiplication on $\mathbf{H}_{\mathcal{A}} \widehat{\otimes} \mathbf{H}_{\mathcal{A}}$ to be

$$(f_{\alpha} \otimes f_{\beta}) * (f_{\gamma} \otimes f_{\delta}) := (\beta, \gamma) (f_{\alpha} * f_{\gamma}) \otimes (f_{\beta} * f_{\delta})$$

We abbreviate $\mathbf{H}_{\mathcal{A}}$ as $\mathbf{H}$ if there is no ambiguity on $\mathcal{A}$, define also the pairing $(\cdot | \cdot) : \mathbf{H} \otimes \mathbf{H} \rightarrow \mathbb{C}$ to be the categorical integration, $\forall f, g \in \mathbf{H}, (f | g) = \int_{[\mathcal{A}]} f * g d\mu := \sum_{M \in \mathrm{Obj}(\mathcal{A}) / \sim} \frac{f(M)g(M)}{|\mathrm{Aut}(M)|}$. It can be easily deduced that this pairing is non-degenerate. Moreover, it is proved by Green that $\mathbf{H}$ has a bi-algebra structure, we integrate to the following technical theorem

Theorem 2.1.4. [Gre95] *Let $(\mathcal{A}, \mathbf{H}, *, \Delta)$ as above, and $\mathrm{gl. dim}(\mathcal{A}) \leq 1$, then we have:*

- (i) $(\mathbf{H}, \Delta)$ is an co-associative co-algebra.
- (ii) $\Delta : \mathbf{H} \rightarrow \mathbf{H} \widehat{\otimes} \mathbf{H}$ is an algebra homomorphism, i.e. $\Delta(f * g) = \Delta(f) * \Delta(g)$.
- (iii) $(\cdot | \cdot)$ is a Hopf pairing, i.e. $(f * g | h) = (f \otimes g | \Delta(h))$ so that multiplication and co-multiplication are adjoint under the pairing.

Proof. To prove (i), we need $(\Delta \otimes 1) \circ \Delta$ to be well defined (on $\mathbf{H} \widehat{\otimes} \mathbf{H} \widehat{\otimes} \mathbf{H}$). This follows from the fact that $\mathcal{A}$ is finitary. Notice even though the sum is taken as infinite one, to each resulting term $[M_1] \otimes [M_2] \otimes [M_3]$, only finitely many term of the form $[N] \otimes [M_3]$ can contribute. The rest of proof is reduced to a coefficient comparing and goes similarly as that of 2.1.2, see also [Sch06, Proposition 1.4].

For (ii), we can write down explicitly the formula on both sides (we can choose element in $\mathbf{H}$ as $\mathbf{1}_{[M]}$ without loss of generality), on one hand,

$$\begin{aligned} \Delta(\mathbf{1}_{[M]} * \mathbf{1}_{[N]}) &= \langle M, N \rangle \sum_J \frac{1}{a_M a_N} \mathbf{P}_{M, N}^J \Delta(\mathbf{1}_{[J]}) \\ &= \sum_{K, L} \langle M, N \rangle \langle K, L \rangle \frac{1}{a_M a_N} \sum_J \frac{1}{a_J} \mathbf{P}_{M, N}^J \mathbf{P}_{K, L}^J \mathbf{1}_{[K]} \otimes \mathbf{1}_{[L]}. \end{aligned}$$

On the other hand, we compute

$$\begin{aligned}
& \Delta(\mathbf{1}_{[M]}) * \Delta(\mathbf{1}_{[N]}) \\
&= \left(\sum_{K_2, L_2} \langle K_2, L_2 \rangle \frac{1}{a_M} \mathbf{P}_{K_2, L_2}^M \mathbf{1}_{[K_2]} \otimes \mathbf{1}_{[L_2]} \right) * \left(\sum_{K_1, L_1} \langle K_1, L_1 \rangle \frac{1}{a_N} \mathbf{P}_{K_1, L_1}^N \mathbf{1}_{[K_1]} \otimes \mathbf{1}_{[L_1]} \right) \\
&= \sum_{K_1, L_1, K_2, L_2} \frac{\langle K_1, L_1 \rangle \langle K_2, L_2 \rangle \langle K_1, L_2 \rangle \langle L_2, K_1 \rangle}{a_M a_N} \mathbf{P}_{K_2, L_2}^M \mathbf{P}_{K_1, L_1}^N (\mathbf{1}_{[K_2]} * \mathbf{1}_{[K_1]}) \otimes (\mathbf{1}_{[L_2]} * \mathbf{1}_{[L_1]}) \quad (\text{note the twist of multiplication}) \\
&= \sum_{K, L} \sum_{K_1, L_1, K_2, L_2} \frac{\langle K_1, L_1 \rangle \langle K_2, L_2 \rangle \langle K_1, L_2 \rangle \langle L_2, K_1 \rangle \langle K_2, K_1 \rangle \langle L_2, L_1 \rangle}{a_M a_N a_{K_1} a_{K_2} a_{L_1} a_{L_2}} \mathbf{P}_{K_2, K_1}^K \mathbf{P}_{L_2, L_1}^L \mathbf{P}_{K_2, L_2}^M \mathbf{P}_{K_1, L_1}^N \mathbf{1}_{[K]} \otimes \mathbf{1}_{[L]} \\
&= \sum_{K, L} \frac{\langle M, N \rangle \langle K, L \rangle}{a_M a_N} \sum_{K_1, K_2, L_1, L_2} \frac{1}{(K_2, L_1)} \frac{\mathbf{P}_{K_2, K_1}^K \mathbf{P}_{L_2, L_1}^L \mathbf{P}_{K_2, L_2}^M \mathbf{P}_{K_1, L_1}^N}{a_{K_1} a_{K_2} a_{L_1} a_{L_2}} \mathbf{1}_{[K]} \otimes \mathbf{1}_{[L]}
\end{aligned}$$

The last step follows from the bi-linearity (on $K(\mathcal{A})$) of the Euler form, hence $\langle M, N \rangle = \langle K_2, K_1 \rangle \langle K_2, L_1 \rangle \langle L_2, K_1 \rangle \langle L_2, L_1 \rangle$, the same goes for $\langle K, L \rangle$. We are in the situation of the following diagrams

$$\begin{array}{ccc}
& 0 & \\
& \downarrow & \\
& L & \\
& \downarrow a' & \\
0 \longrightarrow N \xrightarrow{a} J \xrightarrow{b} M \longrightarrow 0 & & \\
& \downarrow b' & \\
& K & \\
& \downarrow & \\
& 0 &
\end{array}
\qquad
\begin{array}{ccccccc}
& 0 & & 0 & & 0 & \\
& \downarrow & & \downarrow & & \downarrow & \\
0 \longrightarrow L_1 & \xrightarrow{u} & L & \xrightarrow{v} & L_2 & \longrightarrow & 0 \\
& \downarrow u' & & & & \downarrow x & \\
& N & & & & M & \\
& \downarrow v' & & & & \downarrow y & \\
0 \longrightarrow K_1 & \xrightarrow{x'} & K & \xrightarrow{y'} & K_2 & \longrightarrow & 0 \\
& \downarrow & & & & \downarrow & \\
& 0 & & & & 0 &
\end{array}
\tag{2.1}$$

need to show for any $K, L \in \mathcal{A}$, we have

$$\sum_J \frac{1}{a_J} \mathbf{P}_{M, N}^J \mathbf{P}_{K, L}^J = \sum_{K_1, K_2, L_1, L_2} \frac{1}{(K_2, L_1)} \frac{\mathbf{P}_{K_2, K_1}^K \mathbf{P}_{L_2, L_1}^L \mathbf{P}_{K_2, L_2}^M \mathbf{P}_{K_1, L_1}^N}{a_{K_1} a_{K_2} a_{L_1} a_{L_2}} \tag{*}$$

We use the similar method as in Lemma 2.1.3, but on larger diagrams. On one hand we have a 'cross' with an object J lies in the center, we therefore use Cr_J to denote the set of all of those crosses (a, b, a', b') centered in J and thus $|\text{Cr}_J| = \mathbf{P}_{M, N}^J \mathbf{P}_{K, L}^J$. For any $\phi \in \text{Aut}(J)$, its action on Cr_J is given by $(a, b, a', b') \mapsto (a \circ \phi, b \circ \phi^{-1}, a' \circ \phi, b' \circ \phi^{-1})$, as in Lemma 2.1.3 we get $\text{Stab}_{\text{Aut}(J)}((a, b, a', b')) \simeq \text{Hom}(\text{Coker}(b'a), \text{Ker}(b'a))$, and the LHS of (*) becomes

$$\sum_J \frac{1}{a_J} \sum_{(a, b, a', b') \in \mathcal{C}_J} \frac{a_J}{|\text{Hom}(\text{Coker}(b'a), \text{Ker}(b'a))|}$$

where we denote by $\mathcal{C}_J$ to be the set of equivalent classes in Cr_J .

Similarly, we use $D(K_1, K_2, L_1, L_2)$ to denote the set of the 'squares' with corners being K_1, K_2, L_1, L_2 respectively. The automorphism groups of $K_i, L_i, i = 1, 2$ act freely on the set $D(K_1, K_2, L_1, L_2)$, thus the RHS of (*) becomes

$$\sum_{K_1, K_2, L_1, L_2} \frac{|\text{Ext}^1(K_2, L_1)|}{|\text{Hom}(K_2, L_1)|} |\mathcal{D}(K_2, K_1, L_2, L_1)|$$

where $\mathcal{D}$ here is the set of equivalent classes. To compare both sides, we fit the 'crosses' into 'squares', concretely, we set

$$K_1 := \text{Im}(b'a), K_2 := \text{Coker}(b'a), L_1 := \text{Ker}(b'a), L_2 := \text{Im}(ba')$$

For an element $(J, a, b, a', b') \in \text{Cr}(J)$, we note that due to the short exactness, $\text{Ker}(b'a) \simeq \text{Ker}(ba')$ and $\text{Coker}(b'a) \simeq \text{Coker}(ba')$. The arrows on the square are automatically obtained from the construction, and the resulting square belongs to the union of D 's. Such construction gives a map $\Phi : \coprod \text{Cr}_J \longrightarrow \coprod D(K_1, K_2, L_1, L_2)$ and $\text{Aut}(J)$ stabilizes the image as $b' \circ \phi^{-1} \circ \phi \circ a = b'a$, so Φ descends to the equivalent classes $\tilde{\Phi} : \coprod \mathcal{C}_J \longrightarrow \coprod \mathcal{D}(K_1, K_2, L_1, L_2)$.

If the fiber of $\tilde{\Phi}$ at any point of $\mathcal{D}(K_1, K_2, L_1, L_2)$ is of cardinal $|\text{Ext}^1(K_2, L_1)|$, then the LHS and RHS of (*) coincides.

We keep the notation in the diagram 2.1, let S be in $D(K_1, K_2, L_1, L_2)$, then S uniquely determined $X := K \times_{K_2} M$ and Y : push-out of (u, u')

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \uparrow & & \uparrow & & \\
 0 & \longrightarrow & K_1 \oplus L_2 & \xrightarrow{x''} & X & \xrightarrow{y''} & K_2 \longrightarrow 0 \\
 & & \uparrow v'' & & \uparrow b'' & & \parallel \\
 0 & \longrightarrow & Y & \xrightarrow{a''} & J & \xrightarrow{t} & K_2 \longrightarrow 0 \\
 & & \uparrow u'' & & \uparrow s & & \\
 & & L_1 & \xlongequal{\quad} & L_1 & & \\
 & & \uparrow & & \uparrow & & \\
 & & 0 & & 0 & &
 \end{array}
 \quad
 \begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \uparrow & & \uparrow & & \\
 0 & \longrightarrow & K_1 \oplus L_2 & \xrightarrow{x''} & X & \xrightarrow{y''} & K_2 \longrightarrow 0 \\
 & & \uparrow v'' & & \uparrow b'' & & \parallel \\
 0 & \longrightarrow & Y & \xrightarrow{a''} & J & \xrightarrow{t} & K_2 \longrightarrow 0
 \end{array}$$

If the data (K_1, K_2, L_1, L_2) corresponds to the image of some cross $J \in \text{Cr}_J$, then we have X and Y fit into the left above 'octahedron'. Conversely, for a fixed J , elements in $\Phi_J^{-1}(S)$ are determined by right above diagram. For the scale of the fiber, can consider the following diagram (need the condition $\text{gl. dim} \leq 1$ here)

$$0 \rightarrow \text{Hom}(K_2, L_1) \rightarrow \text{Hom}(K_2, Y) \rightarrow \text{Hom}(K_2, K_1 \oplus L_2) \rightarrow \text{Ext}^1(K_2, L_1) \rightarrow \text{Ext}^1(K_2, Y) \xrightarrow{\varphi} \text{Ext}^1(K_2, K_1 \oplus L_2) \rightarrow 0$$

For fixed J and fixed extension $\xi = (a'', t)$ of K_2 by Y , there exists $b'' : J \rightarrow X$ such that the diagram commutes if and only if $\varphi(\xi) = \eta := (x', y') \oplus (x, y) \in \text{Ext}^1(K_2, K_1 \oplus L_2)$. For each $\xi \in \text{Ext}^1(K_2, Y)$, the number of all possible extension (a'', t) in the same equivalent class ξ is $\frac{a_J}{|\text{Hom}(K_2, Y)|}$, and once ξ is fixed, $\#\{\eta \in \text{Ext}^1(K_2, K_1 \oplus L_2) | \varphi^{-1}(\eta) = \xi\} = |\text{Hom}(K_2, K_1 \oplus L_2)|$. Hence for each J , the number of possible extensions is

$$\frac{a_J}{|\text{Hom}(K_2, Y)|} |\text{Hom}(K_2, K_1 \oplus L_2)| |\varphi^{-1}(\eta)| \quad (**)$$

Pass to the case $\tilde{\Phi}$, since for each square S , $\text{Stab}_{\text{Aut}(J)}(S) \simeq \text{Hom}(K_2, L_1)$, we know $|\tilde{\Phi}_J^{-1}(S)| = (**) \cdot \frac{|\text{Stab}|}{a_J}$, and as $\tilde{\Phi}^{-1}(S) = \sum_J \tilde{\Phi}_J^{-1}$, the result follows.

For (iii), we compute directly

$$\begin{aligned}
 (\mathbf{1}_{[M]} * \mathbf{1}_{[N]} | \mathbf{1}_{[R]}) &= \frac{1}{|\text{Aut}(R)|} \langle M, N \rangle \cdot |\{L \subseteq R | L \simeq N, R/L \simeq M\}| = \frac{\langle M, N \rangle}{a_R a_M a_N} \mathbf{P}_{M, N}^R \\
 (\mathbf{1}_{[M]} \otimes \mathbf{1}_{[N]} | \Delta(\mathbf{1}_{[R]})) &= \frac{|\{\xi \in \text{Ext}^1(M, N) | X_\xi \simeq R\}| \langle M, N \rangle^{-1}}{|\text{Ext}^1(M, N)| |\text{Aut}(N)| |\text{Aut}(M)|} = \frac{|\text{Hom}(M, N)| \langle M, N \rangle^{-1}}{|\text{Ext}^1(M, N)| a_R a_M a_N} \mathbf{P}_{M, N}^R
 \end{aligned}$$

then both sides coincide. $\square$

In fact the bi-algebra $\mathbf{H}_{\mathcal{A}}$ can be upgraded to a Hopf algebra in certain condition, this is realized by Xiao's antipode. We extend $\mathbf{H}$ a bit to $\tilde{\mathbf{H}}$ whose underlying vector space is $\mathbf{H} \otimes_{\mathbb{C}} \mathbb{C}[K(\mathcal{A})]$ with additional multiplication relation

$$\mathbf{k}_\alpha \mathbf{k}_\beta = \mathbf{k}_{\alpha+\beta}, \quad \mathbf{k}_\alpha \mathbf{1}_{[M]} = (\alpha, M) \mathbf{1}_{[M]} \mathbf{k}_\alpha$$

where $\mathbf{k}_\alpha \in \mathbb{C}[K(\mathcal{A})]$ is the representative of the class α . The extended co-multiplication $\Delta : \tilde{\mathbf{H}} \rightarrow \tilde{\mathbf{H}} \otimes \tilde{\mathbf{H}}$ is given as

$$\begin{aligned}\Delta(\mathbf{k}_\alpha) &:= \mathbf{k}_\alpha \otimes \mathbf{k}_\alpha \\ \Delta(\mathbf{1}_{[R]}) &:= \sum_{M, N} \langle M, N \rangle \frac{1}{a_R} \mathbf{P}_{M, N}^R \mathbf{1}_{[M]} \mathbf{k}_{[N] + \alpha} \otimes \mathbf{1}_{[N]} \mathbf{k}_\alpha\end{aligned}$$

Assume further that our finitary category $\mathcal{A}$ has finite subobject, Let $\mathbb{F}_M^r$ be the set of strict r -step filtration of M , its elements are of the form $M_\bullet = 0 \subsetneq M_1 \subsetneq \dots \subsetneq M_r = M$. Define the **antipode map** to be

$$S : \mathbf{1}_{[M]} \mapsto \mathbf{k}_{[M]}^{-1} \left(\sum_{r \geq 1} (-1)^r \sum_{M_\bullet \in \mathbb{F}_M^r} \left(\prod_{i=1}^r \langle M_{i+1}/M_i, M_i \rangle \right) \frac{a_{M_1} a_{M_2/M_1} \dots a_{M/M_{r-1}}}{a_M} \times \mathbf{1}_{[M/M_{r-1}]} \dots \mathbf{1}_{[M_1]} \right)$$

We state here the results in [Xia97],

Theorem 2.1.5. (i) *The extended Hall algebra $\tilde{\mathbf{H}}$ is a bi-algebra with respect to extended multiplication and co-multiplication.*
(ii) *If $\mathcal{A}$ has finite subobjects, then $\tilde{\mathbf{H}}$ is an Hopf algebra, with $S : \tilde{\mathbf{H}} \rightarrow \tilde{\mathbf{H}}$ being the antipode as Hopf algebra.*

By varying the underlying category $\mathcal{A}$, the resulting Hall algebra $\mathbf{H}_{\mathcal{A}}$ would interact with various fields. For example, the classical Hall algebra was introduced in the profound book [Mac98], as the Macdonald's ring of symmetric function, while study the representations of p -groups, namely $\mathcal{A}$ is chosen as the category of abelian p -groups (we will come back to this classical Hall algebra later). If $\mathcal{A}$ is $\text{Rep}_{\vec{Q}}^{\mathbb{F}_q}$, the representation category of Dynkin quiver $\vec{Q}$ over finite field, then it was proved by Ringel that $\mathbf{H}_{\text{Rep}_{\vec{Q}}}$ is isomorphic to the (positive half of) quantum group $\mathbf{U}_q(\mathfrak{g}_{\vec{Q}})$ deformed by q , where $\mathfrak{g}_{\vec{Q}}$ is the Lie algebra generated by the underlying Dynkin diagram. In such construction, the PBW basis is related to an orientation of the quiver. Furthermore, in [Lus90], Lusztig constructed a geometric version of Hall algebra through the moduli stack of representation of $\vec{Q}$, such geometric construction successfully gives a canonical basis of quantum groups which does not depend on the orientation of the quiver. We do not expand the story here, see [Lus] for a comprehensive introduction to quantum groups and canonical basis, and [Sch09] for an explanation to Lusztig's construction and examples.

Roughly speaking, Hall algebra captures the structure of extensions, one geometric analog is the diagram

$$\text{Bun}_M \xleftarrow{q} \text{Bun}_P \xrightarrow{p} \text{Bun}_G$$

In the sequel, we will concentrate on the Hall algebra on curves and variants (take $\mathcal{A}$ to be $\text{Coh}(X)$), and see how to express the terms in automorphic representation (eg. automorphic forms, Hecke actions, Eisenstein series, Langlands dualities...) by using Hall algebras.

2.2 Hall algebra of curves

Without specification, by a curve X we mean a smooth, geometrically connected projective curve over $\mathbf{k} = \mathbb{F}_q$. We denote the genus of X by g .

Let the underlying category be $\text{Coh}(X)$, denote the associated Hall algebra to be $\mathbf{H}_X$, called the **Hall algebra of curve** X . As any coherent sheaf on X can be uniquely decomposed into torsion part and vector bundle part, we set the rank of the coherent sheaf to be the rank of its vector bundle part, then $K(\text{Coh}(X))$ is graded by the pair of rank and degree, and the Euler form is given by

$$(q \text{ power of }) \langle \mathcal{F}, \mathcal{G} \rangle = \dim \text{Hom}(\mathcal{F}, \mathcal{G}) - \dim \text{Ext}^1(\mathcal{F}, \mathcal{G}) = (1 - g)r_{\mathcal{F}}r_{\mathcal{G}} + (r_{\mathcal{F}}d_{\mathcal{G}} - r_{\mathcal{G}}d_{\mathcal{F}})$$

By definition $\mathbf{H}_X := \bigoplus_{r, d} \text{Fun}_0(\text{Coh}_{r, d}(X), \mathbb{C})$, and $\tilde{\mathbf{H}}_X = \mathbf{H}_X \otimes \mathbf{K}$ is the extended Hall algebra, $\mathbf{K} := K[\text{Coh}(X)]$.

Consider the torsion-free part, $\text{Bun}(X) \subseteq \text{Coh}(X)$ is an exact subcategory, and we have sub-vector space $\mathbf{H}_X^{\text{Bun}} := \bigoplus_{r, d} \text{Fun}_0(\text{Bun}_{r, d}(X), \mathbb{C})$, this is in fact a sub-algebra of $\mathbf{H}_X$, but not a sub-bialgebra, as the multiplication Δ is not stable on $\mathbf{H}_X^{\text{Bun}}$.

On the other hand, we have $\text{Tor}(X) = \text{Coh}_0(X) \subseteq \text{Coh}(X)$ being a Serre abelian subcategory. This gives rise to a sub-bialgebra $\mathbf{H}_X^{\text{Tor}} \subseteq \mathbf{H}_X$ as well as $\tilde{\mathbf{H}}_X^{\text{Tor}} \subseteq \tilde{\mathbf{H}}_X$. Since $\text{Tor}(X) \simeq \prod_{x \in |X|} \text{Coh}_{0,\{x\}}(X)$, where $\text{Coh}_{0,x} \subseteq \text{Tor}$ is the Serre subcategory of torsion sheaves supported on $x \in |X|$, $|X|$ denotes the set of (Zariski) closed points. We have $\mathbf{H}_X^{\text{Tor}} \simeq \bigotimes_{x \in |X|} \mathbf{H}_x^{\text{Tor}}$, with $\mathbf{H}_x^{\text{Tor}}$ the Hall algebra generated by $\text{Coh}_{0,\{x\}}(X)$.

Now we have obtained two subalgebras of $\mathbf{H}_X$, it is not hard to observe that $\mathbf{H}_X \simeq \mathbf{H}_X^{\text{Bun}} \otimes \mathbf{H}_X^{\text{Tor}}$ due to the uniqueness of torsion decomposition of coherent sheaves. Such decomposition of Hall algebra is however not commutative, namely $[f_{\text{Tor}}, g_{\text{Bun}}] \neq 0$.

We can define an action (**Hecke action**) of $\mathbf{H}_X^{\text{Tor}}$ on $\mathbf{H}_X^{\text{Bun}}$ by

$$T_f(g) := \pi(f * g)$$

where $f \in \mathbf{H}_X^{\text{Tor}}$, $g \in \mathbf{H}_X^{\text{Bun}}$ and $\pi : \mathbf{H}_X \rightarrow \mathbf{H}_X^{\text{Bun}}$ is the restricting map. Before going to details on automorphic forms, we first take a glimpse on the side of Hall algebra of torsion sheaves.

2.2.1 Classical Hall algebra

Let $\mathfrak{O}$ be a discrete valuation ring, with the residue field $\kappa = \mathfrak{O}/\mathfrak{m} \simeq \mathbb{F}_q$, for q some prime power.

Choose $\mathcal{A} := \text{Rep}^{\text{fl}}(\mathfrak{O})$ to be the category of the finite length module of $\mathfrak{O}$. By Krull-Schmitt theorem, we know that each finite length $\mathfrak{O}$ -module is isomorphic to $\bigoplus_{i=1}^n \mathfrak{O}/\mathfrak{m}^{\lambda_i}$. We can choose $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, so that the modules corresponds to a unique partition $\underline{\lambda} = (\lambda_1, \dots, \lambda_n)$, we denote it by $I_{\underline{\lambda}}$. Also, let $\mu_i := \dim_{\kappa}(\mathfrak{m}^{i-1}I_{\underline{\lambda}}/\mathfrak{m}^iI_{\underline{\lambda}})$, we have that $\underline{\mu} = (\mu_1, \mu_2, \dots)$ being the conjugate partition to $\underline{\lambda}$.

The Grothendieck group $K(\text{Rep}^{\text{fl}}(\mathfrak{O}))$ is isomorphic to $\mathbb{Z}$ since $I_{\lambda_i} = \mathfrak{O}/\mathfrak{m}^{\lambda_i}$ is λ_i -steps extension of κ , and have same class as $\kappa^{\oplus \lambda_i}$. Thus $K(\text{Rep}^{\text{fl}}(\mathfrak{O})) \rightarrow \mathbb{Z}$ sends $I_{\underline{\lambda}} \mapsto |\underline{\lambda}|$, the length of partition $\underline{\lambda}$.

Thus the computation of Euler form can simply reduce to the direct sum of κ 's, and because $\dim \text{Hom}(\kappa, \kappa) = \dim \text{Ext}^1(\kappa, \kappa) = 1$, we find that Euler form vanishes. Thus the multiplication on the Hall algebra $\mathbf{H}_{\mathcal{A}}$ is computed as

$$[I_{\underline{\mu}}] [I_{\underline{\nu}}] = \sum_{\underline{\lambda}} H_{\underline{\mu}, \underline{\nu}}^{\underline{\lambda}}(\mathfrak{O}) [I_{\underline{\lambda}}]$$

where $H_{\underline{\mu}, \underline{\nu}}^{\underline{\lambda}}(\mathfrak{O}) = \# \{N \subseteq M \text{ submodule} \mid \text{type}(N) = \nu, \text{co-type}(N) = \mu\}$, M is any representative of the $[I_{\underline{\lambda}}]$, the coefficients are called Hall numbers. It is an associative and commutative algebra under the multiplication.

Also the co-multiplication can be computed as

$$\Delta([I_{\underline{\lambda}}]) = \sum_{\underline{\mu}, \underline{\nu}} \frac{a_{\underline{\mu}} a_{\underline{\nu}}}{a_{\underline{\lambda}}} H_{\underline{\mu}, \underline{\nu}}^{\underline{\lambda}} [I_{\underline{\mu}}] \otimes [I_{\underline{\nu}}]$$

where $a_{\underline{\lambda}} = |\text{Aut}(I_{\underline{\lambda}})|$. And the Hopf pairing is simply $([I_{\underline{\mu}}] | [I_{\underline{\nu}}]) = \delta_{\underline{\mu}, \underline{\nu}} \frac{1}{a_{\underline{\mu}}}$.

On the other hand, we can consider the ring of symmetric functions. Let $\Lambda_n := \mathbb{Z}[x_1, \dots, x_n]^{\mathfrak{S}_n}$ be the ring of symmetric polynomials in n -variables. It is graded by the degree of polynomials, $\Lambda_n = \bigoplus_{k \geq 0} \Lambda_n^k$. For $m \geq n$, we have surjective morphism $\rho_{m,n} : \Lambda_m \twoheadrightarrow \Lambda_n$ that keeps $x_1, \dots, x_n$ and sends x_{n+1}, x_m to zero. The graded morphisms $\rho_{m,n}^k : \Lambda_m^k \rightarrow \Lambda_n^k$ are surjective (and bijective if $m \geq n \geq k$).

Define the **ring of symmetric functions** Λ to be the graded limit $\Lambda := \bigoplus_{k \geq 0} \varprojlim_n \Lambda_n^k$, which we also denote by $\mathbb{Z}[x_1, x_2, \dots]^{\mathfrak{S}_{\infty}}$.

There are various basis of Λ described in [Mac98], labelled by partitions.

- One choice of the basis is called the elementary symmetric functions, $e_{\underline{\lambda}} := e_{\lambda_1} \dots e_{\lambda_n}$, where $e_r := \sum_{i_1 < \dots < i_r} x_{i_1} \dots x_{i_r}$, for each projection $\rho_n^k : \Lambda^k \rightarrow \Lambda_n^k$, the image of e_r is $\sum_{1 \leq i_1 < \dots < i_r \leq n} x_{i_1} \dots x_{i_r}$.
- One basis is given by the power of variables, $p_{\underline{\lambda}} = p_{\lambda_1} \dots p_{\lambda_n}$, where $p_n := \sum_i x_i^n$.
- One basis is given by those (infinite) symmetric monomials, $m_{\underline{\lambda}} := \sum_{i_1 < \dots < i_r} a_i x_{i_1}^{\alpha_1} \dots x_{i_r}^{\alpha_r}$, where $\underline{\lambda}$ is a partition and $(\alpha_1, \dots, \alpha_r)$ is a permutation of $\underline{\lambda}$.

Λ can be endowed the structure of bi-algebra, where the co-multiplication is the graded projective limit of $\Delta_n : \mathbb{Z}[x_1, \dots, x_{2n}]^{\mathfrak{S}_{2n}} \rightarrow \mathbb{Z}[x_1, \dots, x_n]^{\mathfrak{S}_n} \otimes \mathbb{Z}[x_{n+1}, \dots, x_{2n}]^{\mathfrak{S}_n}$. The co-multiplication of our previous choices of bases can be explicitly computed,

$$\Delta(e_n) = \sum_{k=0}^n e_k \otimes e_{n-k}; \quad \Delta(p_n) = p_n \otimes 1 + 1 \otimes p_n; \quad \Delta(m_{\underline{\lambda}}) = \sum_{\underline{\mu}} m_{\underline{\mu}} \otimes m_{\underline{\lambda}-\underline{\mu}}$$

where $\underline{\mu} = (\lambda_{i_1}, \dots, \lambda_{i_k}), i_1 < \dots < i_k$ is a subsequence of $\underline{\lambda}$ and $\underline{\lambda} - \underline{\mu} = (\lambda_1, \dots, \widehat{\lambda_{i_1}}, \dots, \widehat{\lambda_{i_2}}, \dots)$.

The Hopf pairing is set to be $(p_n | p_m) = \delta_{n,m} n$, and in general we have the pairing for the basis of power sums

$$(p_{\underline{\lambda}} | p_{\underline{\mu}}) = \delta_{\underline{\lambda}, \underline{\mu}} \prod_i \lambda_i \prod_j \lambda'_j$$

here $\underline{\lambda}'$ is used to denote the conjugate partition of $\underline{\lambda}$.

To relate $\mathbf{H}_{\mathcal{A}}$ and Λ , we do a deformation on Λ . Set $\Lambda_{\hbar} := \Lambda \otimes_{\mathbb{Z}} \mathbb{C}[\hbar, \hbar^{-1}]$ with a deformed Hopf pairing $(p_n | p_n)_{\hbar} = \frac{n}{(\hbar^2)^{-n-1}}$. There is a new basis of $\Lambda_{\hbar}$ parametrized by partitions, called Hall-Littlewood functions. They are taken as the graded limit of

$$P_{\lambda}(x_1, \dots, x_n; \hbar^2) = \frac{1}{v_{\underline{\lambda}}(\hbar^2)} \sum_{w \in \mathfrak{S}_n} w \left(x_1^{\lambda_1} \dots x_n^{\lambda_n} \prod_{i < j} \frac{x_i - \hbar^2 x_j}{x_i - x_j} \right)$$

where $v_{\underline{\lambda}}(t) = \prod_j v_{l_j}(t)$, $v_m(t) = \prod_{i=1}^m \frac{1-t^i}{1-t}$.

Theorem 2.2.1. *Let $\mathcal{A}$ be $\text{Rep}^{\text{fl}}(\mathfrak{D})$ as before, and assume $|\kappa| = q$, then there is an isomorphism of bialgebra*

$$\mathbf{H}_{\mathcal{A}} \xrightarrow{\simeq} \Lambda_{\hbar^2=q}, \quad [I_{(1^r)}] \mapsto \hbar^{r(r-1)} e_r$$

where $\Lambda_{\hbar^2=q} := \Lambda \otimes \mathbb{C}[\hbar, \hbar^{-1}] / (\hbar^2 - q)$.

Moreover, for any partition $\underline{\lambda}$, the above isomorphism gives a map $[I_{\underline{\lambda}}] \mapsto (\hbar)^{-2n(\underline{\lambda})} P_{\underline{\lambda}}(q^{-1})$.

In particular, assume our curve is defined over $\mathbb{F}_q$, and assume $x \in X$ is a closed point. We have

$$\begin{aligned} \mathbf{H}_{\text{Coh}_{0,\{x\}}} &\xrightarrow{\simeq} \Lambda_{\hbar^2=q_x}, & [\mathcal{O}_x^{\oplus r}] &\mapsto q_x^{\frac{-r(r-1)}{2}} e_r, \\ & & [\mathcal{O}_{\underline{\lambda}[x]}] &\mapsto q_x^{-n(\underline{\lambda})} P_{\underline{\lambda}}(q_x^{-1}) \end{aligned}$$

where $q_x = q^{\deg(x)}$, and $\mathcal{O}_x^{\oplus r}; \mathcal{O}_{\underline{\lambda}[x]} := \mathcal{O}_{\lambda_1[x]} \oplus \dots \oplus \mathcal{O}_{\lambda_n[x]}$ are the torsion sheaves supported on x and parametrized by $\underline{\lambda}$.

2.2.2 Recollection of automorphic forms

As the Hall algebra of curve studies the functions $\text{Coh}(X) \rightarrow \mathbb{C}$, we first do some recollection on the theory of automorphic forms (on function fields) in the language of Hall algebras. Some of them will occur naturally in our later discussion on Hall algebras.

Let X be a curve of genus g_X defined over $\mathbb{F}_q$ as before, set $K = \mathbb{F}_q(X)$. Fix an isomorphism $\mathbb{C} \simeq \mathbb{Q}_l$, with l a prime differs from $\text{char}(\mathbf{k})$.

Consider the dictionary that $\text{Bun}_{n,X} = \text{GL}_n(K) \backslash \text{GL}_n(\mathbb{A}_K) / \text{GL}_n(\mathbb{O})$, then an **unramified automorphic form** (associated to GL_n) is a $\text{GL}_n(\mathbb{O}) = \prod_{x \in |X|} \text{GL}_n(\mathcal{O}_x)$ -invariant complex valued function on $\text{GL}_n(K) \backslash \text{GL}_n(\mathbb{A}_K)$. It is thus equivalent to a function $\text{Bun}_{n,X} \rightarrow \mathbb{C}$.

Denote the space of automorphic form (of rank n) as $\text{AF}_n(X)$, and its subspace of functions $\text{Bun}_{n,d} \rightarrow \mathbb{C}$ as $\text{AF}_{n,d}(X)$.

Definition 2.2.2. For an unramified automorphic form $f \in \text{AF}_n(X)$, it is called **cuspidal** if $\int_{\text{U}(K) \backslash \text{U}(\mathbb{A}_K)} f(ug) du = 0$ on $\text{GL}_n(\mathbb{A}_K)$. Denote $\text{AF}_n^{\text{cusp}}(X) \subseteq \text{AF}_n(X)$ the space of cuspidal forms. This is equivalent to, (in the flavour of Hall algebra) $\sum_{\xi \in \text{Ext}^1(\mathcal{E}, \mathcal{E}')} f(X_{\xi}) = 0$, for all pair $(\mathcal{E}, \mathcal{E}') \in \text{Bun}_{m,X} \times \text{Bun}_{n-m,X}$.

It is a known fact as reduction theory that elements in $\mathrm{AF}_n^{\mathrm{cusp}}(X)$ have finite support and the space $\mathrm{AF}_n^{\mathrm{cusp}}(X)$ is finite dimensional. (a concise proof of finite-dimensionality using the language of Hall algebra is given later in Proposition 2.2.19).

Definition 2.2.3. For $x \in X$ a closed point, the **Hecke algebra** of rank n associated to x is $\mathcal{H}_x^n := \mathrm{Fun}(\mathrm{GL}_n(\mathcal{O}_x) \backslash \mathrm{GL}_n(F_x) / \mathrm{GL}_n(\mathcal{O}_x), \mathbb{C})$ with the Hecke action $(T \cdot f)(g) = \int_{\mathrm{GL}_n(F_x)} f(gh_x^{-1})T(h_x)dh_x$ for $T \in \mathcal{H}_x$, $f \in \mathrm{AF}_n$.

Since for each torsion sheaf $\mathcal{F} \in \mathrm{Coh}(X)$, it is supported on finitely many points, thus we can extend the above local Hecke action to **global Hecke action** $\mathrm{Tor}(X) \times \mathrm{AF}_n \rightarrow \mathrm{AF}_n$ by setting $(T_{\mathcal{F}} \cdot f)(g) = \int_{\mathrm{GL}_n(\mathbb{A})} f(gh^{-1})\mathbf{1}_{\mathcal{F}}(h)dh$. Or equivalently, for any $\mathcal{F}$ torsion sheaf, $T_{\mathcal{F}} : \mathrm{AF}_n \rightarrow \mathrm{AF}_n$ by taking $(T_{\mathcal{F}} \cdot f)(\mathcal{E}) = \sum_{\mathcal{E}' \subseteq \mathcal{E}, \mathcal{E}/\mathcal{E}' \simeq \mathcal{F}} f(\mathcal{E}')$.

For $\mathcal{F} \in \mathrm{Coh}(X)$, we define the **dual Hecke action** by $\mathcal{F}$ to be $(T_{\mathcal{F}}^{\vee} \cdot f)(\mathcal{E}) = \sum_{\mathcal{E} \subseteq \mathcal{E}', \mathcal{E}'/\mathcal{E} \simeq \mathcal{F}} f(\mathcal{E}')$.

By definition, the dual Hecke action $T_{\mathcal{F}}^{\vee}$ is fact the dual operator of $T_{\mathcal{F}}$ under the inner product of Hall algebra.

Lemma 2.2.4. Fix $\mathcal{F} \in \mathrm{Tor}(X)$, for automorphic forms $f \in \mathrm{AF}_{n,d}$, $g \in \mathrm{AF}_{n,d+h^0(\mathcal{F})}$ such that f, g have finite support, then

$$(T_{\mathcal{F}} \cdot f \mid g) = (f \mid T_{\mathcal{F}}^{\vee} \cdot g)$$

Notice $\mathrm{Tor}(X) = \prod_{x \in |X|} \mathrm{Coh}_{0,\{x\}}$, the Hall algebra $\mathbf{H}_X^{\mathrm{Tor}} = \bigotimes'_{x \in |X|} \Lambda_x$ is commutative, where we use Λ_x to denote the commutative algebra of symmetric functions $\Lambda_{h^2=q_x}$ in the last section. We omit the X in the notations if there is no ambiguity.

We have

Proposition 2.2.5. The space of unramified automorphic forms of same rank AF_n is the $\mathbf{H}^{\mathrm{Tor}}$ -module under the map $[\mathcal{F}] \rightarrow T_{\mathcal{F}}$, and the subspace $\mathrm{AF}_n^{\mathrm{cusp}}$ is in fact a submodule.

Proof. AF_n as $\mathbf{H}^{\mathrm{Tor}}$ -module is just tautological. And $\mathrm{AF}_n^{\mathrm{cusp}}$ is $\mathbf{H}^{\mathrm{Tor}}$ stable just because the action of $\mathrm{U}(\mathbb{A})$ in the criterion of cuspidal functions is by left, while the Hecke action is by right. $\square$

Definition 2.2.6. Consider the character $\mu : \mathrm{Pic}(X) \rightarrow \mathbb{C}^{\times}$, those eigenforms are $f \in \mathrm{AF}_n$ so that $f(\mathcal{E} \otimes \mathcal{L}) = \mu(\mathcal{L})f(\mathcal{E})$.

We set the space of **Cusp-eigenforms** with respect to μ to be $\mathrm{AF}_n^{\mathrm{cusp}}(\mu)$, elements in which are those cuspidal forms with the above eigen property. On the other hand, one can consider the characters of the Hall algebra $\chi : \mathbf{H}^{\mathrm{Tor}} \rightarrow \mathbb{C}^{\times}$ and the eigenspace with respect to χ , that is $(T_{\mathcal{F}}g)(\mathcal{E}) = \chi([\mathcal{F}])g(\mathcal{E})$. A χ -eigenform uniquely determines a μ -eigenform, since for $f \in \mathrm{AF}_n^{\mathrm{cusp}}(\mu)$

$$\chi([\mathcal{O}_x^{\oplus n}])f(\mathcal{E}) = f(\mathcal{E} \otimes \mathcal{O}(-x)) = \mu(\mathcal{O}(-x))f(\mathcal{E}) \quad \text{if } f \text{ is } \chi\text{-eigenform.}$$

We have the spectral decomposition $\mathrm{AF}_n^{\mathrm{cusp}}(\mu) = \bigoplus_{\chi} \mathrm{AF}_n^{\mathrm{cusp}}(\mu)_{\chi}$, and according to multiplicity-one theorem, we know that the eigenspace so that χ, μ agrees at each place x is at most one-dimensional. We can thus omit the central character μ and call **Cusp-eigenforms** for those cuspidal forms with eigenvalue corresponding to Hall algebra characters.

Let $\mathrm{Cusp}_n \subseteq \mathrm{AF}_n$ be the set of non-zero representative of cusp-eigenforms. For $f \in \mathrm{Cusp}_n$, it is uniquely associated with a character χ_f . Define also $\chi_f^{\vee} : \mathbf{H}^{\mathrm{Tor}} \rightarrow \mathbb{C}^{\times}$ by setting: $\chi_f^{\vee}([\mathcal{O}_x^i]) = \begin{cases} \chi_f([\mathcal{O}_x^{n-i}])\chi_f^{-1}([\mathcal{O}_x^n]), & i \leq n \\ 0, & i > n \end{cases}$

Lemma 2.2.7. (i) For $f \in \mathrm{Cusp}_n$, it is also a eigenform of dual Hecke operator, given by $T_{\mathcal{F}}^{\vee}f = \chi_f^{\vee}([\mathcal{F}])f$.

(ii) For $\mathcal{F} \in \mathrm{Tor}(X)$, we have $\chi_f^{\vee}([\mathcal{F}]) = \overline{\chi_f([\mathcal{F}])}$.

Definition 2.2.8. We define the **L-function** of f as follows: set $\lambda_{1,x}(\chi_f), \dots, \lambda_{n,x}(\chi_f)$ be a sequence depending on the point $x \in |X|$ and χ_f (Here n is the same as the rank of automorphic form) so that it is the solution of the system of equations

$$e_l(\lambda_{1,x}(\chi_f)^{-1}, \dots, \lambda_{n,x}(\chi_f)^{-1}) = q_x^{\frac{-l(n-l)}{2}} \chi([\mathcal{O}_x^{\oplus l}])$$

where e_l is the elementary symmetric function, $q_x = q^{\deg(x)}$ as in Section 2.2.1. In general, we have for any partition $\underline{\mu}$

$$q_x^{-\sum_{i=1}^s (i-1)\mu_i} P_{\underline{\mu}}\left(q_x^{\frac{n-1}{2}} \lambda_{1,x}(\chi_f)^{-1}, \dots, q_x^{\frac{n-1}{2}} \lambda_{n,x}(\chi_f)^{-1}; q_x^{-1}\right) = \chi_f\left(\bigoplus_{i=1}^s \mathcal{O}_x / \mathfrak{m}_x^{\mu_i}\right), \quad \underline{\mu} = (\mu_1 \geq \dots \geq \mu_s) \quad (2.2)$$

Define $L(f, t) := \prod_{x \in X} \prod_{i=1}^n \frac{1}{1 - \lambda_{i,x}(\chi_f) t^{\deg(x)}}$ be the L-function of f (as well as χ_f).

For $f \in \text{Cusp}_n, g \in \text{Cusp}_m$, can define the Rankin-Selberg L-function

$$L \text{ Hom}(f, g, t) := \prod_{x \in X} \prod_{i=1}^n \prod_{j=1}^m \frac{1}{1 - \frac{\lambda_{i,x}(g)}{\lambda_{i,x}(f)} t^{\deg(x)}}$$

By the Langlands correspondence of GL_n , see [Laf02] and [FGV02], f corresponds to a GL_n local system $\mathcal{L}_f$, and $L \text{ Hom}(f, g, t)$ could be understood as the L-function associated with the local system $\underline{\text{Hom}}(\mathcal{L}_f, \mathcal{L}_g)$. We shall give an explicit explanation for this in Theorem 2.2.15. Before that, let us see a description for the space of cuspidal eigenforms, given by so called Witt scheme.

Locally on point $x \in X$, we set the local Witt scheme to be $W_x := \text{Spec}(\mathbf{H}(\text{Coh}_{0,\{x\}})) =: \text{Spec}(\Lambda_x)$. As is shown in Section 2.2.1, Λ_x is a commutative Hopf algebra, this gives W_x structure of formal affine group scheme.

For some $\mathbb{C}$ -algebra R and an R -valued character (an R -point of W_x) $\chi : \Lambda_x \rightarrow R$, can associate a formal series by setting $B_\chi(t) := 1 + \sum_{d \geq 1} \chi(q_x^{\frac{d(d-1)}{2}} [\mathcal{O}_x^{\oplus d}]) t^d \in R[[t]]$. And $B(t)$ sends $\text{Hom}(\Lambda_x, R)$ to $1 + tR[[t]]$, motivates the name of Witt scheme. In fact $B(t)$ yields an isomorphism:

Lemma 2.2.9. [KSV17, Proposition 2.10] *The above group scheme W_x is isomorphic to classical Witt scheme defined as the spectrum of ring of Witt vectors.*

Also, if we consider $b_d^R : W_x(R) \rightarrow R, \chi \mapsto \chi(q_x^{\frac{d(d-1)}{2}} [\mathcal{O}_x^{\oplus d}])$, then those b_d are functions of coordinates.

Thus a rank r truncated Witt scheme $W_x^{\leq r}$, in above sense of coordinates, is defined by $\text{Spec}(\mathbb{C}[b_1, \dots, b_r])$. It is closed in W_x with a strata $W_x^r = \text{Spec}(\mathbb{C}[b_1, \dots, b_r, b_r^{-1}])$, given by characters χ so that the leading coefficient of truncated series $B_\chi^{\leq r}(t) := 1 + \sum_{d \geq 1}^r \chi(q_x^{\frac{d(d-1)}{2}} [\mathcal{O}_x^{\oplus d}]) t^d \in R[t]$ is invertible (for example, $\chi = \chi_{f,x}$ for some $f \in \text{Cusp}_r$, and we restrict to the local character). Then W_x is as the injective limit of $W_x^{\leq r}$.

Note that the Hecke algebra $\mathcal{H}_x^r$ is isomorphic to $\mathbb{C}[T_{x,1}, \dots, T_{x,r}, T_{x,r}^{-1}]$, where $T_{x,i}$ is the representative of generators of double coset $\text{GL}_r(\mathcal{O}_x) \backslash \text{GL}_r(F_x) / \text{GL}_r(\mathcal{O}_x)$ so the map $b_i \mapsto T_{x,i}$ gives an isomorphism between truncated Witt scheme and spectrum of Hecke algebra.

Definition 2.2.10. Assume $\chi : \Lambda_x \rightarrow \mathbb{C}$ is a local character, we say it is of rank n if $b_n(\chi)$ is invertible. For a rank n local character, define the **Euler L-factor** of χ to be

$$L(\chi, t) := \frac{1}{B_\chi^{\leq n}(t)} = \prod_{i=1}^n \frac{1}{1 - \lambda_i(\chi) t}$$

where λ_i forms the roots of $B_\chi^{\leq n}(t)$. In particular, in eigenform case, this gives the local factor of the L-function defined in 2.2.8.

Recall that the classical Witt scheme is a ring scheme, this structure is inherited by W_x as follows:

- For rank n characters, those λ_i mapping χ to roots of $B_\chi^{\leq n}(t)$ gives a coordinate of W_x^n , we indeed have a ramified covering by $\widetilde{W}_x^n := \text{Spec}(\mathbb{C}[\lambda_1^{\pm 1}, \dots, \lambda_n^{\pm 1}]) \rightarrow W_x^n, (\lambda_1, \dots, \lambda_n) \mapsto B(t) = \prod_{i=1}^n (1 - \lambda_i t)$.
- Note that the closure $\overline{W}_x^n$ of locally closed scheme W_x^n is $W_x^{\leq n}$, the above covering is naturally extended to $\widetilde{W}_x^{\leq n} := \text{Spec}(\mathbb{C}[\lambda_1, \dots, \lambda_n]) \rightarrow W_x^{\leq n}$.
- The ring scheme structure of W_x (we denote the outer multiplication by $\boxtimes$) is given by the restriction

$$\boxtimes_{r,s} : W_x^{\leq r} \times W_x^{\leq s} \rightarrow W_x^{\leq rs}, \left(\prod_{i=1}^r (1 - \lambda_i t) \right) \boxtimes \left(\prod_{j=1}^s (1 - \mu_j t) \right) \mapsto \prod_{i=1}^r \prod_{j=1}^s (1 - \lambda_i \mu_j t)$$

For two charaters χ and χ' of Λ_x (do NOT need them to be of same rank), the **Rankin-Selberg Euler L-factor** is defined as

$$L \text{ Hom}(\chi, \chi', t) := L(\chi^\vee \boxtimes \chi', t)$$

where $\chi^\vee$ is defined by $\lambda_i(\chi^\vee) = \lambda_i(\chi)^{-1}$.

Definition 2.2.11. Going from local to global, we set the global Witt scheme (and its r -truncation) of X to be

$$W_X := \text{Spec}(\mathbf{H}^{\text{Tor}}) = \prod_{x \in X} W_x, \quad W_X^r := \prod_{x \in X} W_x^r$$

Remark. Recall our previous identification $W_x^r \simeq \text{Spec}(\mathcal{H}_x^r)$, W_X^r is thus the spectrum of global Hecke algebra of rank r , namely $W_X^r \simeq \text{Spec}(\text{Fun}_0(\text{GL}_r(\mathbb{O}_X) \backslash \text{GL}_r(\mathbb{A}_K) / \text{GL}_r(\mathbb{O}_X), \mathbb{C}) =: \text{Spec}(\Lambda^r)$. We know Λ^r is a convolution algebra with a projection map $\Lambda := \mathbf{H}^{\text{Tor}} \rightarrow \Lambda^r$.

The global Witt scheme is also a ring scheme, see [KSV17, Lemma 2.11, Proposition 2.16] for criterion and demonstration.

A Λ character valued in $\mathbb{C}$ is given by $\chi = (\chi_x)_{x \in X} : \Lambda \rightarrow \mathbb{C}$, for such a character, its **L-series** is given by

$$L(\chi, t) = \prod_{x \in X} L(\chi_x, t^{\deg x})$$

The global Rankin-Selberg L-series is similarly defined as $L \text{Hom}(\chi, \chi', t) = \prod_{x \in X} L \text{Hom}(\chi_x, \chi'_x, t^{\deg x})$. This matches to our previous definition in Definition 2.2.8. Moreover, the Rankin-Selberg L-functions come from the cusp eigenforms satisfy the rationality and functional equations, concluded in [Laf02, Appendice B].

Theorem 2.2.12. (i) Let $f \in \text{Cusp}_n$ and $g \in \text{Cusp}_m$ be cuspidal eigenforms, with corresponding characters χ_f, χ_g , then $L \text{Hom}(\chi_f, \chi_g, t)$ is a t -rational function satisfies the functional equation

$$L \text{Hom}(\chi_f, \chi_g, \frac{1}{qt}) = \mu_{\chi_f \boxtimes \chi_g^\vee}(\Omega_X^1)(q^{\frac{1}{2}}t)^{2(1-g_X)mn} L \text{Hom}(\chi_f, \chi_g, t)$$

where $\mu_{\chi_f} \in \widehat{\text{Pic}(X)}$ denotes the character of $\text{Pic}(X)$ (uniquely) corresponds to the eigen-character associated with f .

(ii) If $m \neq n$, then $L \text{Hom}(\chi_f, \chi_g, t)$ is a polynomial in t , of degree $2(g_X - 1)mn$.

The Proposition 2.2.5 tells us $\text{AF}_n^{\text{cusp}}$ is a $\mathbf{H}^{\text{Tor}}$ module through the projection $\mathbf{H}^{\text{Tor}} \rightarrow \Lambda^n$. We let

$$\Sigma^{(n)} := \text{Spec}(\Lambda^n / \text{Ann}(\text{AF}_n^{\text{cusp}})) \subseteq W_X^n$$

be the support of (quasi-coherent)sheaf on W_X^n given by $\text{AF}_n^{\text{cusp}}$. Then the following gives a schematic description of space of cusp eigenforms.

Proposition 2.2.13. (i) The set of representative of cusp eigenforms Cusp_n is in bijection with $\Sigma^{(n)}(\mathbb{C})$.

(ii) The scheme $\Sigma^{(n)}$ is an algebraic variety.

Proof. To prove (i), note first that $\chi_f, f \in \text{Cusp}$ determines a point in $\Sigma^{(n)}(\mathbb{C})$.

For the converse side, we refer to the method in [KSV17, Proposition 3.4]. Set $Q := (\oplus_d \text{AF}_{n,d}^{\text{cusp}})^\vee$. This is a right $\mathbf{H}^{\text{Tor}}$ module with respect to the left $\mathbf{H}^{\text{Tor}}$ module structure on $\text{AF}_n^{\text{cusp}}$. For each n , associate a universal cuspidal form

$$f_n^{\text{uni}} : \text{Bun}_{n,X} \rightarrow Q, [\mathcal{E}] \mapsto \text{ev}_{[\mathcal{E}]} : (f \in \text{AF}_n^{\text{cusp}} \mapsto f([\mathcal{E}]))$$

It is cuspidal in the sense of Definition 2.2.2, $\Delta(f_n^{\text{uni}}) = 1 \otimes f_n^{\text{uni}} + f_n^{\text{uni}} \otimes 1$ and commutes with the $\mathbf{H}^{\text{Tor}}$ -action $(T_{\mathcal{F}} \cdot f_n^{\text{uni}})(\mathcal{E}) = 1_{[\mathcal{F}]} \cdot f_n^{\text{uni}}(\mathcal{E})$. It is universal in the sense that for any $\mathbf{H}^{\text{Tor}}$ -module M and function $f : \text{Bun}_{n,X} \rightarrow M$ that is cuspidal and commutes with $\mathbf{H}^{\text{Tor}}$ -action, f factors through a unique $\mathbf{H}^{\text{Tor}}$ -module morphism $Q \rightarrow M$ by f_n^{uni} . The existence of such universal cuspidal form is [KSV17, Lemma 3.5]. Take a character $\chi : \mathbf{H}^{\text{Tor}} \rightarrow \mathbb{C}$, $f \in \text{AF}_n$ is a cuspidal eigenform of character χ if and only if satisfies the conditions above to factor through f_n^{uni} . Thus the set non-zero cuspidal eigenform of character χ is identified with $\text{Hom}_{\mathbf{H}^{\text{Tor}}}(Q, \mathbb{C}_\chi)$. Need to consider those non-zero morphisms $Q \rightarrow \mathbb{C}_\chi$. Indeed, Q as Λ^n module is cyclic, namely $Q \simeq \mathbb{C}[\Sigma^{(n)}]$ as $\mathbf{H}^{\text{Tor}}$ module. Hence the Hom set is non-empty if we choose χ from $\Sigma^{(n)}(\mathbb{C})$.

So it remains to prove that Q is cyclic. For $x \in X$, then the Hecke operator $T_{\mathcal{O}_x^{\oplus n}} \in \text{Aut}(\text{AF}_n)$ given by tensoring $\mathcal{O}_X(-x)$ is invertible in Λ^n , with its inverse, it generates a subalgebra $\langle T_{\mathcal{O}_x^n}, T_{\mathcal{O}_x^n}^{-1} \rangle$ of Λ^n which is isomorphic to $\mathbb{C}[z, z^{-1}]$. Since tensoring $\mathcal{O}_X(-x)$ gives an isomorphism $\text{Bun}_{n,d}(X) \rightarrow \text{Bun}_{n,d-n \deg_x}(X)$, the action of $T_{\mathcal{O}_x^{\oplus n}}$ is periodic. Thus $\text{AF}_n^{\text{cusp}}$ and its dual space Q is finitely generated as graded $\mathbb{C}[z, z^{-1}]$ module. Then Q gives a coherent sheaf $\mathcal{Q}$ on $\mathbb{G}_m$. $\mathcal{Q}$ is a vector bundle since it is $\mathbb{G}_m$ equivariant, and Λ^n action on $\mathcal{Q}$ is the bundle endomorphism. To prove that Q is cyclic, it enough to find some fiber of $\mathcal{Q}$ to be Λ^n -cyclic. Consider $\mathcal{Q}_1^\vee$ is the dual of the fiber at the point 1, then it is the space of cuspidal forms $f : \text{Bun}_{n,X} \rightarrow \mathbb{C}$ so that $f(\mathcal{E} \otimes \mathcal{O}(x)) = f(\mathcal{E})$. By multiplicity one theorem, $\mathcal{Q}_1^\vee$ is direct sum of dimension 1 subspaces invariant by Λ^n , and the Λ^n action on each space is by distinct characters. Thus $\mathcal{Q}_1^\vee$ is cyclic with primitive element to be the diagonal labelled by characters. (ii) follows from the previous discussion on vector bundle $\mathcal{Q}$. $\square$

Remark. Note also in above proof, according to the coherence of the sheaf $\mathcal{Q}$, we know that the set of eigenvalues of rank n cusp eigenforms is a finite union of $\mathbb{G}_m$ families (the variety $\Sigma^{(n)}$ is finite union of $\mathbb{G}_m$ orbits).

From a view of Langlands correspondence, can consider $\mathcal{L} \in \text{Loc}_{\text{GL}_n}(X)$ local systems of rank n on X (as $\overline{\mathbb{Q}}_l$ sheaves). Then on each point $x \in X$, there is a Frobenius action $\text{Fr}(x, \mathcal{L})$ on the stalk $\mathcal{L}_x$ by $\text{Fr}_x \in \text{Gal}(\overline{\mathbb{F}_{q_x}}/\mathbb{F}_{q_x})$, define the **L-function for $\mathcal{L}$** to be

$$L(\mathcal{L}, t) := \prod_{x \in X} \frac{1}{\det(1 - \text{Fr}(x, \mathcal{L})t^{\deg_x})}$$

It satisfies the functional equation intertwining $\mathcal{L}$ and its dual local system $\mathcal{L}^\vee$:

$$L(\mathcal{L}^\vee, \frac{1}{qt}) = c_{\mathcal{L}}(q^{\frac{1}{2}}t)^{2(1-g_X)n} L(\mathcal{L}, t)$$

Example 2.2.14. Consider the trivial local system $\overline{\mathbb{Q}}_l$, it has L-function $L(\overline{\mathbb{Q}}_l, t) = \zeta_X(t)$ equals the Weil-zeta function of X .

Theorem 2.2.15. Consider the subspace $\Sigma_{\text{fin}}^{(n)}(\mathbb{C}) \subseteq \Sigma^{(n)}(\mathbb{C})$ consists those with eigen character χ whose corresponding $\mu_\chi \in \widehat{\text{Pic}(X)}$ has finite order. Let $D_0^{(n)}$ be the set of isomorphism classes of local systems $\mathcal{L}$ that $\det(\mathcal{L})$ has finite order. Then there exists a bijection $\Sigma_{\text{fin}}^{(n)}(\mathbb{C}) \longrightarrow D_0^{(n)}$ sending χ to a local system $\mathcal{L}_\chi$ so that for any $\chi \in \Sigma_{\text{fin}}^{(n)}(\mathbb{C})$, $\chi' \in \Sigma_{\text{fin}}^{(m)}(\mathbb{C})$,

$$L \text{Hom}(\chi', \chi, t) = L(\underline{\text{Hom}}(\mathcal{L}_{\chi'}, \mathcal{L}_\chi), t)$$

2.2.3 The Hopf algebra of automorphic forms

We may restrict ourselves to the space AF_{fs} of automorphic forms with finite support, by definition, it is the sub-algebra (but NOT a sub-bi-algebra) $\mathbf{H}^{\text{Bun}}$ of $\mathbf{H}$. The action of $\mathbf{H}^{\text{Tor}}$ on $(\mathbf{H}^{\text{Bun}}, *)$ is given by $T_f \cdot g := \pi(f * g)$, where $\pi : \mathbf{H} \rightarrow \mathbf{H}^{\text{Bun}}$ is the restriction and $f \in \mathbf{H}^{\text{Tor}}$, $g \in \mathbf{H}^{\text{Bun}}$. We also call this action the Hecke action since it is in the same spirit as before.

Then the Hall algebra on curve X is isomorphic to the semi-direct product $\mathbf{H} \simeq \mathbf{H}^{\text{Bun}} \rtimes \mathbf{H}^{\text{Tor}}$ with respect to the Hecke action, this follows as a corollary from

Proposition 2.2.16. Let $f \in \mathbf{H}^{\text{Tor}}$, $g \in \mathbf{H}^{\text{Bun}}$, assume that $\Delta(f) = \sum_i f'_i \otimes f''_i$ (Notice this sum is finite due to the finiteness of support of torsion sheaves), then we have $f * g = \sum_i T_{f'_i}(g) * f''_i$.

Proof. We test the equality for arbitrary coherent sheaves. Let $\mathcal{E} \in \text{Coh}(X)$, then it is uniquely written as $\mathcal{E} = \mathcal{T} \oplus \mathcal{F}$. Compute

$$\begin{aligned}
(f * g \mid \mathbf{1}_{[\mathcal{E}]}) &= q^{-\frac{\langle \mathcal{F}, \mathcal{T} \rangle}{2}} (f * g \mid \mathbf{1}_{[\mathcal{F}]} * \mathbf{1}_{[\mathcal{T}]}) = q^{-\frac{\langle \mathcal{F}, \mathcal{T} \rangle}{2}} (\Delta(f) * \Delta(g) \mid \mathbf{1}_{[\mathcal{F}]} \otimes \mathbf{1}_{[\mathcal{T}]}) \\
&= q^{-\frac{\langle \mathcal{F}, \mathcal{T} \rangle}{2}} \left(\sum_i (f'_i \otimes f''_i) * (g \otimes 1) \mid \mathbf{1}_{[\mathcal{F}]} \otimes \mathbf{1}_{[\mathcal{T}]} \right) = q^{-\frac{\langle \mathcal{F}, \mathcal{T} \rangle}{2}} \left(\sum_i f'_i * g \otimes f''_i \mid \mathbf{1}_{[\mathcal{F}]} \otimes \mathbf{1}_{[\mathcal{T}]} \right) \\
&= q^{-\frac{\langle \mathcal{F}, \mathcal{T} \rangle}{2}} \left(\sum_i T_{f'_i}(g) \otimes f''_i \mid \mathbf{1}_{[\mathcal{F}]} \otimes \mathbf{1}_{[\mathcal{T}]} \right) \text{ (Note that } (f'_i * g \mid [\mathcal{F}]) = (\pi(f'_i * g) \mid [\mathcal{F}])) \\
&= \left(\sum_i T_{f'_i}(g) \otimes f''_i \mid \mathbf{1}_{[\mathcal{F} \oplus \mathcal{T}]} \right)
\end{aligned}$$

Note as $g \in \mathbf{H}^{\text{Bun}}$, any vector bundle could not contribute to the pairing on the torsion part. $\square$

Recall the equivalent definition 2.2.2 of cuspidal automorphic form, we say $f \in \mathbf{H}^{\text{Bun}}$ is cuspidal if $(\pi \otimes \pi)(\Delta(f)) = 1 \otimes f + f \otimes 1$. Let $\mathbf{H}^{\text{cusp}}$ denote the space of cuspidal functions in $\mathbf{H}^{\text{Bun}}$. For class $\gamma = (r, d)$, we use $\mathbf{H}^{\text{Bun}}_\gamma$ (resp. $\mathbf{H}^{\text{cusp}}_\gamma$) to denote the functions (resp. cuspidal functions) support on $\text{Bun}_{r,d}(X)$.

By definition, we know elements of $\mathbf{H}^{\text{Bun}}_{1,d}, \forall d \in \mathbb{Z}$ are all cuspidal. Moreover, one fact motivates the consideration of cuspidal functions is the following

Proposition 2.2.17. *For any $\gamma = (r, d)$, the space of cuspidal functions satisfies*

$$\mathbf{H}^{\text{cusp}}_\gamma = \left(\bigoplus_{\alpha + \beta = \gamma, \alpha, \beta \neq \gamma} \mathbf{H}^{\text{Bun}}_\alpha * \mathbf{H}^{\text{Bun}}_\beta \right)^\perp \cap \mathbf{H}^{\text{Bun}}_\gamma$$

where the orthogonality is with respect to the Hopf pairing. And as a corollary, $\mathbf{H}^{\text{Bun}}$ is generated by primitive elements $\mathbf{H}^{\text{cusp}}$.

Proof. Take $f \in \mathbf{H}^{\text{cusp}}_\gamma$ and $(g, h) \in \mathbf{H}^{\text{Bun}}_\alpha \times \mathbf{H}^{\text{Bun}}_\beta$, then $(f \mid g * h) = (\Delta(f) \mid g \otimes h) = (\pi^{\otimes 2}(\Delta(f)) \mid g \otimes h) = 0$ since f supports on different strata from g and h . $\square$

Proposition 2.2.18. *For any class γ , all functions in $\mathbf{H}^{\text{cusp}}_\gamma$ have finite support.*

Proof. Consider two vector bundles $\mathcal{V}_1$ and $\mathcal{V}_2$, assume they satisfy $\text{Ext}^1(\mathcal{V}_1, \mathcal{V}_2) = 0$. Then for $\mathcal{V} \in \text{Coh}(X)$, the pairing $(\mathbf{1}_{[\mathcal{V}]} \mid \mathbf{1}_{[\mathcal{V}_1]} * \mathbf{1}_{[\mathcal{V}_2]}) = (\Delta(\mathbf{1}_{[\mathcal{V}]}) \mid \mathbf{1}_{[\mathcal{V}_1]} \otimes \mathbf{1}_{[\mathcal{V}_2]})$ is non-zero only when $\mathcal{V} \simeq \mathcal{V}_1 \oplus \mathcal{V}_2$. So that $f \mid \mathbf{1}_{[\mathcal{V}_1 \oplus \mathcal{V}_2]} = 0$ by Proposition 2.2.17. To prove the proposition, it is thus enough to show that for the class γ , only finitely many vector bundles of this class are not of the form as an extension of vector bundles $\mathcal{V}_1$ and $\mathcal{V}_2$ so that $\text{Ext}^1(\mathcal{V}_1, \mathcal{V}_2) = 0$.

Note that if the slopes satisfy $\mu(\mathcal{V}_1) > \mu(\mathcal{V}_2) + 2(g_X - 1)$, then $\text{Ext}^1(\mathcal{V}_1, \mathcal{V}_2) = 0$, thus for fixed γ , there are only finitely many Harder-Narasimhan type $\underline{\alpha} = (\alpha_1, \dots, \alpha_s)$ with $\mu(\alpha_{i+1}) \leq \mu(\alpha_i) + 2(g_X - 1), \forall i = 1, \dots, s-1$. Let $\text{HN}^\alpha \subseteq \text{Bun}_\gamma$ be the stack of vector bundles with HN type α , then $\text{HN}^\alpha \rightarrow \prod_i \text{Bun}^{\text{ss}}_{\alpha_i}$ is a vector bundle stack. Hence $\# \text{Obj}(\text{HN}^\alpha(\mathbb{F}_q)) / \sim$ is finite because $\text{Bun}^{\text{ss}}_\alpha$ is stack of finite type. We can also conclude that $\mathbf{H}^{\text{cusp}}_\gamma$ is a finite dimensional space $\square$

For $f \in \text{Cusp}_n$ a cusp eigenform of rank n with eigen-character χ_f , we introduce the Eisenstein series associated with f to be

$$E_f(z) := \sum_{\mathcal{V} \in \text{Bun}_{n,X}} f(\mathcal{V}) z^{\deg(\mathcal{V})} \mathbf{1}_{[\mathcal{V}]} \in \mathbf{H}[[t, t^{-1}]]$$

Also the generating series of Hecke operators

$$\psi_f(z) = \sum_{\mathcal{F} \in \text{Tor}(X)} \overline{\chi_f}([\mathcal{F}]) z^{h^0(\mathcal{F})} a_{\mathcal{F}} \mathbf{1}_{[\mathcal{F}]} \in \mathbf{H}[[t]], \text{ where } a_{\mathcal{F}} = |\text{Aut}(\mathcal{F})|$$

Lemma 2.2.19. *Let $f_1, \dots, f_s$ be cusp eigenforms (not necessarily of the same rank), then for any $\mathcal{F} \in \text{Coh}(X)$, the product of Eisenstein series values at $\mathcal{F}$*

$$E_{f_1}(z_1) * E_{f_2}(z_2) * \dots * E_{f_s}(z_s)(\mathcal{F})$$

belongs to the ring $\mathbb{C}[z_1^{\pm 1}, \dots, z_s^{\pm 1}] \otimes_{\mathbb{C}[\frac{z_1}{z_2}, \dots, \frac{z_{s-1}}{z_s}]} \mathbb{C}[[\frac{z_1}{z_2}, \dots, \frac{z_{s-1}}{z_s}]]$ in the region $|z_s| \gg \dots \gg |z_1|$.

Proof. This result is due to an easy combinatoric. We can consider the valuation at a vector bundle $\mathcal{V}$, then the degree of sub-sheaves $\mathcal{V}_s \subseteq \mathcal{V}$ of the same rank as f_s is bounded (by the maximum degree of Harder-Narasimhan filtration of $\mathcal{V}$), and for each $\mathcal{V}_s \subseteq \mathcal{V}$, the degree of subsheaves of $\mathcal{V}/\mathcal{V}_s$ is again bounded. Inductively do the process on the filtration of $\mathcal{V}$, we conclude. $\square$

The following stronger theorem is proved by Harder [Har74, Theorem 1.4.4]

Theorem 2.2.20. *Take the same assumption as in Lemma 2.2.19, we have the series $E_{f_1}(z_1) * E_{f_2}(z_2) * \dots * E_{f_s}(z_s)(\mathcal{F})$ converges in the region $|z_s| \gg \dots \gg |z_1|$, to a rational function in $\mathbb{C}(z_1, \dots, z_s)$.*

Based on Harder's theorem, we consider the intertwining of Eisenstein series,

Theorem 2.2.21. *Let $f \in \text{Cusp}_n, g \in \text{Cusp}_m$ be two cusp eigenforms, then*

$$E_g(w) * E_f(z) = q^{mn(1-g_X)} \frac{L \text{Hom}(f, g, \frac{w}{z})}{L \text{Hom}(f, g, \frac{w}{qz})} E_f(z) * E_g(w)$$

Proof. One of the proof is given by principal series intertwiners, we give a brief sketch.

Let $\Xi_{m,n} := \begin{pmatrix} \text{GL}_m(K) & 0 \\ \text{Mat}_{n,m}(\mathbb{A}) & \text{GL}_n(K) \end{pmatrix} \subset \text{GL}_{m+n}(\mathbb{A})$, then we have an isomorphism obtained by Iwasawa decomposition

$$\theta_{m,n} : \Xi_{m,n} \backslash \text{GL}_{m+n}(\mathbb{A}) / \text{GL}_{m+n}(\mathbb{O}) \xrightarrow{\sim} (\text{GL}_m(K) \backslash \text{GL}_m(\mathbb{A}) / \text{GL}_m(\mathbb{O})) \times (\text{GL}_n(K) \backslash \text{GL}_n(\mathbb{A}) / \text{GL}_n(\mathbb{O})) \simeq \text{Bun}_{m,X} \times \text{Bun}_{n,X}$$

This induces an identification of function spaces by

$$\begin{aligned} \widehat{\epsilon}_{m,n} : (\mathbf{H}_m \widehat{\otimes} \mathbf{H}_n) &= \text{Fun}(\text{Bun}_{m,X} \times \text{Bun}_{n,X}) \longrightarrow \text{Fun}(\Xi_{m,n} \backslash \text{GL}_{m+n}(\mathbb{A}) / \text{GL}_{m+n}(\mathbb{O})) \\ (\tilde{f}_m, \tilde{f}_n) &\mapsto (\bar{g} \mapsto \langle \mathcal{V}'', \mathcal{V}' \rangle \tilde{f}_m(\mathcal{V}') \tilde{f}_n(\mathcal{V}'')) \end{aligned}$$

where $\bar{g}$ is the class in the double coset uniquely corresponding to the pair of class $([\mathcal{V}'], [\mathcal{V}'']) \in \text{Bun}_{m,X} \times \text{Bun}_{n,X}$.

$\widehat{\epsilon}_{m,n}$ restrict to the identification of function spaces of finite support: $\epsilon_{m,n} : \mathbf{H}_m \otimes \mathbf{H}_n \xrightarrow{\sim} \text{Fun}_0(\Xi_{m,n} \backslash \text{GL}_{m+n}(\mathbb{A}) / \text{GL}_{m+n}(\mathbb{O}))$. There is another map called pseudo-Eisenstein series map: $\text{Eis}_{m,n} : \text{Fun}_0(\Xi_{m,n} \backslash \text{GL}_{m+n}(\mathbb{A})) \longrightarrow \text{Fun}_0(\text{GL}_{m+n}(K) \backslash \text{GL}_{m+n}(\mathbb{A}))$

by $(\text{Eis}_{m,n}(f))(g) := \sum_{\gamma \in P_{m,n}(K) \backslash \text{GL}_{m+n}(K)} f(\gamma g)$, where $P_{m,n} = \begin{pmatrix} \text{GL}_m & 0 \\ \text{Mat}_{n,m} & \text{GL}_n \end{pmatrix}$ is the opposite standard parabolic.

Since $\text{Eis}_{m,n}$ is equivariant with respect to right action of $\text{GL}_{m+n}(\mathbb{A})$, it induces the map $\overline{\text{Eis}}_{m,n}$ between $\text{GL}_{m+n}(\mathbb{O})$ -invariant subspaces. Then those operators fit into the following diagram

$$\begin{array}{ccc} \mathbf{H}_m \otimes \mathbf{H}_n & \xrightarrow{\quad * \quad} & \mathbf{H}_{m+n} \\ \epsilon_{m,n} \downarrow \simeq & & \uparrow \simeq \\ \text{Fun}_0(\Xi_{m,n} \backslash \text{GL}_{m+n}(\mathbb{A}))^{\text{GL}_{m+n}(\mathbb{O})} & \xrightarrow[\overline{\text{Eis}}_{m,n}]{} & \text{Fun}_0(\text{GL}_{m+n}(K) \backslash \text{GL}_{m+n}(\mathbb{A}))^{\text{GL}_{m+n}(\mathbb{O})} \end{array}$$

We have [KSV17, Proposition 4.10]

Lemma 2.2.22. *The diagram is commutative, i.e. $\overline{\text{Eis}}_{m,n} \circ \epsilon_{m,n}$ is exactly the multiplication $*$ in the Hall algebra.*

To swap the position of GL_m and GL_n , we use the principal series intertwiner introduced in [MW95]

$$M_{m,n} : \mathrm{Fun}_0(\Xi_{m,n} \setminus \mathrm{GL}_{m+n}(\mathbb{A})) \longrightarrow \mathrm{Fun}(\Xi_{n,m} \setminus \mathrm{GL}_{m+n}(\mathbb{A}))$$

$$f \mapsto \left(g \mapsto \int_{Z \in \mathrm{Mat}_{n,m}(\mathbb{A})} f \left(\begin{pmatrix} Z & \mathbf{1}_n \\ \mathbf{1}_m & 0 \end{pmatrix} \right) dZ \right)$$

where the measure is chosen such that $\int_{\mathbb{A}/K} dZ_{ij} = 1$ for each matrix component. Similar to above, by the right-invariance of $\mathrm{GL}_{m+n}(\mathbb{A})$ -action, we have $M_{m,n}$ inducing the map $\mathbf{H}_m \otimes \mathbf{H}_n \longrightarrow \mathbf{H}_n \widehat{\otimes} \mathbf{H}_m$ so that $\mathrm{Eis}_{m,n} = \mathrm{Eis}_{n,m} \circ M_{m,n}$. The expression of $M_{m,n}$ is given by

Lemma 2.2.23. *For sequences of cusp eigenforms $\mathbf{f} = \{f_1, \dots, f_s\}$, $\mathbf{f}' = \{f'_1, \dots, f'_t\}$, in the region that $|z_1| \gg \dots \gg |z_s|$ and $|z'_1| \gg \dots \gg |z'_t|$, $M_{m,n}(E_{\mathbf{f}'}(\mathbf{z}') \otimes E_{\mathbf{f}}(\mathbf{z}))$ converges to*

$$E_{\mathbf{f}}(\mathbf{z}) \otimes E_{\mathbf{f}'}(\mathbf{z}') \cdot \prod_{i=1}^s \prod_{j=1}^t q^{(1-g_X)n_i m_j} \frac{L \mathrm{Hom}(\chi_{f_i}, \chi_{f'_j}; \frac{z_i}{z'_j})}{L \mathrm{Hom}(\chi_{f_i}, \chi_{f'_j}; \frac{z_i}{q z'_j})}$$

where $n = \sum n_i$ for $f_i \in \mathrm{Cusp}_{n_i}$ and $m = \sum m_j$ for $f'_j \in \mathrm{Cusp}_{m_j}$.

By restricting to local factors and to the case of functions on $\mathrm{GL}_2(F_x)$, the integration map is given as [Kap97, (3.4.4)]. We give, rather a translation of $M_{m,n}$ in the setting of Hall algebras, so that a Langlands' formula of constant term in the following form is obtained (see also [KSV17, (4.13)]):

Proposition 2.2.24. *Consider $\mathbf{f} = (f_1, \dots, f_s)$ a sequence of cusp eigenforms, denote $E_{\mathbf{f}}(\mathbf{z}) := E_{f_1}(z_1) * \dots * E_{f_s}(z_s)$ the pseudo Eisenstein series. Assume $f_i \in \mathrm{Cusp}_{n_i}$ and $n := \sum_i n_i$. For (n', n'') such that $n' + n'' = n$, let $\Delta_{(n', n'')}(x)$ denote the coefficient of $z_1^{d_1} \dots z_s^{d_s}$ in the constant term of $E_{\mathbf{f}}(\mathbf{z})$ along the parabolic $P_{n', n''} = \begin{pmatrix} \mathrm{GL}_{n'} & 0 \\ \mathrm{Mat}_{n'', n'} & \mathrm{GL}_{n''} \end{pmatrix}$, then*

$$\Delta_{(n', n'')}(E_{\mathbf{f}}(\mathbf{z})) = \sum_{\substack{(s', s'') \\ s' + s'' = s}} \sum_{\substack{(i_1, \dots, i_{s'}) \\ (j_1, \dots, j_{s''}), \sum i_k = n', \sum j_l = n''}} \prod_{i_k > j_l} q^{n_{i_k} n_{j_l} (1-g_X)} \frac{L \mathrm{Hom}(f_{i_k}, f_{j_l}; \frac{z_{i_k}}{z_{j_l}})}{L \mathrm{Hom}(f_{i_k}, f_{j_l}; \frac{z_{i_k}}{q z_{j_l}})}$$

$$\times \left(E_{f_{i_1}}(z_{i_1}) * \dots * E_{f_{i_{s'}}}(z_{i_{s'}}) \right) \otimes \left(E_{f_{j_1}}(z_{j_1}) * \dots * E_{f_{j_{s''}}}(z_{j_{s''}}) \right)$$

And our principal intertwiner $M_{m,n} : \mathbf{H}_m \otimes \mathbf{H}_n \longrightarrow \mathbf{H}_n \widehat{\otimes} \mathbf{H}_m$ is $u \otimes v \mapsto \pi^{\otimes 2}(\Delta_{(0,m)}(u) * \Delta_{(n,0)}(v))$.

Proof of Proposition 2.2.24. The proof is established by the following Hall algebra relations

Proposition 2.2.25. *Let $f \in \mathrm{Cusp}_n, g \in \mathrm{Cusp}_m$ as 2.2.21, then we have*

- (i) $\Delta \psi_f(z) = \psi_f(z) \otimes \psi_f(z), \quad \Delta E_f(z) = \psi_f(q^{-\frac{n}{2}} z) \otimes E_f(z) + E_f(z) \otimes 1$
- (ii) $\psi_g(w) * E_f(z) = \frac{L \mathrm{Hom}(f, g, q^{\frac{m}{2}} \frac{w}{z})}{L \mathrm{Hom}(f, g, q^{\frac{m}{2}-1} \frac{w}{z})} E_f(z) * \psi_g(w).$

Proof of Proposition 2.2.25. We first compute the coproduct of Hecke-generating series

$$\begin{aligned}\Delta(\psi_f(z)) &= \Delta \left(\sum_{\mathcal{F} \in \text{Tor}(X)} \bar{\chi}_f([\mathcal{F}]) a_{\mathcal{F}} z^{h^0(\mathcal{F})} \mathbf{1}_{[\mathcal{F}]} \right) \\ &= \sum_{\mathcal{F}} \bar{\chi}_f([\mathcal{F}]) z^{h^0(\mathcal{F})} \sum_{\mathcal{F}', \mathcal{F}'' \in \text{Tor}(X)} \langle \mathcal{F}', \mathcal{F}'' \rangle \mathbf{P}_{\mathcal{F}', \mathcal{F}''} \mathbf{1}_{[\mathcal{F}']} \otimes \mathbf{1}_{[\mathcal{F}'']}\end{aligned}$$

Notice that all $\mathcal{F}, \mathcal{F}', \mathcal{F}''$ are torsion, thus $\langle \mathcal{F}', \mathcal{F}'' \rangle = 1$, and $\left| \frac{\mathbf{P}_{\mathcal{F}', \mathcal{F}''}^{\mathcal{F}}}{a_{\mathcal{F}'} a_{\mathcal{F}''}} \right| = 1 = \# \{ \mathcal{G} \subseteq \mathcal{F} | \mathcal{G} \simeq \mathcal{F}''; \mathcal{F}/\mathcal{G} \simeq \mathcal{F}' \}$

$$\begin{aligned}&= \left(\sum_{\mathcal{F}'} \bar{\chi}_f([\mathcal{F}']) z^{h^0(\mathcal{F}')} a_{\mathcal{F}'} \mathbf{1}_{[\mathcal{F}']} \right) \otimes \left(\sum_{\mathcal{F}''} \bar{\chi}_f([\mathcal{F}'']) z^{h^0(\mathcal{F}'')} a_{\mathcal{F}''} \mathbf{1}_{[\mathcal{F}'']} \right) \\ &= \psi_f(z) \otimes \psi_f(z)\end{aligned}$$

as wanted. We compute next the coproduct of Eisenstein series:

$$\begin{aligned}\Delta(E_f(z)) &= \sum_{\mathcal{V} \in \text{Bun}_{n,X}} f(\mathcal{V}) z^{\deg(\mathcal{V})} \Delta(\mathbf{1}_{[\mathcal{V}]}) \\ &= \sum_{\mathcal{V}} \sum_{\mathcal{F}, \mathcal{G}} \langle \mathcal{G}, \mathcal{F} \rangle \frac{1}{a_{\mathcal{V}}} \mathbf{P}_{\mathcal{G}, \mathcal{F}}^{\mathcal{V}} f(\mathcal{V}) z^{\deg(\mathcal{V})} \mathbf{1}_{[\mathcal{G}]} \otimes \mathbf{1}_{[\mathcal{F}]}\end{aligned}$$

here $\mathcal{F}$ is regarded as the subsheaf of $\mathcal{V}$, thus is torsion free. For $\mathcal{G}$, we can decompose it into $\mathcal{G} = \mathcal{G}_{\text{tor}} \oplus \mathcal{G}_{\text{cotor}}$. Consider

$$\begin{array}{ccccccc} & & 0 & \longrightarrow & 0 & \longrightarrow & \mathcal{G}_{\text{tor}} \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{F} & \longrightarrow & \mathcal{V} & \longrightarrow & \mathcal{G} \longrightarrow 0 \\ & & \downarrow i & & \parallel & & \downarrow \\ 0 & \longrightarrow & \bar{\mathcal{F}} & \longrightarrow & \mathcal{V} & \xrightarrow{p} & \mathcal{G}_{\text{cotor}} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & \text{Coker}(i) & \longrightarrow & 0 & \longrightarrow & 0 \end{array}$$

where $\bar{\mathcal{F}}$ is the kernel of p , and by snake lemma, we see $\text{Coker}(i) \simeq \mathcal{F}$, thus the extension of $\mathcal{G}$ by $\mathcal{F}$ can be decomposed into two short exact sequences $0 \longrightarrow \bar{\mathcal{F}} \longrightarrow \mathcal{V} \longrightarrow \mathcal{G}_{\text{cotor}} \longrightarrow 0$ and $0 \longrightarrow \mathcal{F} \longrightarrow \bar{\mathcal{F}} \longrightarrow \mathcal{G}_{\text{tor}} \longrightarrow 0$. By definition, we have

$$\frac{1}{a_{\mathcal{G}} a_{\mathcal{F}}} \mathbf{P}_{\mathcal{G}, \mathcal{F}}^{\mathcal{V}} = \sum_{\bar{\mathcal{F}}} \frac{1}{a_{\mathcal{G}_{\text{cotor}}} a_{\bar{\mathcal{F}}}} \cdot \frac{1}{a_{\mathcal{G}_{\text{tor}}} a_{\mathcal{F}}} \mathbf{P}_{\mathcal{G}_{\text{cotor}}, \bar{\mathcal{F}}}^{\mathcal{F}} \mathbf{P}_{\bar{\mathcal{F}}, \mathcal{F}}^{\mathcal{G}_{\text{tor}}}$$

Moreover, consider the map $\pi^{\otimes 2} \circ \Delta : \text{AF}_n \longrightarrow \bigoplus_{i+j=n} \text{AF}_i \otimes \text{AF}_j$ is given by

$$f \mapsto \left((\mathcal{W}', \mathcal{W}'') \mapsto \sum_{\mathcal{V} \in \text{Bun}_n} \frac{1}{a_{\mathcal{V}}} \mathbf{P}_{\mathcal{W}', \mathcal{W}''}^{\mathcal{V}} f(\mathcal{V}) \right)$$

Hence

$$\begin{aligned}
\Delta(E_f(z)) &= \sum_{\mathcal{V}} \sum_{\substack{\mathcal{F}, \bar{\mathcal{F}}, \mathcal{G}_{cotor} \in \text{Bun}(X) \\ \mathcal{G}_{tor} \in \text{Tor}(X)}} q^{\text{rk}(\mathcal{G}_{cotor})h^0(\mathcal{G}_{tor})} f(\mathcal{V}) \langle \mathcal{G}_{cotor} \oplus \mathcal{G}_{tor}, \mathcal{F} \rangle \frac{1}{a_{\mathcal{V}}} \frac{1}{a_{\mathcal{F}}} \mathbf{P}_{\mathcal{G}_{tor}, \mathcal{F}}^{\bar{\mathcal{F}}} \mathbf{P}_{\mathcal{G}_{cotor}, \bar{\mathcal{F}}} \times \\
&\quad z^{\deg(\mathcal{V})} \mathbf{1}_{[\mathcal{G}_{cotor} \oplus \mathcal{G}_{tor}]} \otimes \mathbf{1}_{[\mathcal{F}]} \\
&= \sum_{\mathcal{F}, \bar{\mathcal{F}}, \mathcal{G}_{cotor}, \mathcal{G}_{tor}} q^{\text{rk}(\mathcal{G}_{cotor})h^0(\mathcal{G}_{tor})} (\pi^{\otimes 2} \circ \Delta(f)(\bar{\mathcal{F}}, \mathcal{G}_{cotor})) \frac{1}{a_{\bar{\mathcal{F}}}} \mathbf{P}_{\mathcal{G}_{tor}, \mathcal{F}}^{\bar{\mathcal{F}}} \langle \mathcal{G}_{cotor} \oplus \mathcal{G}_{tor}, \mathcal{F} \rangle \times \\
&\quad z^{\deg(\bar{\mathcal{F}}) + \deg(\mathcal{G}_{cotor})} \mathbf{1}_{[\mathcal{G}_{cotor} \oplus \mathcal{G}_{tor}]} \otimes \mathbf{1}_{[\mathcal{F}]}
\end{aligned}$$

By our assumption, f is cuspidal, thus $\pi^{\otimes 2} \circ \Delta(f)(\bar{\mathcal{F}}, \mathcal{G}_{cotor}) = 0$ unless one of them is zero.

- If $\bar{\mathcal{F}} = 0$, then $\mathcal{V} \simeq \mathcal{G}_{cotor}$ and $\mathcal{G}_{tor} = 0$, the summation gives

$$\sum_{\mathcal{G}} f(\mathcal{G}) z^{\deg(\mathcal{G})} (\mathbf{1}_{[\mathcal{G}]} \otimes 1) = E_f(z) \otimes 1$$

- If $\mathcal{G}_{cotor} = 0$, then the summation becomes

$$\begin{aligned}
&\sum_{\mathcal{F}, \bar{\mathcal{F}}, \mathcal{G}_{tor}} f(\bar{\mathcal{F}}) \frac{1}{a_{\bar{\mathcal{F}}}} \mathbf{P}_{\mathcal{G}_{tor}, \mathcal{F}}^{\bar{\mathcal{F}}} \langle \mathcal{G}_{tor}, \mathcal{F} \rangle z^{\deg(\mathcal{F}) + h^0(\mathcal{G}_{tor})} \mathbf{1}_{[\mathcal{G}_{tor}]} \otimes \mathbf{1}_{[\mathcal{F}]} \\
&= \sum_{\mathcal{F} \in \text{Bun}_{n,X}} \sum_{\mathcal{T} \in \text{Tor}(X)} (T_{\mathcal{T}}^{\vee} f)(\mathcal{F}) a_{\mathcal{T}} q^{\frac{-n}{2} h^0(\mathcal{T})} z^{\deg(\mathcal{F}) + h^0(\mathcal{T})} \mathbf{1}_{[\mathcal{T}]} \otimes \mathbf{1}_{[\mathcal{F}]} \quad (\text{by definition of } T_{\mathcal{T}}^{\vee}) \\
&= \left(\sum_{\mathcal{T} \in \text{Tor}(X)} \bar{\chi}_f([\mathcal{T}]) a_{\mathcal{T}} (q^{\frac{-n}{2}} z)^{h^0(\mathcal{T})} \mathbf{1}_{[\mathcal{T}]} \right) \otimes \left(\sum_{\mathcal{F} \in \text{Bun}_{n,X}} f(\mathcal{F}) z^{\deg(\mathcal{F})} \mathbf{1}_{[\mathcal{F}]} \right) \quad (\text{by Lemma 2.2.7}) \\
&= \psi_f(q^{\frac{-n}{2}} z) \otimes E_f(z)
\end{aligned}$$

For (ii), since $\text{Ext}^1(\mathcal{V}, \mathcal{T}) = 0$ for $\mathcal{V}$ locally free, $\mathcal{T}$ torsion, we have $\mathbf{1}_{[\mathcal{V}]} * \mathbf{1}_{[\mathcal{F}]} = \langle \mathcal{V}, \mathcal{F} \rangle [\mathcal{V} \oplus \mathcal{F}]$. We can first compute

$$\begin{aligned}
E_f(z) * \psi_g(w) &= \left(\sum_{\mathcal{V} \in \text{Bun}_{n,X}} f(\mathcal{V}) z^{\deg(\mathcal{V})} \mathbf{1}_{[\mathcal{V}]} \right) * \left(\sum_{\mathcal{F} \in \text{Tor}(X)} \bar{\chi}_g([\mathcal{F}]) w^{h^0(\mathcal{F})} a_{\mathcal{F}} \mathbf{1}_{[\mathcal{F}]} \right) \\
&= \sum_{\mathcal{F} \in \text{Tor}(X)} \sum_{\mathcal{V} \in \text{Bun}_{n,X}} \bar{\chi}_g([\mathcal{F}]) f(\mathcal{V}) a_{\mathcal{F}} q^{\frac{n}{2} h^0(\mathcal{F})} w^{h^0(\mathcal{F})} z^{\deg(\mathcal{V})} \mathbf{1}_{[\mathcal{V} \oplus \mathcal{F}]} \quad (2.3)
\end{aligned}$$

Switching the order of multiplication, we know for $\mathcal{F} \in \text{Tor}(X)$, $\mathcal{V} \in \text{Bun}_{n,X}$

$$\begin{aligned}
\mathbf{1}_{[\mathcal{F}]} * \left(\sum_{\mathcal{V} \in \text{Bun}_{n,X}} f(\mathcal{V}) \mathbf{1}_{[\mathcal{V}]} \right) &= \sum_{\mathcal{V}} \sum_{\widetilde{\mathcal{W}} \in \text{Coh}(X)} f(\mathcal{V}) \langle \mathcal{F}, \mathcal{V} \rangle \frac{1}{a_{\mathcal{F}} a_{\mathcal{V}}} \mathbf{P}_{\mathcal{F}, \mathcal{V}}^{\widetilde{\mathcal{W}}} \mathbf{1}_{[\widetilde{\mathcal{W}}]} \\
&= \sum_{\mathcal{V}} \sum_{\substack{\mathcal{W} \in \text{Bun}_{n,X} \\ \mathcal{F}' \in \text{Tor}(X)}} f(\mathcal{V}) \langle \mathcal{F}, \mathcal{V} \rangle \frac{1}{a_{\mathcal{F}} a_{\mathcal{V}}} \mathbf{P}_{\mathcal{F}, \mathcal{V}}^{\mathcal{W} \oplus \mathcal{F}'} \mathbf{1}_{[\mathcal{W} \oplus \mathcal{F}']} \quad (\text{take unique decomposition of coherent sheaf})
\end{aligned}$$

The above fits into the diagram

$$\begin{array}{ccccccc}
& & \mathcal{F}' & \xlongequal{\quad} & \mathcal{F}' & & \\
& & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathcal{V} & \longrightarrow & \mathcal{W} \oplus \mathcal{F}' & \longrightarrow & \mathcal{F} \longrightarrow 0 \\
& & \parallel & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathcal{V} & \longrightarrow & \mathcal{W} & \longrightarrow & \mathcal{F}'' \longrightarrow 0
\end{array}$$

To compute $\mathbf{P}_{\mathcal{V}, \mathcal{F}}^{\mathcal{W} \oplus \mathcal{F}'}$, notice we are in the case of the cross $J(\mathcal{V}, \mathcal{F}, \mathcal{F}', \mathcal{W})$ as in the proof of Theorem 2.1.4. One can show that

$$\frac{\mathbf{P}_{\mathcal{F}, \mathcal{V}}^{\mathcal{W} \oplus \mathcal{F}'}}{a_{\mathcal{F}} a_{\mathcal{V}}} = \sum_{\mathcal{F}' \subseteq \mathcal{F}} \frac{\mathbf{P}_{\mathcal{F}/\mathcal{F}', \mathcal{V}}^{\mathcal{W}}}{a_{\mathcal{F}}^2 a_{\mathcal{V}}^2} a_{\mathcal{F}'} q^{nh^0(\mathcal{F}')}$$

We then have

$$\begin{aligned}
\mathbf{1}_{[\mathcal{F}]} * \left(\sum_{\mathcal{V} \in \text{Bun}_{n, X}} f(\mathcal{V}) \mathbf{1}_{[\mathcal{V}]} \right) &= \sum_{\mathcal{F}' \subseteq \mathcal{F}} \sum_{\mathcal{W}} \sum_{\substack{\mathcal{V} \subseteq \mathcal{W} \\ \mathcal{W}/\mathcal{V} \simeq \mathcal{F}/\mathcal{F}'}} \frac{\mathbf{P}_{\mathcal{W}/\mathcal{V}, \mathcal{V}}^{\mathcal{W}}}{a_{\mathcal{V}}^2 a_{\mathcal{F}}^2} a_{\mathcal{F}'} f(\mathcal{V}) \mathbf{1}_{[\mathcal{W} \oplus \mathcal{F}']} q^{nh^0(\mathcal{F}')} \langle \mathcal{F}, \mathcal{V} \rangle \\
&= \sum_{\mathcal{F}' \subseteq \mathcal{F}} \sum_{\mathcal{W}} (T_{\mathcal{F}/\mathcal{F}'} f)(\mathcal{W}) \frac{\mathbf{P}_{\mathcal{F}/\mathcal{F}', \mathcal{V}}^{\mathcal{W}}}{a_{\mathcal{V}}^2 a_{\mathcal{F}}^2} a_{\mathcal{F}'} \mathbf{1}_{[\mathcal{W} \oplus \mathcal{F}']} q^{\frac{n}{2} h^0(\mathcal{F}')}
\end{aligned} \tag{2.4}$$

Using formula 2.4, we can compute

$$\psi_g(w) * E_f(z) = \sum_{\mathcal{F} \in \text{Tor}(X)} \bar{\chi}_g([\mathcal{F}]) a_{\mathcal{F}} \sum_{\mathcal{F}' \subseteq \mathcal{F}} \sum_{\mathcal{W} \in \text{Bun}_{n, X}} \chi_f([\mathcal{F}/\mathcal{F}']) f(\mathcal{W}) q^{\frac{nh^0(\mathcal{F}') - nh^0(\mathcal{F}/\mathcal{F}')}{2}} \frac{\mathbf{P}_{\mathcal{F}/\mathcal{F}', \mathcal{V}}^{\mathcal{W}}}{a_{\mathcal{V}}^2 a_{\mathcal{F}}^2} a_{\mathcal{F}'} \times$$

Note that $\mathcal{F}$ is the extension of $\mathcal{F}''$ by $\mathcal{F}'$, and $\mathbf{P}_{\mathcal{F}/\mathcal{F}'}^{\mathcal{W}} = a_{\mathcal{V}} a_{\mathcal{F}/\mathcal{F}'}$

$$\begin{aligned}
&= \sum_{\mathcal{W}} \sum_{\mathcal{F}', \mathcal{F}'' \in \text{Tor}(X)} \chi_f([\mathcal{F}'']) \bar{\chi}_g([\mathcal{F}'']) \bar{\chi}_g([\mathcal{F}']) f(\mathcal{W}) \frac{a_{\mathcal{F}'} a_{\mathcal{F}''}}{a_{\mathcal{V}} a_{\mathcal{F}}} q^{\frac{nh^0(\mathcal{F}') - nh^0(\mathcal{F}/\mathcal{F}')}{2}} \times \\
&= \left(\sum_{\mathcal{W}} \sum_{\mathcal{F}'} \bar{\chi}_g([\mathcal{F}']) f(\mathcal{W}) q^{\frac{nh^0(\mathcal{F}')}{2}} a_{\mathcal{F}'} z^{\deg(\mathcal{W})} w^{h^0(\mathcal{F}')} \mathbf{1}_{[\mathcal{W} \oplus \mathcal{F}']} \right) \\
&\quad \times \left(\sum_{\mathcal{F}''} \chi_f([\mathcal{F}'']) \bar{\chi}_g([\mathcal{F}'']) q^{\frac{-nh^0(\mathcal{F}'')}{2}} z^{-h^0(\mathcal{F}'')} w^{h^0(\mathcal{F}'')} a_{\mathcal{F}''} \right)
\end{aligned}$$

Comparing to 2.3, this equals to

$$= (E_f(z) * \psi_g(w)) \times \Lambda_{f, g} \left(\frac{w}{q^{\frac{n}{2}} z} \right)$$

where we set

$$\Lambda_{f, g}(t) := \sum_{\mathcal{F} \in \text{Tor}(X)} \chi_f([\mathcal{F}]) \bar{\chi}_g([\mathcal{F}]) a_{\mathcal{F}} t^{h^0(\mathcal{F})}$$

and $\Lambda_{f, g}(t)$ can be computed explicitly. For each $\mathcal{F} \in \text{Tor}(X)$, $\mathcal{F} = \bigoplus_{x \in X} \mathcal{F}_{x, \underline{\mu}}$, where $\underline{\mu} = (\mu_1 \geq \dots \geq \mu_s)$ is the unique partition of $\mathcal{F}_x$. And $\Lambda_{f, g}(t) = \prod_{x \in X} \Lambda_{f, g, x}(t^{\deg(x)})$, where $\Lambda_{f, g, x}(t) := \sum_{\underline{\mu} \in \mathcal{P}} \chi_f([\mathcal{F}_{x, \underline{\mu}}]) \bar{\chi}_g([\mathcal{F}_{x, \underline{\mu}}]) a_{\mathcal{F}_{x, \underline{\mu}}} t^{|\underline{\mu}|}$. By 2.2,

$$q_x^{-\sum (i-1)\mu_i} P_{\underline{\mu}} \left(q_x^{\frac{n-1}{2}} \lambda_{1, x}(\chi_f)^{-1}, \dots, q_x^{\frac{n-1}{2}} \lambda_{n, x}(\chi_f)^{-1}; q_x^{-1} \right) = \chi_f \left(\bigoplus_{i=1}^s \mathcal{O}_x / \mathfrak{m}_x^{\mu_i} \right) = \chi_f([\mathcal{F}_{x, \underline{\mu}}])$$

The similar for $\bar{\chi}_g(\left[\mathcal{F}_{x,\mu}\right])$. Summing up the partitions, we have the Macdonald's formula

$$\Lambda_{f,g,x}(t) = \prod_{i=1}^n \prod_{j=1}^m \frac{1 - q x^{\frac{n+m}{2}-1} \frac{\lambda_{j,x}(g)}{\lambda_{i,x}(f)} t}{1 - q x^{\frac{n+m}{2}} \frac{\lambda_{j,x}(g)}{\lambda_{i,x}(f)} t}; \quad \Lambda_{f,g}(t) = \frac{L \operatorname{Hom}(f, g, q^{\frac{n+m}{2}}; t)}{L \operatorname{Hom}(f, g, q^{\frac{n+m}{2}-1}; t)}$$

We conclude that $\psi_g(w) * E_f(z) = \frac{L \operatorname{Hom}(f, g, q^{\frac{m}{2}} \frac{w}{z})}{L \operatorname{Hom}(f, g, q^{\frac{m}{2}-1} \frac{w}{z})} E_f(z) * \psi_g(w)$, and thus the Proposition is proved. $\square$

Now we consider the intertwining of Eisenstein series, as rational functions, we have

$$\begin{aligned} M_{m,n}(E_g(w) \otimes E_f(z)) &= \pi^{\otimes 2} ((\psi_g(q^{-\frac{m}{2}} w) \otimes E_g(w)) * (E_f(z) \otimes 1)) \\ &= q^{(1-g_X)mn} \pi^{\otimes 2} (\psi_g(q^{-\frac{m}{2}} w) * E_f(z) \otimes E_g(w)) \quad (\text{Notice that it is a twisted multiplication on } \mathbf{H} \otimes \mathbf{H}) \\ &= q^{(1-g_X)mn} \frac{L \operatorname{Hom}(f, g; \frac{w}{z})}{L \operatorname{Hom}(f, g; \frac{w}{qz})} \pi^{\otimes 2} ((E_f(z) * \psi_g(q^{-\frac{m}{2}} w)) \otimes E_g(w)) \\ &= q^{(1-g_X)mn} \frac{L \operatorname{Hom}(f, g; \frac{w}{z})}{L \operatorname{Hom}(f, g; \frac{w}{qz})} E_f(z) \otimes E_g(w) \end{aligned}$$

The last step uses the fact that $\mathbf{1}_{[\mathcal{V}]} * \mathbf{1}_{[\mathcal{T}]} = \mathbf{1}_{[\mathcal{V} \oplus \mathcal{T}]}$ in $\mathbf{H}$ for $\mathcal{V} \in \operatorname{Bun}(X)$ and $\mathcal{T} \in \operatorname{Tor}(X)$. By reducing to the case when $s = 2$, the Proposition 2.2.24 and hence the theorem is proved. $\square$

Further, using the intertwiner $M_{m,n}$, it can be shown that $\mathbf{H}$ is made into a braided bialgebra, we refer to [KSV17, Theorem 4.9] for details. $\square$

2.2.4 Spherical Hall algebras

In the previous sections, we have seen that the Hall algebra $\mathbf{H}_X$ of a curve encodes the space of automorphic forms and the Hecke actions. Moreover, $\mathbf{H}_X$ is generated by the subspace of cuspidal forms as an algebra. In this section, we consider the 'principal' cusp eigenforms and compute the related Rankin-Selberg L-functions.

Definition 2.2.26. Set characteristic functions on $\operatorname{Bun}(X)$ and $\operatorname{Tor}(X)$ to be

$$\mathbf{1}_{0,d} := \sum_{\substack{\mathcal{T} \in \operatorname{Tor}(X) \\ \deg(\mathcal{T})=d}} \mathbf{1}_{[\mathcal{T}]}(d \geq 1); \quad \mathbf{1}_{r,d} = \sum_{\mathcal{F} \in \operatorname{Coh}_{r,d}(X)} \mathbf{1}_{[\mathcal{F}]}; \quad \mathbf{1}_{r,d}^{\operatorname{Bun}} := \sum_{\mathcal{V} \in \operatorname{Bun}_{r,d}(X)} \mathbf{1}_{[\mathcal{V}]}; \quad \mathbf{1}_{r,d}^{\operatorname{ss}} := \sum_{\mathcal{V} \in \operatorname{Bun}_{r,d}^{\operatorname{ss}}(X)} \mathbf{1}_{[\mathcal{V}]}.$$

Define the **spherical Hall algebra** to be the sub-algebra of $\mathbf{H}$, generated by $\left\{ \mathbf{1}_{1,k}^{\operatorname{ss}} = \mathbf{1}_{1,k}^{\operatorname{Bun}}, \mathbf{1}_{1,d} \right\}_{k \in \mathbb{Z}, d \geq 1}$, denoted by $\mathbf{H}^{\operatorname{sph}}$.

Notice that $\mathbf{1}_{0,d} \in \mathbf{H}^{\operatorname{Tor}}$ since for each fixed degree d , the number of closed points on X whose degree $< d$ is finite and $\mathbf{1}_{1,k}^{\operatorname{Bun}} = \mathbf{1}_{1,k}^{\operatorname{ss}} \in \mathbf{H}^{\operatorname{Bun}}$ since all vector bundles of rank one are semistable and $\operatorname{Bun}_{1,d}(X) = \operatorname{Pic}^d(X)$ is finite, while for higher rank, $\mathbf{1}_{r,d}$ is in the topological completion of Hall algebra, which we denote as $\bar{\mathbf{H}}$.

Set $\bar{\mathbf{H}}^{\operatorname{sph}}$ to be the completion of $\mathbf{H}^{\operatorname{sph}}$ in $\bar{\mathbf{H}}$, in which the functions are as the limit of the approximation by minimal slopes, namely

$$\bar{H}_{\alpha=(r,d)}^{\operatorname{sph}} = \left\{ f \in \bar{\mathbf{H}}_{\alpha} \mid \forall n, \exists f_n \in \mathbf{H}^{\operatorname{sph}} \text{ such that } f_n|_{\operatorname{Coh}_{\alpha}^{\geq n}} = f|_{\operatorname{Coh}_{\alpha}^{\geq n}} \right\}$$

where $\operatorname{Bun}^{\geq n}$ is the substack of $\operatorname{Bun}(X)$ (resp. $\operatorname{Coh}^{\geq n} \subseteq \operatorname{Coh}(X)$ subcategory) classifying the vector bundles (resp. coherent sheaves) with minimal slopes $\geq n$. We then have

Proposition 2.2.27. For any class $\alpha = (r, d)$, the function $\mathbf{1}_{r,d}^{\operatorname{Bun}}$ and $\mathbf{1}_{r,d}$ are contained in the completion of spherical Hall algebra $\bar{\mathbf{H}}^{\operatorname{sph}}$. Moreover, $\bar{\mathbf{H}}^{\operatorname{sph}}$ is a subalgebra of $\bar{\mathbf{H}}$.

Proof. We do the induction on the rank r . The case when $r = 1$, $\mathbf{1}_{r,d}^{\text{Bun}} \in \mathbf{H}^{\text{sph}}$ is clear by definition. And we have

$$\mathbf{1}_{1,d} = \sum_{d' \geq 0} q^{-\frac{d'}{2}} \mathbf{1}_{1,d-d'}^{\text{Bun}} * \mathbf{1}_{0,d'} \in \overline{\mathbf{H}}^{\text{sph}}$$

For general pair (r, d) , to show $\mathbf{1}_{r,d}^{\text{Bun}} \in \overline{\mathbf{H}}^{\text{sph}}$, it is enough to show that $\mathbf{1}_{r,d}^{\text{Bun}}|_{\text{Bun}_{r,d}^{\geq n}} \in \mathbf{H}^{\text{sph}}$ for any n (same for $\mathbf{1}_{r,d}$ by restricting to $\text{Coh}_{r,d}^{\geq n}$). Fix r, d, n , where n is the slope so that $n \leq \frac{d}{r}$. Assume that for $r' < r$, $\mathbf{1}_{r',d'}^{\text{Bun}}, \mathbf{1}_{r',d'} \in \overline{\mathbf{H}}^{\text{sph}}$ for arbitrary degree d' , assume also that $\mathbf{1}_{r,d'}^{\text{Bun}} \in \overline{\mathbf{H}}^{\text{sph}}$ for $rn \leq d' < d$. Note that if $d' < rn$, then $\mathbf{1}_{r,d'}^{\text{Bun}}|_{\text{Bun}_{r,d'}^{\geq n}} = 0$, so we can first exclude such situation. Choose l small enough, and consider the multiplication $\mathbf{1}_{r-1,d-l} * \mathbf{1}_{1,l} = \sum_{\mathcal{F} \in \text{Coh}(X)} c_{\mathcal{F}} \mathbf{1}_{[\mathcal{F}]}$ in $\overline{\mathbf{H}}$. For any coherent sheaf $\mathcal{F}$ with $c_{\mathcal{F}} \neq 0$, we can compute this coefficient by counting the extensions of the form $0 \rightarrow \mathcal{L} \rightarrow \mathcal{F} \rightarrow \mathcal{E} \rightarrow 0$, with $\mathcal{L} \in \text{Pic}^l(X)$. By definition, $c_{\mathcal{F}} = q^{-\frac{\langle (r-1, d-l), (1, l) \rangle}{2}} \# \{ \mathcal{L} \subseteq \mathcal{F} \mid \mathcal{L} \in \text{Pic}^l(X) \}$.

As we choose small l , $\dim \text{Hom}(\mathcal{L}, \mathcal{F}) = \langle [\mathcal{L}], [\mathcal{F}] \rangle$ as $\text{Ext}^1(\mathcal{L}, \mathcal{F}) = 0$. We can separate the morphism $\mathcal{L} \rightarrow \mathcal{F}_{\text{tor}}$ and $\mathcal{L} \rightarrow \mathcal{F}_{\text{cotor}}$, while the first one depends only on $\deg(\mathcal{F}_{\text{tor}})$. We can deduce that for any $\mathcal{F}, \mathcal{G} \in \text{Coh}_{r,d}$ with $\deg(\mathcal{F}_{\text{tor}}) = \deg(\mathcal{G}_{\text{tor}})$, we have $c_{\mathcal{F}} = c_{\mathcal{G}}$. Geometrically, consider extension diagram $\text{Coh}_{\alpha-(1,l)} \times \text{Coh}_{1,l} \xleftarrow{q} \overline{\text{Coh}}_{\alpha-(1,l),(1,l)} \xrightarrow{p} \text{Coh}_{\alpha}$, the above argument shows for $\alpha = (r, d')$, the fiber of p on $\text{Bun}_{\alpha}^{\geq n}$ is equi-dimensional. In all, we have the multiplication being

$$\mathbf{1}_{r-1,d-l} * \mathbf{1}_{1,l} |_{\text{Coh}^{\geq n}} = \sum_{l=0}^{d-rn} c_l \mathbf{1}_{r,d-l}^{\text{Bun}} * \mathbf{1}_{0,l}$$

Then by induction hypothesis, we recurse to be consequence that $\mathbf{1}_{r,d}^{\text{Bun}} \in \overline{\mathbf{H}}^{\text{sph}}$ and we see $\mathbf{1}_{r,\bullet} \in \overline{\mathbf{H}}^{\text{sph}}$ by simply joining the torsions. To check that $\overline{\mathbf{H}}^{\text{sph}}$, we only need to show $\Delta(\mathbf{1}_{1,d}) \in \mathbf{H} \widehat{\otimes} \mathbf{H}$ by continuity of Δ and the generating relation. Inverting the formula $\mathbf{1}_{1,d} = \sum_{d' \geq 0} q^{-\frac{d'}{2}} \mathbf{1}_{1,d-d'}^{\text{Bun}} * \mathbf{1}_{0,d'} \in \overline{\mathbf{H}}^{\text{sph}}$, we have

$$\begin{aligned} \mathbf{1}_{r,d}^{\text{Bun}} &= \mathbf{1}_{r,d} - \sum_{d_1 \geq 1} q^{-\frac{rd_1}{2}} (\mathbf{1}_{r,d-d_1} - \sum_{d_2 \geq 1} q^{-\frac{rd_2}{2}} \mathbf{1}_{r,d-d_1-d_2}^{\text{Bun}}) * \mathbf{1}_{0,d_1} \\ &= \mathbf{1}_{r,d} - \sum_{i \geq 1} (-1)^{i+1} \sum_{d_1, \dots, d_i} q^{-\frac{r(\sum_{j=1}^i d_j)}{2}} \mathbf{1}_{1,d-(\sum_{j=1}^i d_j)} * \mathbf{1}_{0,d_i} * \dots * \mathbf{1}_{0,d_1} \end{aligned}$$

We see the above sum (and hence the sum of $\Delta(\mathbf{1}_{r,d}^{\text{Bun}})$) terminates if put restriction on slopes. □

Inspired by the constant term formula as Proposition 2.2.24, we introduce the constant term map to study the sturcture of spherical Hall algebra.

Definition 2.2.28. Let $\mathbf{H}^{\text{sph}, >}$ be the subalgebra of $\mathbf{H}$ generated by $\mathbf{1}_{1,d}^{\text{ss}}$. For $r \geq 1$, define the **constant term map** as

$$\text{CT}_r := \pi^{\otimes r} \circ \Delta_{1, \dots, 1}^{(r)} : \mathbf{H}_r^{\text{sph}, >} \longrightarrow \widehat{\otimes}_r \mathbf{H}_1^{\text{sph}, >}, \quad \text{CT} := \sum_{r \geq 1} \text{CT}_r : \mathbf{H}^{\text{sph}, >} \longrightarrow \bigoplus_{r \geq 1} \left(\mathbf{H}_1^{\text{sph}, >} \right)^{\widehat{\otimes} r}$$

Notice that by the map $\mathbf{1}_{1,d}^{\text{ss}} \mapsto x^d$, we identify $\left(\mathbf{H}_1^{\text{sph}, >} \right)^{\widehat{\otimes} r} \simeq \mathbb{C}[x_1^{\pm 1}, \dots, x_r^{\pm 1}]$, $\left(\mathbf{H}_1^{\text{sph}, >} \right)^{\widehat{\otimes} r} \simeq \mathbb{C}[x_1^{\pm 1}, \dots, x_r^{\pm 1}][[\frac{x_1}{x_2}, \dots, \frac{x_{r-1}}{x_r}]]$.

Lemma 2.2.29. *The constant term map CT is injective.*

Proof. Let f be in the kernel of CT_r , then $\text{CT}_r(f) = \sum_{\mathcal{L}_i \in \text{Pic}^{d_i}} \frac{1}{(q-1)^r} (f | \mathbf{1}_{1,d_1} * \dots * \mathbf{1}_{1,d_r}) \mathbf{1}_{\mathcal{L}_1} \otimes \dots \otimes \mathbf{1}_{\mathcal{L}_r}$. Then the results follows from the fact that $\mathbf{H}_r^{\text{sph}}$ is generated by those $\mathbf{1}_{1,d_i}$, and the Hopf pairing is positive non-degenerate. □

Remark. Using the method in the above proof, we can see in fact $\text{CT}_r : \overline{\mathbf{H}}^{\text{sph}} \longrightarrow (\overline{\mathbf{H}}_1^{\text{sph}})^{\widehat{\otimes} r}$ the extended constant term map is still injective.

We can define the Shuffle algebra structure on $\mathbf{F} := \bigoplus_{r \geq 1} (\mathbf{H}_1^{\text{sp}, >})^{\widehat{\otimes}^r} = \bigoplus_{r \geq 1} \mathbb{C}[x_1^{\pm 1}, \dots, x_r^{\pm 1}][[\frac{x_1}{x_2}, \dots, \frac{x_{r-1}}{x_r}]]$ as follows, for each rational function $h(z) \in \mathbb{C}(z)$, the multiplication on $\mathbf{F}$ is given by

$$P(x_1, \dots, x_s) \odot Q(x_1, \dots, x_t) := \sum_{\sigma \in \text{Sh}_{s,t}} h_{\sigma}(x_1, \dots, x_{s+t}) \sigma(P(x_1, \dots, x_s) Q(x_{s+1}, \dots, x_{s+t}))$$

where $\text{Sh}_{s,t} = \{\omega \in \mathfrak{S}_{s+t} \mid w(i) < w(j) \forall 1 \leq i < j \leq s \text{ or } s < i < j \leq s+t\}$ is the subset of (s, t) -shuffles in $\mathfrak{S}_{s+t}$. For each shuffle $\omega \in \text{Sh}_{s,t}$, set $I_{\omega} = \{(i, j) \mid 1 \leq i < j \leq s+t, \omega^{-1}(i) \geq r \geq \omega^{-1}(j)\}$, then $h_{\sigma}(x_1, \dots, x_{s+t}) := \prod_{(i,j) \in I_{\sigma}} h(\frac{x_i}{x_j})$. Denote the shuffle algebra associated with $h(z)$ by $\mathbf{F}_{h(z)}$, we define also the coproduct on it

$$\Delta : \mathbf{F}_{h(z)} \longrightarrow \mathbf{F}_{h(z)} \widehat{\otimes} \mathbf{F}_{h(z)}, \quad \Delta_{s,t} : x_1^{i_1} \dots x_{s+t}^{i_{s+t}} \mapsto x_1^{i_1} \dots x_s^{i_s} \otimes x_1^{i_{s+1}} \dots x_t^{i_{s+t}}$$

It is proved in [SV12, Proposition 1.6] that $\mathbf{H}^{\text{sp}, >}$ is as a subalgebra of some shuffle algebra.

Proposition 2.2.30. *The constant term map sends $\mathbf{H}^{\text{sp}, >}$ to $\mathbf{F}$ can be regarded as an algebra morphism $\mathbf{H}^{\text{sp}, >} \longrightarrow \mathbf{F}_{h_X(z)}$, where $h_X(z) := q^{1-g} \frac{Z_X(z)}{Z_X(q^{-1}z)}$. And it is compatible with the co-multiplication in the sense that $\text{CT}^{\otimes 2} \circ (\pi)^{\otimes 2} \circ \Delta = \Delta \circ \text{CT}$.*

We now consider the essential Eisenstein series of rank r as the following

$$E_r(z) := \sum_{d \in \mathbb{Z}} \mathbf{1}_{r,d} z^d; \quad E_r^{\text{Bun}}(z) := \sum_{d \in \mathbb{Z}} \mathbf{1}_{r,d} z^d (r \geq 1 \text{ in this case}). \quad (2.5)$$

Some properties of $E_r(z)$ are instant. By the uniqueness of torsion-locally free decomposition of coherent sheaves, we know that

$$E_r(z) = E_r^{\text{Bun}}(z) E_0(q^{-\frac{r}{2}} z)$$

Notice there is a twist on the torsion. Also, by the commutativity of Hecke operators, we have

$$E_0(z) * E_0(w) = E_0(w) * E_0(z) \quad \text{in } \mathbf{H}[z, w]$$

We have also the intertwining relations, residue expressions of essential Eisenstein series, due to Harder [Har74]. Those are actually the special case of what we discussed in previous section, but in concrete expression, we include them in the following

Theorem 2.2.31. (i) $E_0(z) * E_r^{\text{Bun}}(w) = \prod_{i=0}^{r-1} (Z_X(q^{-\frac{r}{2}+i} \frac{z}{w})) E_r^{\text{Bun}}(w) * E_0(z)$, where $Z_X(t)$ is the Weil zeta function associated with X , and $Z_X(q^{\bullet} \frac{z}{w})$ is expanded in the region $|z| \ll |w|$.

$$(ii) \Delta(E_r(z)) = \sum_{s+t=r} q^{\frac{(g-1)st}{2}} E_s(q^{\frac{t}{2}} z) \otimes E_t(q^{-\frac{s}{2}} z),$$

$$\Delta(E_r^{\text{Bun}}(z)) = \sum_{s+t=r} q^{\frac{(g-1)st}{2}} E_s^{\text{Bun}}(q^{\frac{t}{2}} z) * E_0(q^{\frac{t-s}{2}} z) * E_0^{-1}(q^{-\frac{r}{2}} z) \otimes E_t^{\text{Bun}}(q^{-\frac{s}{2}} z).$$

$$(iii) E_r^{\text{Bun}}(q^{\frac{1-r}{2}} z_1) = C_r \times \text{Res}_{\frac{z_r}{z_{r-1}} = \frac{z_{r-1}}{z_{r-2}} = \dots = \frac{z_2}{z_1} = q^{-1}} (E_1^{\text{Bun}}(z_r) * \dots * E_1^{\text{Bun}}(z_1)),$$

where C_r is a constant depends only on r , and Res is a iterated residue.

Remark. Notice that the coproduct (ii) in Theorem 2.2.31 is in a slightly different form than that of 2.3, as when we consider $E_r(z)$, it is the generating series of $\mathbf{1}_{\text{Coh}_r(X)}$, which is not finitely supported and bringing into the torsion.

We will see that $C_r = q^{\frac{(1-g_X)r(r-1)}{4}} \text{Vol}(\text{Bun}_{r,\bullet}(X)) \text{Vol}(\text{Bun}_{1,\bullet}(X))^{-r}$

Proof. For (i), we first consider the case when $r = 1$, the Hecke action is given by

$$T_{\mathbf{1}_{0,t}}(\mathbf{1}_{1,d}^{\text{ss}}) = q^{-\frac{l}{2}} |S^l(X)(\mathbb{F}_q)| \mathbf{1}_{1,l+d}^{\text{ss}} \quad (2.6)$$

To see this, we choose an arbitrary testing line bundle $\mathcal{L} \in \text{Pic}^0(X)$, consider the generating series

$$\begin{aligned}
\sum_{l \geq 1} T_{1_{0,l}}(\mathbf{1}_{1,-l}^{\text{ss}})(\mathcal{L})z^l &= \sum_{\lambda: |X| \rightarrow \mathbb{N}} \prod_{x \in |X|} T_{1_{\mathcal{O}_x^{\deg_x}}}(1_{1,-l}^{\text{ss}})(\mathcal{L})z^{\lambda_x \deg_x} \quad \left(l = \sum_x \lambda_x \deg_x \right) \\
&= \prod_{x \in |X|} \sum_{n \geq 0} (q^{-\frac{1}{2}}z)^{n \deg_x} = \prod_{x \in |X|} \frac{1}{1 - (q^{-\frac{1}{2}}z)^{\deg_x}} \\
&= \exp \left(\sum_{x \in |X|} \sum_{l \geq 1} \frac{(q^{-\frac{1}{2}}z)^{\deg_x} l}{l} \right) = \exp \left(\sum_{l \geq 1} |X(\mathbb{F}_{q^l})| \frac{(q^{-\frac{1}{2}}z)^l}{l} \right) \\
&= Z_X(q^{-\frac{1}{2}}z) = \sum_{n \geq 1} |S^n X(\mathbb{F}_q)| (q^{-\frac{1}{2}}z)^n
\end{aligned}$$

Then 2.6 holds in the case of $T_{1_{0,l}}(\mathbf{1}_{1,-l}^{\text{ss}})$, in general, we have that $T_{1_{\mathcal{O}_x^{\deg_x}}}(1_{1,d}^{\text{ss}}) = q^{-\frac{1}{2}n \deg_x} 1_{1,d+n \deg_x}$, use Proposition 2.2.19, we know $1_{0,l} E_1^{\text{Bun}}(w) = \sum_{l'+l''=l} T_{1_{0,l'}}(E_1^{\text{Bun}}(w)) * 1_{0,l''}$, the case of $r = 1$ then holds.

For greater r , it is enough to prove $E_0(z) * E_r(w) = \prod_{i=0}^{r-1} (Z_X(q^{-\frac{r}{2}+i} \frac{z}{w})) E_r(w) * E_0(z)$ because of the decomposition $E_r = E_r^{\text{Bun}} E_0$ and the commutativity of E_0 's. To show these two sides coincide, use the injectivity of constant term map (remark of Lemma 2.2.29), we only need to see if their images coincide. We have for two sides:

$$\begin{aligned}
\Delta_{1,\dots,1}^{(r)}(E_0(z) * E_r(w)) &= \Delta^{(r)}(E_0(z)) \Delta_{1,\dots,1}^{(r)}(E_r(w)) = E_0(z)^{\otimes r} * \left(q^{\frac{r(r-1)}{2}(g_X-1)} E_1(q^{\frac{r-1}{2}} w) \otimes E_1(q^{\frac{r-3}{2}} w) \otimes \dots \otimes E_1(q^{\frac{1-r}{2}} w) \right) \\
\Delta_{1,\dots,1}^{(r)}(E_r(w) * E_0(z)) &= \left(q^{\frac{r(r-1)}{2}(g_X-1)} E_1(q^{\frac{r-1}{2}} w) \otimes \dots \otimes E_1(q^{\frac{1-r}{2}} w) \right) * E_0(z)^{\otimes r}
\end{aligned}$$

The results follows from using case $r = 1$ to permute E_1 and E_0 .

For (ii), the first equation follows from a direct computation of the coproduct. we remark here that in general Hall algebra, $\Delta(1_\gamma) = \sum_{\alpha+\beta=\gamma} \frac{1}{\langle \alpha, \beta \rangle} 1_\alpha \otimes 1_\beta$. The second one is from the first by considering $\Delta(E_r(z)) = \Delta(E_r^{\text{Bun}}(z)) * \Delta(E_0(z))$.

For (iii), we also compare the constant term map to show the equality. One shall first note that the iterated residue involves a choice of order. For $f(z_1, \dots, z_r) \in \mathbb{C}(z_1, \dots, z_r)$ a rational function, To take the iterated residue $\text{Res}_{\frac{z_r}{z_{r-1}}=a_r, \dots, \frac{z_2}{z_1}=a_2}$ of $f(z_1, \dots, z_r)$, we first regarded a rational function in $\mathbb{C}(z_1, \dots, z_{r-1})(z_r)$ so that the first step residue at $z_r = a_r z_{r-1}$ is taken value in $\mathbb{C}(z_1, \dots, z_{r-1})$. Moving forward, in the end, $\text{Res}_{\frac{z_r}{z_{r-1}}=a_r, \dots, \frac{z_2}{z_1}=a_2} (f(z_1, \dots, z_r)) = \text{Res}_{z_2=a_2 z_1} \circ \dots \circ \text{Res}_{z_r=a_r z_{r-1}} f(z_1, \dots, z_r) \in \mathbb{C}(z_1)$ is a rational function in one variable. From now on, we abbreviate $E_{r_k}^{\text{Bun}}(z_k) * \dots * E_{r_1}^{\text{Bun}}(z_1)$ (resp. $E_{r_k}(z_k) * \dots * E_{r_1}(z_1)$) as $E_{r_k, \dots, r_1}^{\text{Bun}}(z_k, \dots, z_1)$ (resp. $E_{r_k, \dots, r_1}(z_k, \dots, z_1)$) for convenience, if there is no ambiguity. We compute

$$\text{CT}_r(E_{1,\dots,1}^{\text{Bun}}(z_r, \dots, z_1)) = \pi^{\otimes r} \left[\Delta_{1,\dots,1}^{(r)}(E_1^{\text{Bun}}(z_r)) * \dots * \Delta_{1,\dots,1}^{(r)}(E_1^{\text{Bun}}(z_1)) \right]$$

where

$$\begin{aligned}
\Delta_{1,\dots,1}^{(r)}(E_1^{\text{Bun}}(z_r)) &= \Delta_{1,\dots,1}^{(r-1)} \left(E_1^{\text{Bun}}(z_r) \otimes 1 + E_0(q^{\frac{1}{2}} z_r) * E_0(q^{-\frac{1}{2}} z_r)^{-1} \otimes E_1^{\text{Bun}}(z_r) \right) \\
&= \dots = \left[E_1^{\text{Bun}}(z_r) \otimes 1^{\otimes(r-1)} + E_0(q^{\frac{1}{2}} z_r) E_0(q^{-\frac{1}{2}} z_r)^{-1} \otimes E_1^{\text{Bun}}(z_r) \otimes 1^{\otimes(r-2)} \right. \\
&\quad \left. + \dots + \left(E_0(q^{\frac{1}{2}} z_r) E_0(q^{-\frac{1}{2}} z_r)^{-1} \otimes \dots \otimes E_0(q^{\frac{1}{2}} z_r) E_0(q^{-\frac{1}{2}} z_r)^{-1} \right) \otimes E_1^{\text{Bun}}(z_r) \right]
\end{aligned}$$

Using the projection map π to eliminate the torsion term of the coproduct, the constant term map has the image

$$\sum_{\sigma \in \mathfrak{S}_r} \left[\prod_{\substack{i > j \\ \sigma(i) < \sigma(j)}} \left(q^{1-g_X} \frac{Z_X(\frac{z_i}{z_j})}{Z_X(\frac{z_i}{q z_j})} \right) \cdot \overrightarrow{\otimes}_i E_1^{\text{Bun}}(z_{\sigma(i)}) \right], \quad \text{where } \overrightarrow{\otimes}_i \text{ is along the direction } (z_r, \dots, z_1) \quad (2.7)$$

For fixed $\iota: \bar{\mathbb{Q}}_l \xrightarrow{\sim} \mathbb{C}$, we choose $\sigma_k, k = 1, \dots, 2g_X$ be Weil numbers, so that $Z_X(t) = \frac{P(t)}{(1-t)(1-qt)}$, with $P(t) = \prod_{k=1}^{2g_X} (1-\sigma_k t) =$

$\prod_{k=1}^{g_X} (1 - \sigma_k t) (1 - \frac{q}{\sigma_k} t)$ a polynomial of degree $2g_X$. We can see

$$\text{Res}_{|t=q^{-1}} \left(\frac{Z_X(t)}{Z_X(q^{-1}t)} \right) = \left(\prod_{k=1}^{2g_X} (1 - \sigma_k^{-1}) \right) \frac{1}{Z_X(q^{-2})(1 - q^{-1})}$$

Before counting the iterated residue, we state the following obvious fact

Lemma 2.2.32. *The iterated residue $\underline{\text{Res}}$ commutes with the constant term map in the region $|z_r| \ll \dots \ll |z_1|$*

Hence, to compute $\text{CT}_r(\underline{\text{Res}}(E_{1,\dots,1}^{\text{Bun}}(\mathbf{z})))$, we can compute the residue on formula 2.7. Notice only when $\sigma = (r, r-1, \dots, 1)$ is the longest element in $\mathfrak{S}_r$, can 2.7 contribute to the whole iterated residue. In such case

$$\underline{\text{Res}}_{\frac{z_r}{z_{r-1}}=\dots=\frac{z_2}{z_1}=q^{-1}} (\text{CT}_r(E_{1,\dots,1}^{\text{Bun}}(z_r, \dots, z_1))) = C'_r \times E_1^{\text{Bun}}(z_1) \otimes E_1^{\text{Bun}}(q^{-1}z_1) \otimes \dots \otimes E_1^{\text{Bun}}(q^{1-r}z_1) \quad (2.8)$$

$$C'_r = \prod_{i>j} q^{1-g_X} \frac{Z^*(q^{j-i})}{Z^*(q^{j-i-1})}, \quad \text{where } Z^*(t) = \begin{cases} \frac{\prod_{k=1}^{2g_X} (1 - \sigma_k^{-1})}{1 - q^{-1}}, & t = q^{-1} \\ Z_X(t), & t \in q^{-2-\mathbb{N}} \end{cases} \quad (2.9)$$

On the other hand, use the result of (ii), we have

$$\text{CT}_r(E_r^{\text{Bun}}(z_1)) = q^{\frac{r(r-1)}{4}(g_X-1)} E_1^{\text{Bun}}(q^{\frac{r-1}{2}} z_1) \otimes E_1^{\text{Bun}}(q^{\frac{r-1}{2}-1} z_1) \dots \otimes E_1^{\text{Bun}}(q^{\frac{1-r}{2}} z_1) \quad (2.10)$$

Combining formula 2.7 and 2.10, we conclude that

$$E_r^{\text{Bun}}(q^{\frac{1-r}{2}} z_1) = C_r \underline{\text{Res}}_{\frac{z_r}{z_{r-1}}=\dots=\frac{z_2}{z_1}=q^{-1}} (E_{1,\dots,1}^{\text{Bun}}(z_r, \dots, z_1)), \quad C_r = \left(q^{\frac{3}{4}r(r-1)(1-g_X)} \frac{Z^*(q^{-1})^{r-1}}{\prod_{i=2}^r Z_X(q^{-i})} \right)^{-1}$$

Hence (iii) is proved. □

As an application of Theorem 2.2.31, we compute the volume of the stack $\text{Bun}_{r,\bullet}(X)$ (called the Siegel formula) as follows

Theorem 2.2.33.

$$\text{Vol}(\text{Bun}_{r,d}) = \frac{q^{(g_X-1)(r^2-1)}}{q-1} |\text{Jac}_X(\mathbb{F}_q)| \times \prod_{i=2}^r Z_X(q^{-i}) \quad (2.11)$$

Proof. By definition,

$$\text{Vol}(\text{Bun}_{r,d}(X)) = \int_{\text{Bun}_{r,d}(X)} 1 d\mu := \sum_{\mathcal{V} \in \text{Bun}_{r,d}(X)/\sim} \frac{1}{a_{\mathcal{V}}} = (\mathbf{1}_{r,d}^{\text{Bun}} | \mathbf{1}_{r,d}^{\text{Bun}}) = (\mathbf{1}_{r,d}^{\text{Bun}} | \mathbf{1}_{r,d})$$

$$\begin{aligned} \text{Hence, } \text{Vol}(\text{Bun}_{r,0}(X)) &= (E_r^{\text{Bun}}(q^{\frac{1-r}{2}} z) | \mathbf{1}_{r,0}) = C_r (\underline{\text{Res}}(E_{1,\dots,1}^{(r),\text{Bun}}(\mathbf{z})) | \mathbf{1}_{r,0}) = C_r \underline{\text{Res}}(E_{1,\dots,1}^{\text{Bun}}(\mathbf{z}) | \mathbf{1}_{r,0}) \\ &= C_r \underline{\text{Res}} \left(\bigotimes E_1^{\text{Bun}}(z_i) | \Delta_{1,\dots,1}^{(r)}(\mathbf{1}_{r,0}) \right) \end{aligned}$$

Using the formula of coproduct $\Delta(\mathbf{1}_{\gamma}) = \sum_{\alpha+\beta} q^{-\frac{1}{2}\langle \alpha, \beta \rangle} \mathbf{1}_{\alpha} \otimes \mathbf{1}_{\beta}$, we have

Lemma 2.2.34. $\Delta_{1,\dots,1}^{(r)}(\mathbf{1}_{r,0}) = q^{\frac{1}{4}r(r-1)(g_X-1)} \sum_{\sum_{i=1}^r d_i=0} d \in \mathbb{Z}^r q^{\rho \cdot d} \mathbf{1}_{1,d_1} \otimes \dots \otimes \mathbf{1}_{1,d_r}$, where $\rho = (r-1, r-3, \dots, 1-r)$.

As a consequence,

$$\begin{aligned}
\text{Vol}(\text{Bun}_{r,0}(X)) &= C_r q^{\frac{1}{4}r(r-1)(g_X-1)} \underline{\text{Res}}_{\mathbf{z}} \left[\sum_{\substack{\underline{d} \in \mathbb{Z}^r \\ \sum_{i=1}^r d_i=0}} q^{\rho \cdot \underline{d}} (\mathbf{1}_{1,d_i}^{\text{Bun}} | \mathbf{1}_{1,d_i}) \mathbf{z}^{\underline{d}} \right] \\
&= C_r q^{\frac{1}{4}r(r-1)(g_X-1)} \frac{|\text{Jac}_X(\mathbb{F}_q)|^r}{(q-1)^r} \underline{\text{Res}}_{\mathbf{z}} \left[\sum_{\substack{\underline{d} \in \mathbb{Z}^r \\ \sum_{i=1}^r d_i=0}} q^{\rho \cdot \underline{d}} \mathbf{z}^{\underline{d}} \right] \left(\frac{|\text{Jac}_X(\mathbb{F}_q)|}{(q-1)} = \text{Vol}(\text{Pic}^d(X)) = (\mathbf{1}_{1,d}^{\text{Bun}} | \mathbf{1}_{1,d}) \right)
\end{aligned} \tag{2.12}$$

Note also

$$\frac{|\text{Jac}_X(\mathbb{F}_q)|}{Z^*(q^{-1})} = \frac{q-1}{q} \prod_{k=1}^{g_X} \frac{(1-\sigma_k)(1-\frac{q}{\sigma_k})}{(1-\frac{1}{\sigma_k})(1-\frac{\sigma_k}{q})} = q^{g_X-1} \cdot (q-1)$$

We have

$$\text{Vol}(\text{Bun}_{r,0}(X)) = q^{(r^2-1)(g_X-1)} \frac{|\text{Jac}_X(\mathbb{F}_q)|}{q-1} \prod_{i=2}^r Z_X(q^{-i}) \underline{\text{Res}}_{\mathbf{z}} \left[\sum_{\substack{\underline{d} \in \mathbb{Z}^r \\ \sum_{i=1}^r d_i=0}} q^{\rho \cdot \underline{d}} \mathbf{z}^{\underline{d}} \right] \tag{2.13}$$

However, The function in the previous formula to take the residue is wild, and can not be expanded as rational function in the region $|z_r| \ll \dots \ll |z_1|$. To obtain the volume formula, the technique is to approximate the whole stack by gradually increased substacks. Namely, we consider the coproduct

$$u_n := \underline{\text{Res}} \left(E_{1,\dots,1}^{\text{Bun}}(\mathbf{z}) | \mathbf{1}_{\text{Coh}_{r,0}^{\geq n}} \right) = \underline{\text{Res}} \left(\sum_{\mathcal{V} \in \text{Coh}_{r,0}^{\geq n}(\mathbb{F}_q)} \frac{1}{a_{\mathcal{V}}} (E_1^{\text{Bun}}(z_r) * \dots * E_1^{\text{Bun}}(z_1)(\mathcal{V})) \right)$$

where $\text{Coh}^{\geq n}$ is the substack of Coh containing those classes of coherent sheaves whose minimal slopes are $\geq n$. For u_n , since there are only finitely many possible types of Harder-Narasimhan filtrations, it is a finite sum of rational functions. Moreover

Proposition 2.2.35. *For any coherent sheaf $\mathcal{V} \in \text{Coh}_{r,0}^{\geq n}$, the rational function $E_{1,\dots,1}^{\text{Bun}}(\mathbf{z})(\mathcal{V})$ is regular outside of hyperplanes $\frac{z_i}{z_j} = q^{-1}$, $\frac{z_i}{z_j} = \sigma_k$ and $z_i = 0$, with at most simple poles at $\frac{z_i}{z_j} = q^{-1}$. As a consequence, the residue at $\frac{z_r}{z_{r-1}} = \dots = \frac{z_2}{z_1}$ of $(E_{1,\dots,1}^{\text{Bun}}(\mathbf{z}) | \mathbf{1}_{\text{Coh}_{r,0}^{\geq n}})$ is*

$$\lim_{\substack{d_r \gg \dots \gg d_1 \\ d_i=0}} \left(\mathbf{1}_{1,d_r}^{\text{ss}} * \dots * \mathbf{1}_{1,d_1}^{\text{ss}} | \mathbf{1}_{\text{Coh}_{r,0}^{\geq n}} \right) q^{-\rho \underline{d}}, \quad \rho = (r-1, r-3, \dots, 1-r), \underline{d} = (d_r, \dots, d_1).$$

Proof. Due to the degree zero property, the non-vanishing value of the $E_{1,\dots,1}^{\text{Bun}}(\mathcal{V})$ on vector bundles $\mathcal{V}$ is the linear combination of $\prod_{\substack{(i,j) \\ i>j}} \left(\frac{q z_i}{z_j} \right)^{d_{ij}}$, where the scalar q comes from the scalar of multiplication. The form of residue comes from the following lemma:

Lemma 2.2.36. *Let $F(y_1, \dots, y_s) \in \mathbb{C}(y_1, \dots, y_s)$ be a rational function which has at most simple poles along the hypersurfaces $\{y_{i_1} \dots y_{i_k} \cdot q = 1\}$ and other possible poles concentrated in the hypersurfaces $\{y_{j_1} \dots y_{j_l} \cdot \sigma_k^{-1} = 1\}$ or $\{y_i = 0\}$. Then consider the expansion of $F(y_1, \dots, y_s)$ in $\mathbb{C}[[y_1, \dots, y_s]] \cdot (y_1, \dots, y_s)^{-D}$ (for some $D \in \mathbb{N}$) as $\sum_{\underline{\lambda}} U_{\underline{\lambda}} y_1^{\lambda_1} \dots y_s^{\lambda_s}$, we have the residue $\text{Res}_{y_1=q^{-1}} \circ \dots \circ \text{Res}_{y_s=q^{-1}} (F(y_1, \dots, y_s))$ equals to the limit $\lim_{\lambda_s \rightarrow \infty} \lim_{\lambda_{s-1} \rightarrow \infty} \dots \lim_{\lambda_1 \rightarrow \infty} (q^{-\sum \lambda_i} U_{\underline{\lambda}})$*

Proof. It is enough to prove for the case $s = 1$. Consider $F(y) = \frac{a}{1-xy} + \sum_{i \in I} \frac{b_i}{(1-\sigma_i y)^{m_i}} + G(y)$, where $G(y) \in \mathbb{C}[z^{\pm 1}]$. Expand $F(y) = \sum_{n \geq -D} U_n y^n$ we find that $\lim_{n \rightarrow \infty} q^{-n} U_n = R = \text{Res}_{y=q^{-1}} (F(y))$, simply because the radius of σ is less than q . For the case of multiple variants, regard $\mathbb{C}(z_1, \dots, z_s)$ as $\mathbb{C}(z_1, \dots, z_{s-1})(z_s)$ and repeat the previous step. $\square$

Back to our case of residue of Eisenstein series, take $y_i := \frac{z_{i+1}}{z_i}$, consider in the rational function $\left(E_{1,\dots,1}^{\text{Bun}}(z_r, \dots, z_1) \mid \mathbf{1}_{\text{Coh}_{r,0}^{\geq n}}\right)$, $y_{r-1}^{\lambda_1} \dots (y_1)^{\lambda_{r-1}} = z_r^{\lambda_1} z_{r-1}^{\lambda_2 - \lambda_1} z_1^{-\lambda_{r-1}}$, apply the Lemma, we get

$$\text{Res}_{\underline{z}} \left(E_{1,\dots,1}^{\text{Bun}}(\underline{z}) \mid \mathbf{1}_{\text{Coh}_{r,0}^{\geq 0}} \right) = \lim_{\substack{d_r \gg \dots \gg d_1 \\ \sum d_i = 0}} \left(\mathbf{1}_{1,d_r}^{\text{ss}} * \dots * \mathbf{1}_{1,d_1}^{\text{ss}} \mid \mathbf{1}_{\text{Coh}_{r,0}^{\geq n}} \right) q^{-\rho \underline{d}}$$

The Proposition is proved. $\square$

To get the formula of Theorem 2.2.33, it is suffice to show

$$\lim_{n \rightarrow -\infty} \lim_{\substack{d_r \gg \dots \gg d_1 \\ \sum d_i = 0}} \frac{\left(\mathbf{1}_{1,d_r}^{\text{ss}} * \dots * \mathbf{1}_{1,d_1}^{\text{ss}} \mid \mathbf{1}_{\text{Coh}_{r,0}^{\geq n}} \right)}{|\text{Jac}_X(\mathbb{F}_q)|^r} (q-1)^r q^{-\rho \underline{d} + \frac{r(r-1)}{4}(1-g_X)} = 1 \quad (2.14)$$

Geometrically, we can consider the extension of stacks $\text{Coh}_{r,0} \xleftarrow{q_{\underline{d}}} \widetilde{\mathcal{FL}}_{\underline{d}} \xrightarrow{p_{\underline{d}}} \text{Bun}_{1,d_r} \times \dots \times \text{Bun}_{1,d_1}$, where $p_{\underline{d}}$ is a vector bundle stack, and $q_{\underline{d}}$ is the composition of the proper projection map of $\widetilde{\mathcal{FL}}_{\underline{d}} \rightarrow \text{Bun}_{r,0}$ with the inclusion $i : \text{Bun}_{r,0} \hookrightarrow \text{Coh}_{r,0}$. The numerator of 2.14 is translated as the counting for all possible sequences of line bundles $(\mathcal{L}_r, \dots, \mathcal{L}_1)$ of degree $d_r, \dots, d_1$, the size of $p_{\underline{d}}^{-1}(\mathcal{L}_r, \dots, \mathcal{L}_1) \cap q_{\underline{d}}^{-1}(\text{Coh}_{r,0}^{\geq n}(\mathbb{F}_q))$ modulo the automorphism, which has size $(q-1)^r$. Also, the coefficient in q coincides with the (minus of) relative dimension of vector bundle stack $p_{\underline{d}}$. In all, the problem reduced to

$$\begin{aligned} & \lim_{n \rightarrow -\infty} \lim_{\substack{d_r \gg \dots \gg d_1 \\ \sum d_i = 0}} V_{\geq n, \underline{d}} \\ &:= \lim_{n \rightarrow -\infty} \lim_{\substack{d_r \gg \dots \gg d_1 \\ \sum d_i = 0}} \left(\frac{q^{-\dim(p_{\underline{d}})}}{|\text{Jac}_X(\mathbb{F}_q)|^r} \times \sum_{\mathcal{L}_i \in \text{Pic}^{d_i}(X)} \left| p_{\underline{d}}^{-1}(\mathcal{L}_r, \dots, \mathcal{L}_1) \cap q_{\underline{d}}^{-1}(\text{Coh}_{r,0}^{\geq n}(\mathbb{F}_q)) \right| \right) \end{aligned} \quad (2.15)$$

$$= \lim_{n \rightarrow -\infty} \lim_{\substack{d_r \gg \dots \gg d_1 \\ \sum d_i = 0}} \left(\frac{1}{\text{Vol}(\widetilde{\mathcal{FL}}_{\underline{d}})} \times \sum_{(\mathcal{L}_i)} \left| p_{\underline{d}}^{-1}(\mathcal{L}_r, \dots, \mathcal{L}_1) \cap q_{\underline{d}}^{-1}(\text{Coh}_{r,0}^{\geq n}(\mathbb{F}_q)) \right| \right) = 1 \quad (2.16)$$

For each n , can also define the value $V_{< n, \underline{d}}$ by swapping $\text{Coh}_{r,0}^{\geq n}$ with its complementary stack $\text{Coh}_{r,0}^{< n}$ in $\text{Coh}_{r,0}(X)$. Then $V_{\geq n, \underline{d}} + V_{< n, \underline{d}} = 1$ (also the limit on n) by definition. Use the same argument, one has $V_{< n, \underline{d}} \propto \text{Vol}(\text{Bun}_{r,0}^{< n})$, we reduce to

Proposition 2.2.37. $\lim_{n \rightarrow -\infty} \text{Vol}(\text{Bun}_{r,0}^{< n}) = 0$, this is same to say $\text{Vol}(\text{Bun}_{r,0}) < \infty$.

Proof. To be added. $\square$

Finally, the Siegel formula for $\text{Bun}_{r,\bullet}$ is verified. We shall encounter it again in next Chapter (in a motivic version, see 3.2.11) $\square$

2.2.5 K-theoretic Hall algebras

To be added.

Chapter 3

Higher Rank Zeta Functions

3.1 Higher rank zeta functions for curves

3.1.1 Definition

Throughout the section, we set X to be a smooth, geometrically connected projective curve over finite field $\mathbb{F}_q$, let $g_X \in \mathbb{N}$ be its genus. To begin with, we recall the Weil zeta function associated with X is defined as

$$Z_X(t) = \prod_{x \in |X|} \frac{1}{1 - t^{\deg_x}} = \exp \left(\sum_{n=1}^{\infty} \frac{|X(\mathbb{F}_{q^n})|}{n} t^n \right).$$

The properties of $Z_X(t)$ can be concluded as (a special case of) Weil conjectures, as following

Theorem 3.1.1 (Artin, Schmidt, Weil). *Set $\zeta_X(s) := Z_X(q^{-s})$, then we have*

- (i) (Rationality) $Z_X(t)$ is a rational function and can be written as $\frac{P(t)}{(1-t)(1-qt)}$, where $P(t) \in \mathbb{Z}[t]$ is an integral polynomial of degree $2g_X$.
- (ii) (Functional equation) $Z_X(\frac{1}{qt}) = (qt^2)^{1-g_X} Z_X(t)$, or in the form of $\zeta_X(s)$, $\zeta_X(s) = q^{(1-2s)(g_X-1)} \zeta_X(1-s)$.
- (iii) (Riemann hypothesis) The polynomial $P(t)$ as in (i) can be factored in $\mathbb{C}[t]$ as $\prod_{i=1}^{2g_X} (1 - \alpha_i t)$, with $|\alpha_i| = q^{\frac{1}{2}}$. Equivalently, $\zeta_X(s)$ has zeros concentrated in the line $\text{Re}(s) = \frac{1}{2}$.

The Weil conjectures have general form for arbitrary smooth projective varieties over $\mathbb{F}_q$. We do not expand here, see for example [Del74], [Del80] and [KW01] for details.

For the case of curves, we notice that the combination of closed points contributes to an effective divisor, thus we can further write

$$\zeta_X(s) = \sum_{D \geq 0} \frac{1}{(q^{\deg(D)})^s} = \sum_{\mathcal{D}} \sum_{\substack{D \in \mathcal{D} \\ D \geq 0}} \frac{1}{(q^{\deg(D)})^s} = \sum_{\mathcal{D}} \frac{q^{h^0(X, \mathcal{O}(\mathcal{D}))}}{q-1} (q^{-s})^{d(\mathcal{D})}$$

where in above formula, $\mathcal{D}$ is the equivalent class of effective divisors. Hence

$$\zeta_X(s) = \sum_{d \geq 0} \sum_{\mathcal{L} \in \text{Pic}^d(X)} \frac{q^{h^0(X, \mathcal{L})} - 1}{q-1} (q^{-s})^{d(\mathcal{L})}, \quad Z_X(t) = \sum_{d \geq 0} \sum_{\mathcal{L} \in \text{Pic}^d(X)} \frac{q^{h^0(X, \mathcal{L})} - 1}{q-1} t^{d(\mathcal{L})} \quad (3.1)$$

Based on 3.1, there was purposed a generalization of Weil zeta function (we can see from 3.1 it is a counting on the line bundles) to higher ranks in [Wen05]. Such generalization is due to an observation $\text{Pic}^d(X) = \text{Bun}_{1,d}^{\text{ss}}(X)$, and $q-1 = |\text{Aut}(\mathcal{L})|$.

Definition 3.1.2. We set:

$$\widetilde{Z}_{X,r}(t) = \sum_{d \geq 0} \sum_{\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} t^d, \quad \widetilde{\zeta}_{X,r}(s) := \widetilde{Z}_{X,r}(q^{-s}) \quad (3.2)$$

We call $\widetilde{Z}_{X,r}$ and $\widetilde{\zeta}_{X,r}$ defined above the **mixed degree higher rank zeta functions**.

In parallel with Theorem 3.1.1, we shows some zeta facts of such generalized zeta functions.

Proposition 3.1.3 (Rationality). *For each rank r , the mixed higher rank zeta function $\widetilde{\zeta}_{X,r}(s)$ converges for $\text{Re}(s) > 1$, and admits a meromorphic extension to the whole complex plane. Moreover, let $Q := q^r, T := t^r$, then $\widetilde{Z}_{X,r}(t)$ has the form $\frac{\widetilde{P}_{X,r}(t)}{(1-T)(1-QT)}$, where $\widetilde{P}_{X,r}(t) \in \mathbb{Q}[t]$ is a polynomial.*

Proof. Since $h^0(\mathcal{E}) = \dim \text{Hom}(\mathcal{O}_X, \mathcal{E})$ and $h^1(\mathcal{E}) = \dim \text{Ext}^1(\mathcal{O}_X, \mathcal{E}) = \dim \text{Hom}(\mathcal{E}, \Omega_X)$, we have immediately that $h^1(\mathcal{E}) = 0$ if $\mu(\mathcal{E}) - \mu(\Omega_X) > g_X - 1$, which holds for those bundles with $\deg(\mathcal{E}) \geq r(2g_X - 2) + 1$. Then the summation becomes

$$\begin{aligned} \widetilde{\zeta}_{r,X}(s) &= \left(\sum_{d=0}^{r(2g_X-2)} + \sum_{d \geq r(2g_X-2)+1} \right) \sum_{\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} (q^{-s})^d \\ &= \sum_{\substack{\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}} \\ 0 \leq d \leq r(2g_X-2)}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} (q^{-s})^d + \sum_{\substack{\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}} \\ d > r(2g_X-2)}} \frac{q^{r(1-g_X)+d} - 1}{|\text{Aut}(\mathcal{E})|} (q^{-s})^d \end{aligned}$$

For degree $d \in \mathbb{Z}$, set the variables

$$\alpha_{X,r}(d) := \sum_{\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|}; \quad \beta_{X,r}(d) := \sum_{\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}}} \frac{1}{|\text{Aut}(\mathcal{E})|} = \text{Vol}(\text{Bun}_{r,d}^{\text{ss}}). \quad (3.3)$$

Then we have the periodicity $\beta_{X,r}(d) = \beta_{X,r}(d + rn), \forall n \in \mathbb{Z}$, as we can choose some line bundle $\mathcal{L}_1$ of degree one, then $\text{Bun}_{r,d}^{\text{ss}} \longrightarrow \text{Bun}_{r,d+rn}^{\text{ss}}, \mathcal{E} \mapsto \mathcal{E} \otimes \mathcal{L}_1^{\otimes n}$ is an isomorphism, and $\text{Aut}(\mathcal{E}) \simeq \text{Aut}(\mathcal{E} \otimes \mathcal{L}_1^{\otimes n})$. Hence for $\text{Re}(s) > 1$,

$$\begin{aligned} \sum_{\substack{\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}} \\ d > r(2g_X-2)}} \frac{q^{r(1-g_X)+d} - 1}{|\text{Aut}(\mathcal{E})|} (q^{-s})^d &= \sum_{d \geq r(2g_X-2)+1} \beta_{r,d} \left(q^{r(1-g_X)+(1-s)d} - q^{-sd} \right) \\ &= \sum_{i=1}^r \beta_{X,r}(i) \left(q^{r(1-g_X)} \cdot \frac{q^{(1-s)(r(2g_X-2)+i)}}{1 - q^{(1-s)r}} - \frac{q^{-s(r(2g_X-2)+i)}}{1 - q^{-sr}} \right) \end{aligned}$$

The first part in the summation of $\widetilde{\zeta}_{r,X}(s)$ is a finite sum, and the second part now is provided to be convergent for $\text{Re}(s) > 1$, the meromorphic continuation is easily seen from the previous formula. Also, substitute q^{-s} with t , we get the rationality of $\widetilde{Z}_{X,r}$, we are done. $\square$

Remark. Notice we do not know yet the degree of $\widetilde{P}_{X,r}(t)$, this however follows from the functional equation as follows.

Proposition 3.1.4 (Functional equation). *Set $\widetilde{Z}_{X,r}(t) := t^{r(1-g_X)} \widetilde{Z}_{X,r}(t)$, and the corresponding $\widetilde{\zeta}_{X,r}(s) = (q^{-s})^{r(1-g_X)} \widetilde{\zeta}_{X,r}(s)$, we have the equality*

$$\widetilde{\zeta}_{X,r}(s) = \widetilde{\zeta}_{X,r}(1-s) \quad (3.4)$$

Proof. Instead of considering the stack of semistable vector bundle of fixed degree, we consider for a fixed line bundle $\mathcal{L}$, the isomorphism classes of rank r vector bundles with determinant $\mathcal{L}$, denoted as $\text{Bun}_{r,\mathcal{L}}^{\text{ss}}$. We use the periodicity comes from the isomorphisms $\text{Bun}_{r,\mathcal{L}}^{\text{ss}} \longrightarrow \text{Bun}_{r,\mathcal{L} \otimes \Omega_X^{rn}}^{\text{ss}}, \mathcal{E} \mapsto \mathcal{E} \otimes \Omega_X$ and $\text{Bun}_{r,\mathcal{L}}^{\text{ss}} \longrightarrow \text{Bun}_{r,\mathcal{L}^{-1} \otimes \Omega_X^n}^{\text{ss}}, \mathcal{E} \mapsto \mathcal{E}^\vee \otimes \Omega_X$.

Then we can always choose $\mathcal{L}$ with degree $0 \leq \deg(\mathcal{L}) \leq r(g_X - 1)$. For such line bundle $\mathcal{L}$, set

$$M_{X,r}^{\mathcal{L}} := \bigcup_{n \in \mathbb{Z}} \left(\text{Bun}_{r,\mathcal{L} \otimes \Omega_X^{rn}}^{\text{ss}} \cup \text{Bun}_{r,\mathcal{L}^{-1} \otimes \Omega_X^{rn}}^{\text{ss}} \right)$$

to be the union of all isomorphic periods of $\text{Bun}_{r,\mathcal{L}}^{\text{ss}}$, and the zeta function can be divided into

$$\widetilde{\zeta}_{X,r}^{\mathcal{L}}(s) = \sum_{\substack{\mathcal{L} \in \text{Pic}(X) \\ 0 \leq \deg(\mathcal{L}) \leq r(g_X - 1)}} \widetilde{\zeta}_{X,r}^{\mathcal{L}}(s)$$

And to show the functional equation, it suffices to show $\widetilde{\zeta}_{X,r}^{\mathcal{L}}(s) = \widetilde{\zeta}_{X,r}^{\mathcal{L}}(1-s)$. We have $\widetilde{\zeta}_{X,r}^{\mathcal{L}}(s) = \sum_{\substack{\mathcal{E} \in M_{X,r}^{\mathcal{L}} \\ \deg(\mathcal{E}) \geq 0}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} (q^{-s})^{\chi(\mathcal{E})}$ written as

$$\widetilde{\zeta}_{X,r}^{\mathcal{L}}(s) = \underbrace{\sum_{\substack{\mathcal{E} \in M_{X,r}^{\mathcal{L}} \\ r(2g_X - 2) \geq \deg(\mathcal{E}) \geq 0}} \frac{q^{h^0(\mathcal{E})}}{|\text{Aut}(\mathcal{E})|} (q^{-s})^{\chi(\mathcal{E})}}_I + \underbrace{\left(\sum_{\substack{\mathcal{E} \in M_{X,r}^{\mathcal{L}} \\ \deg(\mathcal{E}) > r(2g_X - 2)}} \frac{q^{(1-s)\chi(\mathcal{E})}}{|\text{Aut}(\mathcal{E})|} - \sum_{\substack{\mathcal{E} \in M_{X,r}^{\mathcal{L}} \\ \deg(\mathcal{E}) \geq 0}} \frac{1}{|\text{Aut}(\mathcal{E})|} (q^{-s})^{\chi(\mathcal{E})} \right)}_{II}$$

$$\begin{aligned} \text{For I, } I &= \frac{1}{2} \left(\sum_{\substack{\mathcal{E} \in M_{X,r}^{\mathcal{L}} \\ r(2g_X - 2) \geq \deg(\mathcal{E}) \geq 0}} \frac{q^{h^0(\mathcal{E})}}{|\text{Aut}(\mathcal{E})|} (q^{-s})^{\chi(\mathcal{E})} + \sum_{\substack{\mathcal{E}^\vee \in M_{X,r}^{\mathcal{L}} \\ 0 \geq \deg(\mathcal{E}^\vee) \geq r(2-2g_X)}} \frac{q^{h^0(\mathcal{E}^\vee \otimes \Omega_X)}}{|\text{Aut}(\mathcal{E}^\vee \otimes \Omega_X)|} (q^{-s})^{\chi(\mathcal{E}^\vee \otimes \Omega_X)} \right) \\ &= \frac{1}{2} \left(\sum_{\substack{\mathcal{E} \in M_{X,r}^{\mathcal{L}} \\ r(2g_X - 2) \geq \deg(\mathcal{E}) \geq 0}} \frac{q^{h^0(\mathcal{E})}}{|\text{Aut}(\mathcal{E})|} (q^{-s})^{\chi(\mathcal{E})} + \sum_{\substack{\mathcal{E}^\vee \in M_{X,r}^{\mathcal{L}} \\ 0 \geq \deg(\mathcal{E}^\vee) \geq r(2-2g_X)}} \frac{q^{h^1(\mathcal{E}^\vee \otimes \Omega_X) + \chi(\mathcal{E}^\vee \otimes \Omega_X)}}{|\text{Aut}(\mathcal{E}^\vee \otimes \Omega_X)|} q^{-s(\chi(\mathcal{E}^\vee \otimes \Omega_X))} \right) \\ &= \frac{1}{2} \sum_{\substack{\mathcal{E} \in M_{X,r}^{\mathcal{L}} \\ r(2g_X - 2) \geq \deg(\mathcal{E}) \geq 0}} \frac{q^{h^0(\mathcal{E})}}{|\text{Aut}(\mathcal{E})|} \left((q^{-s})^{\chi(\mathcal{E})} + (q^{s-1})^{\chi(\mathcal{E})} \right) \quad (\text{symmetric for } s \mapsto 1-s) \end{aligned}$$

$$\begin{aligned} \text{For II, } II &= \left(\sum_{\substack{\mathcal{E} \in \text{Bun}_{r,\mathcal{L} \otimes \Omega_X^{rn}}^{\text{ss}} \\ d(\mathcal{L}) + rn(2g_X - 2) > r(2g_X - 2)}} \frac{q^{(1-s)\chi(\mathcal{E})}}{|\text{Aut}(\mathcal{E})|} - \sum_{\substack{\mathcal{E}' \in \text{Bun}_{\mathcal{L}^{-1} \otimes \Omega_X^{rn}}^{\text{ss}} \\ -d(\mathcal{L}) + rn(2g_X - 2) \geq 0}} \frac{q^{-s\chi(\mathcal{E}')}}{|\text{Aut}(\mathcal{E}')|} \right) \\ &+ \left(\sum_{\substack{\mathcal{E} \in \text{Bun}_{r,\mathcal{L}^{-1} \otimes \Omega_X^{rn}}^{\text{ss}} \\ -d(\mathcal{L}) + rn(2g_X - 2) > r(2g_X - 2)}} \frac{q^{(1-s)\chi(\mathcal{E})}}{|\text{Aut}(\mathcal{E})|} - \sum_{\substack{\mathcal{E}' \in \text{Bun}_{\mathcal{L} \otimes \Omega_X^{rn}}^{\text{ss}} \\ d(\mathcal{L}) + rn(2g_X - 2) \geq 0}} \frac{q^{-s\chi(\mathcal{E}')}}{|\text{Aut}(\mathcal{E}')|} \right) \end{aligned}$$

Notice our choice of $\mathcal{L}$ so that $0 \leq \deg(\mathcal{L}) \leq r(g_X - 1)$, we have the counting

$$\begin{aligned} \Pi &= \left(\sum_{n=1}^{\infty} \sum_{\mathcal{E} \in \text{Bun}_{r, \mathcal{L} \otimes \Omega_X^n}^{\text{ss}}} \frac{q^{(1-s)[r(1-g_X)+d(\mathcal{L})+rn(2g_X-2)]}}{|\text{Aut}(\mathcal{E})|} - \sum_{n=1}^{\infty} \sum_{\mathcal{E}' \in \text{Bun}_{r, \mathcal{L}^{-1} \otimes \Omega_X^n}^{\text{ss}}} \frac{q^{-s(-d(\mathcal{L})+rn(2g_X-2)+r(1-g_X))}}{|\text{Aut}(\mathcal{E}')|} \right) \\ &+ \left(\sum_{n=2}^{\infty} \sum_{\mathcal{E} \in \text{Bun}_{r, \mathcal{L}^{-1} \otimes \Omega_X^n}^{\text{ss}}} \frac{q^{(1-s)[r(1-g_X)-d(\mathcal{L})+rn(2g_X-2)]}}{|\text{Aut}(\mathcal{E})|} - \sum_{n=0}^{\infty} \sum_{\mathcal{E}' \in \text{Bun}_{r, \mathcal{L} \otimes \Omega_X^n}^{\text{ss}}} \frac{q^{-s(d(\mathcal{L})+rn(2g_X-2)+r(1-g_X))}}{|\text{Aut}(\mathcal{E}')|} \right) \\ &= \beta_{X,r}(\mathcal{L}) \left[\frac{q^{(1-s)(d(\mathcal{L})-r(g_X-1))}}{q^{(s-1)r(2g_X-2)} - 1} + \frac{q^{s(d(\mathcal{L})-r(g_X-1))}}{q^{-sr(2g_X-2)} - 1} \frac{q^{(s-1)(d(\mathcal{L})-r(g_X-1))}}{q^{(s-1)r(2g_X-2)} - 1} + \frac{q^{-s(d(\mathcal{L})-r(g_X-1))}}{q^{-sr(2g_X-2)} - 1} \right] \end{aligned}$$

where $\beta_{X,r}(\mathcal{L}) = \beta_{X,r}(\mathcal{L} \otimes \Omega_X^{rn}) = \text{Vol}(\text{Bun}_{r, \mathcal{L}}^{\text{ss}})$, and we observe that Π is symmetric for $s \mapsto 1-s$. $\square$

Corollary 3.1.5. (i) Write $\widetilde{Z}_{X,r}(t)$ as $\frac{\widetilde{P}_r(t)}{(1-t^r)(1-(qt)^r)}$, then $\widetilde{P}_r(t) \in \mathbb{Q}[t]$ is a polynomial of degree $2rg_X$.

(ii) Denote by $\{\sigma_{r,i}\}_{i=1,\dots,2rg_X}$ the roots of $P_r(t)$ in $\mathbb{C}$, then we can choose an order of i so that $\sigma_{r,i} \cdot \sigma_{r,(2rg_X-i)} = q$.

Within the rationality and functional equation, we may wonder if the Riemann Hypothesis holds. However, evidences show that we need to modify our zeta functions by counting only the vector bundles of degrees divide by rank r .

For example, in the previous definition of zeta functions, we see

$$\widetilde{\zeta}_{r,X}(s) = \sum_{d=0}^{r(2g_X-2)} \alpha_{X,r}(d)(q^{-s})^d + \sum_{i=1}^r \beta_{X,r}(i) \left(q^{r(1-g_X)} \cdot \frac{q^{(1-s)(r(2g_X-2)+i)}}{1-q^{(1-s)r}} - \frac{q^{-s(r(2g_X-2)+i)}}{1-q^{-sr}} \right)$$

which involves the data of volume of semistable stacks ranging all possible degrees from 1 to r . Instead, we define:

Definition 3.1.6. For any $r \geq 1$, We associate X with the **pure higher rank zeta function** of rank r , defined as

$$Z_{X,r}(t) = \sum_{m \geq 0} \sum_{\mathcal{E} \in \text{Bun}_{r, rm}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} t^m$$

Set also

$$\widehat{Z}_{X,r}(t) := t^{1-g} Z_{X,r}(t); \quad \widehat{\zeta}_{X,r}(s) := \widehat{Z}_{X,r}(q^{-rs})$$

We have the similar expression: $\widehat{\zeta}_{X,r}(s) = \sum_{m \geq 0} \sum_{\mathcal{E} \in \text{Bun}_{r, rm}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} (q^{-s})^{\chi(\mathcal{E})}$.

Proposition 3.1.7 (Rationality). We have $Z_{X,r}(t) = \frac{P_r(T)}{(1-T)(1-QT)}$ with $T = t^r$, $Q = q^r$ and $P_r(T) \in \mathbb{Q}[T]$ is of degree $2g_X$.

Proof. We use the similar method to divide the summation by degree, in the case of pure zeta functions

$$\begin{aligned}\widehat{Z}_{X,r}(t^r) &= \left(\sum_{m=0}^{2g_X-2} + \sum_{m>2g_X-2} \right) \sum_{\mathcal{E} \in \text{Bun}_{r,rm}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} t^{X(\mathcal{E})} \\ &= \left(\sum_{m=0}^{(g_X-1)-1} \sum_{\substack{\mathcal{E} \in \text{Bun}_{r,rm}^{\text{ss}} \\ \text{Bun}_{r,r(2g_X-2-m)}^{\text{ss}}}} + \sum_{\mathcal{E} \in \text{Bun}_{r,r(g_X-1)}^{\text{ss}}} \right) \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} t^{X(\mathcal{E})} + \sum_{m>2g_X-2} \sum_{\mathcal{E} \in \text{Bun}_{r,0}^{\text{ss}}} \frac{q^{r(m+(1-g_X))} - 1}{|\text{Aut}(\mathcal{E})|} t^{r(m+(1-g_X))}\end{aligned}$$

(By the vanishing of $h^1(\mathcal{E})$ for large slopes, and by periodicity, we can count for $\text{Bun}_{r,0}^{\text{ss}}$)

$$\begin{aligned}&= \sum_{m=0}^{(g_X-1)-1} \left(\sum_{\mathcal{E} \in \text{Bun}_{r,rm}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} t^{r(m+1-g_X)} + \sum_{\mathcal{E}' \in \text{Bun}_{r,rm}^{\text{ss}}} \frac{q^{h^0(\mathcal{E}')-r(m+1-g_X)} - 1}{|\text{Aut}(\mathcal{E}' \otimes \Omega_X)|} t^{r(g_X-1-m)} \right) \\ &+ \sum_{\mathcal{E} \in \text{Bun}_{r,r(g_X-1)}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} t^0 + \sum_{\mathcal{E} \in \text{Bun}_{r,0}^{\text{ss}}} \frac{1}{|\text{Aut}(\mathcal{E})|} \left(\frac{(qt)^{rg}}{1 - (qt)^r} - \frac{t^{rg}}{1 - t^r} \right)\end{aligned}$$

(we can compute $h^0(\Omega_X \otimes \mathcal{E}^\vee) = \chi(\Omega_X \otimes \mathcal{E}^\vee) + h^1(\Omega_X \otimes \mathcal{E}^\vee) = r(g_X - 1) - rm + h^0(\mathcal{E})$)

$$\begin{aligned}&= \sum_{m=0}^{(g_X-1)-1} \left(\left[\sum_{\mathcal{E} \in \text{Bun}_{r,rm}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} (t^{r(m+1-g_X)} + (qt)^{r(g_X-1-m)}) \right] + \sum_{\mathcal{E}' \in \text{Bun}_{r,rm}^{\text{ss}}} \frac{(qt)^{r(g_X-1-m)} - t^{r(g_X-1-m)}}{|\text{Aut}(\mathcal{E}')|} \right) \\ &+ \sum_{\mathcal{E} \in \text{Bun}_{r,r(g_X-1)}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} t^0 + \sum_{\mathcal{E} \in \text{Bun}_{r,0}^{\text{ss}}} \frac{1}{|\text{Aut}(\mathcal{E})|} \left(\frac{(qt)^{rg}}{1 - (qt)^r} - \frac{t^{rg}}{1 - t^r} \right) \\ &= \sum_{m=0}^{(g_X-1)-1} \alpha_{X,r}(rm) \left(t^{r(m+1-g_X)} + \left(\frac{1}{qt} \right)^{r(m+1-g_X)} \right) + \alpha_{X,r}(r(g_X - 1)) \\ &+ \beta_{X,r}(0) \left(\frac{(qt)^{rg}}{1 - (qt)^r} - \frac{t^{rg}}{1 - t^r} + \sum_{m=0}^{(g_X-1)-1} \left[(qt)^{r(g_X-1-m)} - t^{r(g_X-1-m)} \right] \right).\end{aligned}$$

where the summation in the bracket after coefficient $\beta_{X,r}(0)$ is

$$\frac{(qt)^{rg}}{1 - (qt)^r} - \frac{t^{rg}}{1 - t^r} + \frac{((qt)^{r(2-g_X)} - (qt)^r) \cdot (qt)^{r(g_X-1)}}{1 - (qt)^r} - \frac{(q^{r(2-g_X)} - t^r) \cdot t^{r(g_X-1)}}{1 - t^r} = \frac{(qt)^r}{1 - (qt)^r} - \frac{t^r}{1 - t^r}$$

In all, we have the pure zeta function to be

$$\widehat{\zeta}_{X,r}(s) = \sum_{m=0}^{(g_X-1)-1} \alpha_{X,r}(rm) \left(q^{-sr(m+1-g_X)} + q^{(s-1)r(m+1-g_X)} \right) + \alpha_{X,r}(r(g_X - 1)) + \beta_{X,r}(0) \left(\frac{q^{(1-s)r}}{1 - q^{(1-s)r}} - \frac{q^{-sr}}{1 - q^{-sr}} \right) \quad (3.5)$$

For simplicity, we set $T := t^r$ and $Q := q^r$ as before, then the expression of $Z_{X,r}(t)$ can be written as

$$Z_{X,r}(T) = \sum_{m=0}^{(g_X-1)-1} \alpha_{X,r}(rm) (T^m + Q^{g_X-1-m} T^{2g_X-2-m}) + \alpha_{X,r}(r(g_X - 1)) + (Q - 1) \beta_{X,r}(0) \frac{T^g}{(1 - T)(1 - QT)}. \quad (3.6)$$

The rationality follows. $\square$

Remark. The functional equation is not obvious from the above expression, we will give a demonstration later in Proposition 3.3.2.

Conjecture 3.1.8 (Riemann Hypothesis). *The zeros of $\zeta_{X,r}(s)$ concentrated in the line $\text{Re}(s) = \frac{1}{2}$.*

Example 3.1.9. We give an example when $X = E$ is an elliptic curve and $r = 2$.

Write $Z_{X,r}(t) = \frac{P_{X,r}(T)}{(1-T)(1-QT)}$, as in the expression of rationality, we get $P_{E,2}(T) = \alpha_{E,2}(0)(1-T)(1-QT) + (Q-1)\beta_{E,2}(0)T$. The number $\beta_{E,2}(d)$ can be generally determined later, see Theorem 3.2.18. We borrow the expression, so that

$$\beta_{E,2}(0) = \left(\frac{1}{1-q^2} \text{Vol}(\text{Bun}_{1,E})^2 + \text{Vol}(\text{Bun}_2)\right) = \frac{N^2}{(1-q^2)(q-1)^2} + \frac{N}{q-1}Z_E(q^{-2}) = \frac{N}{q-1}\left(1 + \frac{N}{q^2-1}\right)$$

where we use N to denote $|\text{Jac}_E(\mathbb{F}_q)| = |E(\mathbb{F}_q)|$.

And for any rank 2 semistable bundle $\mathcal{V}$ on E , it is known either $\mathcal{V} \simeq \mathcal{O}_E \oplus \mathcal{L}$ for $\mathcal{L}$ a degree 0 line bundle or $\mathcal{V}$ is indecomposable, as a (the only) non-trivial extension of $\mathcal{O}_E$ by $\mathcal{O}_E$, denote this indecomposable bundle as I . Thus

$$\begin{aligned} \alpha_{E,2}(0) &= \frac{q^{h^0(\mathcal{O} \oplus \mathcal{O})} - 1}{|\text{Aut}(\mathcal{O} \oplus \mathcal{O})|} + \frac{q^{h^0(I)} - 1}{|\text{Aut}(I)|} + \sum_{\mathcal{L} \in \text{Pic}^0(E) \setminus \mathcal{O}} \frac{q^{h^0(\mathcal{O} \oplus \mathcal{L})}}{|\text{Aut}(\mathcal{O} \oplus \mathcal{L})|} \\ &= \frac{q^2 - 1}{(q^2 - 1)(q^2 - q)} + \frac{q - 1}{q(q - 1)} + (N - 1) \frac{q - 1}{(q - 1)^2} = \frac{N}{q - 1} \end{aligned}$$

We then have $Z_{E,2}(t) = \alpha_{E,2}(0) \frac{1+(N-2)T+QT^2}{(1-T)(1-QT)}$. Using the Riemann Hypothesis for elliptic curve (Hasse's theorem), we can deduce $\Delta = (N-2)^2 - 4Q < 0$, so that the roots have absolute value $\frac{\sqrt{4Q}}{2Q} = \frac{1}{\sqrt{Q}}$, this concludes to Riemann Hypothesis in rank 2 case for elliptic curves.

3.2 Motivic Interpretation

3.2.1 Motivic classes of algebraic stacks and motivic zeta functions

Definition 3.2.1. We fix $\mathbf{k}$ a ground field. We can attach $\mathbf{k}$ the ring of algebraic varieties $K_0(\text{Var}_{\mathbf{k}})$, in which the generators are given by symbols $[X]$, for X is representative of isomorphism class of varieties over $\mathbf{k}$. The additive relation is given by

$$[X] = [X - Z] + [Z], \quad \text{for } Z \text{ a closed subvariety of } X.$$

And the multiplicative relation is given by the Cartesian product over $\mathbf{k}$, $[X \times_{\mathbf{k}} Y] = [X] \cdot [Y]$.

It is clear that $K_0(\text{Var}_{\mathbf{k}})$ is a commutative ring with unit $1 = [\mathbf{k}]$, called the **(naïve) Grothendieck ring of varieties over $\mathbf{k}$** , the symbol $[X]$ is called the **motivic class** of X . Denote also by $\mathbb{L}$ the motivic class of the affine line $\mathbb{A}^1$.

Remark. The name of motivic come from the fact that there is a canonical ring homomorphism $K_0(\text{Var}_{\mathbf{k}}) \longrightarrow K_0(\text{Mot}_{\sim}(\mathbf{k}))$, the later is the Grothendieck ring of pure motives (as a tensor ring) It sends $\mathbb{L}$ to $[\mathbb{Q}(-1)]$ the class of Tate twist.

Example 3.2.2. We can determine immediately the motivic classes of common varieties

$$[\mathbb{A}^n] = \mathbb{L}^n, \quad [\mathbb{P}^n] - [\mathbb{P}^{n-1}] = [\mathbb{A}^n], \text{ hence } [\mathbb{P}^n] = 1 + \mathbb{L} + \mathbb{L}^2 + \dots + \mathbb{L}^n$$

$$[\text{GL}_n] = [\text{GL}_n/B] \cdot [T] \cdot [U] = [\mathbb{P}^n] \cdot [\mathbb{G}_m]^n \cdot [\mathbb{A}^1]^{\frac{n(n-1)}{2}} = (\mathbb{L}^n - 1) \dots (\mathbb{L}^n - \mathbb{L}^{n-1}), \text{ choose } B \text{ to be standard upper triangular matrices.}$$

Moreover, using Bruhat decomposition, we have

Proposition 3.2.3. Let G be a connected, split semisimple algebraic group over $\mathbf{k}$, then the motivic class of G is

$$[G] = \mathbb{L}^{\dim G} \prod_{i=1}^r (1 - \mathbb{L}^{-d_i}), \quad r = \text{rank of } G = \dim T, T : \text{maximal torus.}$$

where $d_i - 1 = m_i = \deg(f_i)$ is the exponent of $\mathfrak{g}$, so that $Z(\mathfrak{g}) \simeq \mathbf{k}[f_1, \dots, f_r]$.

Proof. We choose a maximal torus T and a Borel $B = TU$ with unipotent radical U . By the connectedness and split assumption, we have $[T] = [\mathbb{G}_m]^{r(G)}$ and $[G] = [G/B] \cdot [T] \cdot [U]$. Since $[U]$ is the iterated extension of additive groups, we know $[U] = \mathbb{L}^{\frac{1}{2}(\dim G - r)}$, where the power of $\mathbb{L}$ is $\dim U = \frac{1}{2}(\dim G - r)$. By Bruhat decomposition, we have also $[G/B] = \sum_{w \in W} \mathbb{L}^{l(w)}$, W is the Weyl group of G and $l(w)$ is the length. The proposition then follows from the fact that $(\mathbb{L} - 1)^r \sum_{w \in W} \mathbb{L}^{l(w)} = \prod_{i=1}^r (\mathbb{L}^{d_i} - 1)$. $\square$

Next, we extend the ring $K_0(\text{Var}_{\mathbf{k}})$ to encode the motivic classes of 'good' stacks.

Definition 3.2.4. Let $K_0(\text{Var}_{\mathbf{k}})[\frac{1}{\mathbb{L}}]$ be the localization ring. For fixed $n \in \mathbb{N}$, set the level structure $F^n(K_0(\text{Var}_{\mathbf{k}})[\frac{1}{\mathbb{L}}])$ to be the abelian group generated by $\frac{[X]}{[\mathbb{L}^m]}$ with $\dim X - m \leq -n$. This gives a filtration on the localized ring. We define $\widehat{K}_0(\text{Var}_{\mathbf{k}})$ to be the completion of $K_0(\text{Var}_{\mathbf{k}})[\frac{1}{\mathbb{L}}]$ with respect to the filtration, called the dimension completion.

In particular, we observe that $[\text{GL}_n]$ is invertible in $\widehat{K}_0(\text{Var}_{\mathbf{k}})$ since $(\mathbb{L}^n - 1)^{-1} = \frac{1}{\mathbb{L}^n(1 - \frac{1}{\mathbb{L}^n})} = \mathbb{L}^{-n}(\sum_{k=0}^{\infty} \mathbb{L}^{-kn}) \in \widehat{K}_0(\text{Var}_{\mathbf{k}})$. That enables us to consider the motivic classes of quotient stacks. We consider those algebraic stacks locally of finite type and for its geometric points, their stabilizers are linear algebraic groups. And we require also for such stacks that they are essentially of finite type, in the sense of [BD07], where by definition if $\mathcal{X}$ is essentially of finite type, we mean that there is a countable stratification $\mathcal{X} = \bigcup_{i=1}^{\infty} \mathcal{Z}_i$, with each $\mathcal{Z}_i$ is of finite type, locally closed, in the form of global quotient stack $[X_i / \text{GL}_{n_i}]$ (quotient stack of X_i by GL_{n_i} , we abuse the notation here) and $\dim \mathcal{Z}_i \rightarrow -\infty$ as $i \rightarrow \infty$. We use the result [Kre99, Proposition 3.5.9]:

Lemma 3.2.5. *Let $\mathcal{X}$ be a stack, then $\mathcal{X}$ admits a stratification by global quotient stacks if and only if it has linear algebraic groups as geometric stabilizers.*

Definition 3.2.6. We call an algebraic stack $\mathcal{X}$ satisfies the condition of Lemma 3.2.5 a **local quotient stack**.

Let X be a variety with an action of GL_n , define for the global quotient stack $\mathcal{X} = [X / \text{GL}_n]$, the class $[\mathcal{X}] \in \widehat{K}_0(\text{Var}_{\mathbf{k}})$ to be $\frac{[X]}{[\text{GL}_n]}$. And for $\mathcal{X} := \bigcup_{i=1}^{\infty} \mathcal{Z}_i$ essentially of finite type, define its class to be $[\mathcal{X}] := \sum_{i=1}^{\infty} \frac{[X_i]}{[\text{GL}_{n_i}]}$, where $\mathcal{Z}_i = [X_i / \text{GL}_{n_i}]$.

Lemma 3.2.7. *The value of $\frac{[X]}{[\text{GL}_n]}$ does NOT depend on the choice of local presentation, hence the above motivic class $[\mathcal{X}]$ is well-defined, also the motivic class of local quotient stacks.*

Proof. Assume that there is a (finite) strata $\mathcal{X} = \bigcup_{i=1}^N \mathcal{Z}_i$, then each stratum $\mathcal{Z}_i \simeq [Z_i / \text{GL}_{n_i}]$, let $X_i := X \times_{\mathcal{X}} \mathcal{Z}_i$ represents the preimage of $\mathcal{Z}_i$ in the presentation X . Set also Y_i to be the cartesian product $Y_i := Z_i \times_{\mathcal{Z}_i} X_i$, so that $Y_i \rightarrow Z_i$ is a GL_{n_i} -torsor, and $Y_i \rightarrow X_i$ is a GL_{n_i} -torsor. Since the GL_n torsors are Zariski locally trivial, for any $P \rightarrow S$ an GL_n torsor with S irreducible, $\exists U \subseteq S$ dense open so that $P_U \rightarrow U$ is trivial, then we can reduce to $S - U$ and use Noetherian induction to show $[P] = [\text{GL}_n] \cdot [S]$. We have thus:

$$\sum_{i=1}^N \mathcal{Z}_i = \sum_{i=1}^N \frac{[Z_i]}{[\text{GL}_{n_i}]} = \sum_{i=1}^N \frac{[Y_i]}{[\text{GL}_n] \cdot [\text{GL}_{n_i}]} = \sum_{i=1}^N \frac{[X_i]}{[\text{GL}_n]} = \frac{[X]}{[\text{GL}_n]} = [\mathcal{X}]. \quad \square$$

We see also $\widehat{K}_0(\text{Var}_{\mathbf{k}})$ includes the classes of all Deligne-Mumford stacks of finite types.

Example 3.2.8. Let G be the same group as in Proposition 3.2.3, we have $[BG] = \mathbb{L}^{-\dim G} \prod_{i=1}^r (1 - \mathbb{L}^{-d_i})^{-1}$.

Observe that for a curve X over $\mathbb{F}_q$, we can write the associated Weil zeta function as $Z_X(t) = \sum_{n=0}^{\infty} S^n X(\mathbb{F}_q) t^n$, where $S^n X$ is the symmetric power of X . As an analogy, we can define the motivic zeta function for X over field $\mathbf{k}$ as

$$Z_{X, \text{mot}}(t) := \sum_{n=0}^{\infty} [S^n X] t^n \in \widehat{K}_0(\text{Var}_{\mathbf{k}})[[t]] \quad (3.7)$$

Such zeta function is firstly introduced in [Kap00] under a generalization of the language of motivic integration, initially invented in [DL99] for p -adic case. Concretely, for a commutative ring A , we can define a **motivic measure** to be ring homomorphism:

$$\mu : K_0(\text{Var}_{\mathbf{k}}) \longrightarrow A$$

Fix μ a motivic measure valued in A . For $Z \in \text{Var}(\mathbf{k})$ and $W \subseteq Z$ a closed subscheme, set $\mathbf{1}_W : Z(\bar{\mathbf{k}}) \longrightarrow A$ to be the characteristic function of $W(\bar{\mathbf{k}})$, where $\bar{\mathbf{k}}$ is the separable closure of $\mathbf{k}$. By a constructible function we mean $f : Z(\bar{\mathbf{k}}) \longrightarrow A =$

$\sum_{i \text{ finite}} a_i \mathbf{1}_{W_i}$ with finite support. Define the **motivic integration** of constructible function f to be $\int_Z f(z) d\mu := \sum_i a_i \mu([W_i])$. Some examples of motivic measure could be:

- When $\mathbf{k} = \mathbb{F}_q$, $A = \mathbb{Z}$, we can take $\mu : X \mapsto X(\mathbb{F}_q)$ to be the point counting function. We note this could be extended to $\widehat{K}_0(\text{Var}_{\mathbb{F}_q}) \longrightarrow \mathbb{Z}(q^{-1})$ by inverting $q = |\mathbb{A}^1(\mathbb{F}_q)| = \mu(\mathbb{L})$.
- When $\mathbf{k} = \mathbb{C}$ (resp. $\mathbf{k} = \mathbb{F}_q$), $A = \mathbb{Z}[q^{\frac{1}{2}}]$, we can take μ to be the (resp. ℓ -adic) mixed Poincaré polynomial $X \mapsto \sum_i (-1)^i q^{\frac{i}{2}} \chi(\text{gr}_i^W(H^\bullet(X(\mathbb{C}), \mathbb{C})))$ (resp. $H_{\text{et}}^\bullet(X, \bar{\mathbb{Q}}_\ell)$).

The motivic measure can also be extended to GL_n -torsors in the same reason that GL_n -torsors are Zariski locally trivial, applying the motivic measure as rational point counting (resp Poincaré series) for the case $\mathbf{k} = \mathbb{F}_q$, we can get the volume (resp Poincaré series) of $\text{Bun}_{n,d}(X)$. However, it is not true for any algebraic group G and $P \rightarrow X$ a G -torsor that $[P] = [X] \cdot [G]$, to extend the motivic measure to those stacks essentially of finite type and with linear stabilizers, we use

Lemma 3.2.9. *For G an algebraic group over $\mathbf{k}$, let $\widehat{K}_0^G(\text{Var}_{\mathbf{k}})$ be the quotient of $\widehat{K}_0(\text{Var}_{\mathbf{k}})$ by the ideal generated by $[P] - [X] \cdot [G]$, with $X \in \text{Var}_{\mathbf{k}}$, $P \rightarrow X$ a G -torsor, then*

(i) *Let $\mathcal{X}$ be a stack essentially of finite type with linear stabilizers, and $P \rightarrow \mathcal{X}$ a G -torsor, then P is essentially of finite type with linear stabilizers if and only if $[P] = [\mathcal{X}] \cdot [G]$ in $\widehat{K}_0^G(\text{Var}_{\mathbf{k}})$.*

(ii) *Assume G is connected, then the motivic measure of Poincaré series or points counting factor through $\widehat{K}_0^G(\text{Var}_{\mathbf{k}})$.*

Proposition 3.2.10. (i) *We have $Z_{X,\text{mot}}(t)$ being rational in t , and in fact $(1-t)(1-\mathbb{L}t)Z_{X,\text{mot}}(t)$ is a polynomial of degree $2g_X$.*

(ii) *There is a functional equation $Z_{X,\text{mot}}(\frac{1}{\mathbb{L}t}) = (\mathbb{L}t^2)^{1-g_X} Z_{X,\text{mot}}(t)$.*

Proof. Let $p_n : S^n X \rightarrow \text{Pic}^n(X)$, $(x_1, \dots, x_m) \mapsto \mathcal{O}_X(\sum_i x_i^{\deg(x_i)})$ be the projection, for $\mathcal{L} \in \text{Pic}^n(X)$, $p_n^{-1}(\mathcal{L}) \simeq \mathbb{P}(H^0(X, \mathcal{L}))$, using previous computation, we have

$$Z_{X,\text{mot}}(t) = \sum_{n \geq 0} \sum_{\mathcal{L} \in \text{Pic}^n} [\mathbb{P}(H^0(X, \mathcal{L}))] t^n = \sum_{n \geq 0} \sum_{\mathcal{L} \in \text{Pic}^n} \frac{\mathbb{L}^{h^0(\mathcal{L})} - 1}{\mathbb{L} - 1} t^n = \sum_{n \in \mathbb{Z}} \sum_{\mathcal{L} \in \text{Pic}^n} \frac{\mathbb{L}^{h^0(\mathcal{L})} - 1}{\mathbb{L} - 1} t^n$$

The last equality is due to the vanishing of $H^0(X, \mathcal{L})$ for line bundles of degree < 0 . The rationality of $Z_{X,\text{mot}}(t)(1-t)(1-\mathbb{L}t)$ comes from

$$Z_{X,\text{mot}}(t) = G(t) + \sum_{n \geq 2g_X - 1} \sum_{\mathcal{L} \in \text{Pic}^n} \frac{\mathbb{L}^{1-g_X+n} - 1}{\mathbb{L} - 1} t^n = G(t) + \sum_{n \geq 2g_X - 1} \sum_{\mathcal{L} \in \text{Pic}^n} (1 + \mathbb{L} + \dots + \mathbb{L}^{n-g_X}) t^n$$

We have also

$$\begin{aligned} Z_{X,\text{mot}}(\frac{1}{\mathbb{L}t}) &= \sum_{n \in \mathbb{Z}} \sum_{\mathcal{L} \in \text{Pic}^{-n}} \frac{\mathbb{L}^{h^0(\mathcal{L})} - 1}{\mathbb{L} - 1} (\mathbb{L}t)^n \\ \mathbb{L}^{1-g_X} t^{2-2g_X} Z_{X,\text{mot}}(t) &= \sum_{n \in \mathbb{Z}} \left(\sum_{\mathcal{L} \in \text{Pic}^{2-2g_X+n}} \frac{\mathbb{L}^{h^0(\mathcal{L})} - 1}{\mathbb{L} - 1} \right) \mathbb{L}^{1-g_X} t^n \end{aligned}$$

From the bijection $\text{Pic}^{-n} \longrightarrow \text{Pic}^{2g_X-2+n}$, $\mathcal{L} \mapsto \mathcal{L}^\vee \otimes \Omega_X$, Serre duality and Riemann-Roch for $\mathcal{L} \in \text{Pic}^{-n}(X)$: $h^0(\mathcal{L}) - h^0(\mathcal{L}^\vee \otimes \Omega_X) = \chi(\mathcal{L}) = 1 - g_X - n$, we have

$$\mathbb{L}^{1-g_X} t^{2-2g_X} Z_{X,\text{mot}}(t) = \sum_{n \in \mathbb{Z}} \left(\sum_{\mathcal{L} \in \text{Pic}^{-n}} \frac{\mathbb{L}^{h^0(\mathcal{L}^\vee \otimes \Omega_X)} - 1}{\mathbb{L} - 1} \right) \mathbb{L}^{1-g_X} t^n = \sum_{n \in \mathbb{Z}} \left(\sum_{\mathcal{L} \in \text{Pic}^{-n}} \frac{\mathbb{L}^{h^0(\mathcal{L})+n} - \mathbb{L}^{1-g_X}}{\mathbb{L} - 1} \right) t^n$$

This produces a term of difference $\frac{1}{\mathbb{L}-1} \sum_{n \in \mathbb{Z}} [\text{Jac}_X] (\mathbb{L}^n - \mathbb{L}^{1-g_X}) t^n = [\text{Jac}_X] \left(\frac{\mathbb{L}^{1-g_X}}{\mathbb{L}-1} \delta(t) - \frac{1}{\mathbb{L}-1} \delta(\mathbb{L}t) \right)$, with $\delta(t) = \sum_{n \in \mathbb{Z}} t^n$. We conclude with [Kap00, Lemma 1.3.4]. $\square$

Remark. The rationality of motivic zeta function for non-singular connected projective curves can be generalized to arbitrary one-dimensional variety X as we can separate connected components and for each one, we can find $\tilde{X}$ smooth projective containing X as dense open, with $\tilde{X} \setminus X$ being zero dimensional. It is expected by Kapranov the general rationality for higher dimensional varieties, however, the rationality fails even for surfaces, see for example [LL04].

We now can give a motivic version of Siegel's formula of $\text{Bun}_{r,d}(X)$

Theorem 3.2.11 ([BD07]). *We have the motivic class in $\widehat{K}_0(\text{Var}_{\mathbf{k}})$:*

$$[\text{Bun}_{r,d}] = \frac{\mathbb{L}^{(gx-1)(r^2-1)}}{\mathbb{L}-1} \cdot [\text{Jac}_X] \prod_{i=2}^r Z_{X,\text{mot}}(\mathbb{L}^{-i}) \quad (3.8)$$

Notice that when $\mathbf{k} = \mathbb{F}_q$, apply the motivic measure of points counting, we get exactly the Siegel formula 2.2.33 .

Conjecture 3.2.12. *Generally, for a split semisimple group G , it is conjectured in [BD07] that in $\widehat{K}_0^G(\text{Var}_{\mathbf{k}})$, one has*

$$[\text{Bun}_G] = \pi_1(G) \mathbb{L}^{(gx-1)\dim G} \prod_{i=1}^r Z_{X,\text{mot}}(\mathbb{L}^{-d_i})$$

The case when $X = \mathbb{P}^1$ is proved in loc. cit.

Using the Harder-Narasimhan stratification, we have the motivic class of $\text{Bun}_{r,d}$ in terms of classes of HN-strata. We use:

Proposition 3.2.13. *Let $\mathcal{X}$ be a local quotient stack, $\mathcal{E}_0, \mathcal{E}_1$ vector bundle on $\mathcal{X}$ of constant rank. Then the motivic class of vector bundle stack is $[\mathcal{E}_1/\mathcal{E}_0] = [\mathcal{X}] \mathbb{L}^{\text{rk}(\mathcal{E}_1) - \text{rk}(\mathcal{E}_0)} \in \widehat{K}_0(\text{Var}_{\mathbf{k}})$ (again, we abuse the notation of quotient stack and motivic class).*

Proof. When $\mathcal{X} = X$ is a scheme, then there is a stratification $X = \bigcup_i X_i$ such that on each X_i , $\mathcal{E}_0, \mathcal{E}_1$ are trivial and the map $\phi : \mathcal{E}_0 \rightarrow \mathcal{E}_1$ is locally of constant rank on X_i . Then $[\mathcal{E}_1/\mathcal{E}_0]$ restricts on X_i is $[X_i] \mathbb{L}^{\text{rk}(\mathcal{E}_1) - \text{rk}(\mathcal{E}_0)}$.

In general, it enough to prove for global quotient stack $\mathcal{X} = [X/\text{GL}_n]$. Since automorphism group of any object in $[\mathcal{E}_1/\mathcal{E}_0]$ is affine, by Lemma 3.2.5, we know $[\mathcal{E}_1/\mathcal{E}_0]$ is a local quotient stack, we write it as $\bigcup_i [Y_i/\text{GL}_{n_i}]$. $\pi : X \rightarrow [X/\text{GL}_n] = \mathcal{X}$ is a presentation, the fiber product $X \times_{\mathcal{X}} [\mathcal{E}_1/\mathcal{E}_0] \simeq \bigcup_i X \times_{\mathcal{X}} [Y_i/\text{GL}_{n_i}]$, set $p : X \rightarrow \mathcal{X}$, we know it equals to $[p^*\mathcal{E}_1/p^*\mathcal{E}_0]$ over X . While $Y_i \times_{\mathcal{X}} X \rightarrow Y_i$ is GL_{n_i} -torsor and $Y_i \times_{\mathcal{X}} X \rightarrow X \times_{\mathcal{X}} [Y_i/\text{GL}_{n_i}]$ is GL_{n_i} -torsor. We can compute in $\widehat{K}_0(\text{Var}_{\mathbf{k}})$,

$$\begin{aligned} [\mathcal{E}_1/\mathcal{E}_0] &= \sum_i \frac{[Y_i]}{[\text{GL}_{n_i}]} = \sum_i \frac{[Y_i][\text{GL}_n]}{[\text{GL}_{n_i}][\text{GL}_n]} = \sum_i \frac{[Y_i \times_{[X/\text{GL}_n]} X]}{[\text{GL}_{n_i}][\text{GL}_n]} \\ &= \frac{1}{[\text{GL}_n]} \sum_i [X \times_{\mathcal{X}} [Y_i/\text{GL}_{n_i}]] = \frac{[p^*\mathcal{E}_1/p^*\mathcal{E}_0]}{[\text{GL}_n]}, \quad \text{use motivic class of vector bundle stack on } X \\ &= \frac{[X]}{[\text{GL}_n]} \mathbb{L}^{\text{rk}(\mathcal{E}_1) - \text{rk}(\mathcal{E}_0)} = [\mathcal{X}] \mathbb{L}^{\text{rk}(\mathcal{E}_1) - \text{rk}(\mathcal{E}_0)} \end{aligned} \quad \square$$

As a corollary to be added, we know for a HN-type partition $\underline{\alpha} \in \mathcal{P}_{\alpha} = (\alpha_1, \dots, \alpha_s)$ of $\alpha = (r, d)$,

$$[\text{Bun}_{\alpha}^{\underline{\alpha}}] = \mathbb{L}^{\sum_{i < j} -\langle \alpha_i, \alpha_j \rangle} \prod_{i=1}^s [\text{Bun}_{\alpha_i}^{\text{ss}}]$$

3.2.2 Motivic higher rank zeta functions

Follow [MR14], we give a motivic interpretation of Weng's higher rank zeta function. As a consequence, we can get a concrete expression of higher zeta functions in terms of usual Weil zeta function.

Let X be a smooth projective and geometrically connected curve over $\mathbf{k}$, note that for $\mathbf{k} = \mathbb{F}_q$, $Z_{X,r}(t)$ defined before is a generating series of $\frac{q^{h^0(\mathcal{E})}-1}{|\text{Aut}(\mathcal{E})|}$, we thus want to include the data of sections in defining the motive of stability conditions.

Definition 3.2.14. Consider the pair $(\mathcal{E}, s)$ with $\mathcal{E}$ a vector bundle on X , and $s \in H^0(X, \mathcal{E})$ is a section of $\mathcal{E}$. The pairs form a

$\mathbf{k}$ -linear category with morphism $f = (f_0, f_1) : (\mathcal{E}, s) \rightarrow (\mathcal{E}', s')$ is a commutative diagram:

$$\begin{array}{ccc} \mathcal{O}_X & \xrightarrow{s} & \mathcal{E} \\ f_0 \downarrow & & \downarrow f_1 \\ \mathcal{O}_X & \xrightarrow{s'} & \mathcal{E}' \end{array}$$

For $\tau \in \mathbb{R}$ (slope) and the class $(r, d) \in \mathbb{N}^* \times \mathbb{Z}$, we say that the pair $(\mathcal{E}, s)$ is τ -**semistable** (resp. τ -stable) if

- (i) For any $\mathcal{F} \subseteq \mathcal{E}$ subbundle, $\mu(\mathcal{F}) \leq \tau$ (resp. $\mu(\mathcal{F}) < \tau$).
- (ii) For any $\mathcal{F} \subseteq \mathcal{E}$ subbundle with $s \in H^0(X, \mathcal{F})$, we have $\mu(\mathcal{E}/\mathcal{F}) \geq \tau$ (resp. $\mu(\mathcal{E}/\mathcal{F}) > \tau$).

Set $\text{Bun}_{r,d}^{\tau, \text{ss}}$ to be the moduli stack of τ -semistable pairs, we can consider the forgetful map $\text{Bun}_{r,d}^{\tau, \text{ss}} \rightarrow \text{Bun}_{r,d}^{\text{ss}}, (\mathcal{E}, s) \mapsto \mathcal{E}$ as the condition (i) indicates the semistability of $\mathcal{E}$, and the forgetful map is representable since the fibers parametrize the morphism of sheaves. For condition (ii), we can see that section is a closed condition, while the slope restriction is an open condition, thus we can conclude that $\text{Bun}_{r,d}^{\tau, \text{ss}}$ is an algebraic stack of finite type and locally closed in $\text{Bun}_{r,d}^{\text{ss}}$.

Also we denote the $\mathbf{M}_{r,d}^{\tau}$ the coarse moduli space of τ -semistable pairs, which are studied in [HL92]. When τ is (r, d) -generic, i.e $\tau \neq \frac{d}{r}, \tau \notin \frac{\mathbb{Z}}{r'}, \forall 1 \leq r' < r$ so that we can deduce τ -semistable is τ -stable under generic condition, we have $\text{Bun}_{r,d}^{\tau, \text{ss}} \rightarrow \mathbf{M}_{r,d}^{\tau}$ is a $\mathbb{G}_m$ -gerbe since for any $(\mathcal{E}, s)$ is τ -stable, $\text{End}(\mathcal{E}, s) \simeq \mathbf{k}$.

When $\mathbf{k}$ is of characteristic zero and algebraically closed, under the generic assumption, we have the motivic class $[\text{Bun}_{r,d}^{\tau, \text{ss}}] = \frac{[\mathbf{M}_{r,d}^{\tau}]}{\mathbb{L}-1} \in \widehat{K}_0(\text{Var}_{\mathbf{k}})$. When $\mathbf{k} \simeq \mathbb{F}_q$ is finite, define the motivic class of $\text{Bun}_{r,d}^{\tau, \text{ss}}$ to be

$$\sum_{(\mathcal{E}, s) \in M_{r,d}^{\tau}} \frac{1}{[\text{Aut}(\mathcal{E}, s)]}$$

where $M_{r,d}^{\tau}$ is the set of isomorphism classes of τ -semistable pairs.

If we take $\tau = \frac{d}{r}$, then it is clear that $(\mathcal{E}, s)$ is τ -semistable if and only if $\mathcal{E}$ is semistable vector bundle. Thus when $\mathbf{k} = \mathbb{F}_q$:

Lemma 3.2.15. *We have for finite field, the motivic class is $[\text{Bun}_{r,d}^{\tau, \text{ss}}] = \sum_{(\mathcal{E}, s) \in M_{r,d}^{\tau}} \frac{1}{[\text{Aut}(\mathcal{E}, s)]} = \frac{1}{\mathbb{L}-1} \sum_{\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}}(X)} \frac{\mathbb{L}^{h^0(\mathcal{E})} - 1}{[\text{Aut}(\mathcal{E})]}$.*

Proof. For any vector bundle $\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}}$, consider $G_{\mathcal{E}} := \mathbb{G}_m \times \text{Aut}(\mathcal{E})$ acting on $H^0(X, \mathcal{E}) \setminus \{0\}$ by $(k, \varphi) \cdot f : x \mapsto k(\varphi f(x))$, for a section s of $\mathcal{E}$, the stabilizer $\text{Stab}_s(G_{\mathcal{E}})$ is thus $\text{Aut}(\mathcal{E}, s)$. Thus for a fixed vector bundle $\mathcal{E} \in \text{Bun}_{r,d}^{\text{ss}}$, we have

$$\sum_{(\mathcal{E}, s)} \frac{1}{[\text{Aut}(\mathcal{E}, s)]} = \frac{[\mathbb{L}]^{h^0(\mathcal{E})} - 1}{[G_{\mathcal{E}}]} = \frac{[\mathbb{L}]^{h^0(\mathcal{E})} - 1}{(\mathbb{L} - 1)[\text{Aut}(\mathcal{E})]}.$$

□

As a corollary, the motivic (pure) higher rank zeta function is defined as

$$Z_{X,r}^{\text{mot}}(t) := \sum_{m \geq 0} \sum_{\mathcal{E} \in \text{Bun}_{r,rm}^{\text{ss}}} \frac{\mathbb{L}^{h^0(\mathcal{E})} - 1}{[\text{Aut}(\mathcal{E})]} t^m = (\mathbb{L} - 1) \sum_{k \geq 0} [\text{Bun}_{r,d}^{k, \text{ss}}] t^k \quad (3.9)$$

Before introducing the wall-crossing method in [MR14], we introduce the motivic version of Zagier's inversion formula, which can be thought as the motivic generalization of [LR95].

Let $R := \widehat{K}_0(\text{Var}_{\mathbf{k}})[\mathbb{L}^{\pm \frac{1}{2}}]$, define the quantum affine plane $\mathbb{A}_0$ to be the completion $R[x_1, x_2^{\pm 1}]$ with respect to the multiplication $x^{\alpha} * x^{\alpha'} = \mathbb{L}^{\langle \alpha, \alpha' \rangle} x^{\alpha + \alpha'}$, where $\alpha = (r, d)$ is the Chern character taking value in $\mathbb{N} \times \mathbb{Z}$, with $x^{\alpha} := x_1^r x_2^d$.

It is defined in [Rei06] a map $I : \mathbf{H}_X \rightarrow R[x_1, x_2^{\pm 1}]$ (called integration map) by $\mathbf{1}_{[\mathcal{E}]} \mapsto -\mathbb{L}^{\frac{1}{2} \langle \mathbf{1}, \mathcal{E} \rangle} \frac{x^{\text{cl}(\mathcal{E})}}{[\text{Aut}(\mathcal{E})]}$ (we identify q with $\mathbb{L}$), it is an algebra homomorphism and can be extended to the completions $\widehat{I} : \widehat{\mathbf{H}}_X \rightarrow \mathbb{A}_0$. The essential characteristic function in Hall algebra is mapped to motivic class of vector bundle stacks, namely for a class $\alpha = (r, d)$

$$\widehat{I} : \mathbf{1}_{\alpha}^{\text{Bun}} \mapsto -\mathbb{L}^{\frac{(1-g_X)r^2}{2}} [\text{Bun}_{r,d}] x^{\alpha}, \quad \mathbf{1}_{\alpha}^{\text{ss}} \mapsto -\mathbb{L}^{\frac{(1-g_X)r^2}{2}} [\text{Bun}_{r,d}^{\text{ss}}] x^{\alpha}$$

Use the same notation as [MR14], we set $b_r = \mathbb{L}^{\frac{(1-g_X)r(r-1)}{2}} [\text{Bun}_{r,d}]$, by the formula in 3.2.11, $b_r = \frac{[\text{Jac}_X]}{\mathbb{L}-1} \prod_{i=1}^{r-1} \widehat{Z}_{X, \text{mot}}(\mathbb{L}^{-i})$. Hence $\widehat{I}(\mathbf{1}_{\alpha}) = -\mathbb{L}^{\frac{(1-g_X)r}{2}} b_r x^{\alpha}$. Set also $\beta_{\alpha} := \mathbb{L}^{\frac{(1-g_X)r(r-1)}{2}} [\text{Bun}_{r,d}^{\text{ss}}]$, we have similarly $\widehat{I}(\mathbf{1}_{\alpha}^{\text{ss}}) = -\mathbb{L}^{\frac{(1-g_X)r}{2}} \beta_{\alpha} x^{\alpha}$.

Lemma 3.2.16. *In the Hall algebra $\widehat{\mathbf{H}}_X$, we have the relation*

$$\mathbf{1}_\alpha^{\text{Bun}} = \sum_{\underline{\alpha} \in \mathcal{P}_\alpha} q^{\frac{1}{2} \sum_{i > j} \langle \alpha_i, \alpha_j \rangle} \mathbf{1}_{\alpha_s}^{\text{ss}} * \dots * \mathbf{1}_{\alpha_1}^{\text{ss}}$$

where $\underline{\alpha} = (\alpha_1, \dots, \alpha_s) \in \mathcal{P}_\alpha$ is the set of partition of Harder-Narasimhan type of α .

Proof. It is easy to check the formula for any vector bundle of class α , and the formal sum of all Harder-Narasimhan types in well-defined in the sense of 2.2.35. $\square$

Using the above Lemma and the map $\widehat{I}$, we obtain the combinatoric formula for motivic class of vector bundles:

Proposition 3.2.17. *For $\alpha = (r, d)$*

$$b_r = \sum_{\underline{\alpha} \in \mathcal{P}_\alpha} \mathbb{L}^{-\sum_{i < j} (r_i d_j - r_j d_i)} \beta_{\alpha_s} \dots \beta_{\alpha_1}, \quad \alpha_i = (r_i, d_i). \quad (3.10)$$

It is the same as the computation using local quotient stack. Moreover, the inversion formula of above is given by

Theorem 3.2.18.

$$\beta_\alpha = \sum_{\substack{r_1, \dots, r_k > 0 \\ \sum r_i = r}} \left(\prod_{i=1}^{k-1} \frac{\mathbb{L}^{(r_i + r_{i+1}) \{ (r_1 + \dots + r_i) d / r \}}}{1 - \mathbb{L}^{r_i + r_{i+1}}} \right) b_{r_1} \dots b_{r_k} \quad (3.11)$$

where $\{c\} = c \bmod \mathbb{Z}$ for $c \in \mathbb{Q}$ takes the value in $[0, 1)$.

3.2.3 Stable pairs and wall crossing

To encode the data of sections, we extend the coherent sheaf category into its 1-simplicial category $\text{Coh}(X)^{\Delta_1}$.

Concretely, company with Definition 3.2.14, we consider the abelian category of triples $E = (\mathcal{E}_0, \mathcal{E}_1, s_E)$, where $\mathcal{E}_a \in \text{Coh}(X)$, $a = 0, 1$ and $s_E \in \text{Hom}(\mathcal{E}_0, \mathcal{E}_1)$. The pros and cons of studying triple is, we have more information on maps while the global dimension is increased. In fact, let E, F be two triples, then there is a long exact sequence, comes from a quiver representation theoretic argument.

$$0 \longrightarrow \text{Hom}(E, F) \longrightarrow \oplus_{a=0,1} \text{Hom}(\mathcal{E}_a, \mathcal{F}_a) \longrightarrow \text{Hom}(\mathcal{E}_0, \mathcal{F}_1) \longrightarrow \text{Ext}^1(E, F) \longrightarrow \quad (3.12)$$

$$\oplus_{a=0,1} \text{Ext}^1(\mathcal{E}_a, \mathcal{F}_a) \longrightarrow \text{Ext}^1(\mathcal{E}_0, \mathcal{F}_1) \longrightarrow \text{Ext}^2(E, F) \longrightarrow 0 \quad (3.13)$$

We copy some terminology in [MR14] to introduce their wall-crossing method on stability of triples.

Definition 3.2.19. A **framed category** is a pair $(\mathcal{A}, \nu)$ where $\mathcal{A}$ is abelian and $\nu : K_0(\mathcal{A}) \rightarrow \mathbb{Z}$ is a map takes value in $\mathbb{N}$. Let $\mathcal{A}_k$ be the subcategory of objects E with $\nu(E) = k$. In particular, objects in $\mathcal{A}_0$ are called **unframed objects**, and objects in $\mathcal{A}_1$ are called **framed objects**.

A **torsion pair** in an abelian category $\mathcal{C}$ is a pair $(\mathbb{T}, \mathbb{F})$ of strict full subcategories of $\mathcal{C}$ satisfying

- (i) $\text{Hom}(T, F) = 0$ for any $T \in \mathbb{T}, F \in \mathbb{F}$.
- (2) For any $X \in \mathcal{C}$ there exists an extension in $\mathcal{C}$ $0 \longrightarrow T \longrightarrow X \longrightarrow F$ such that $T \in \mathbb{T}$ and $F \in \mathbb{F}$.

Assume also for a framed category $(\mathcal{A}, \nu)$ and $E \in \mathcal{A}$ there exists a maximal (with respect to inclusion) unframed subobject $E_1 \subseteq E$ and a maximal quotient unframed object $E \rightarrow E_2$.

Definition 3.2.20. Let $(\mathbb{T}, \mathbb{F})$ be a torsion pair of the unframed category $\mathcal{A}_0$, then a framed object $E \in \mathcal{A}_1$ is called

- (i) $(\mathbb{T}, \mathbb{F})$ -stable if $E_1 \in \mathbb{F}, E_2 \in \mathbb{T}$.
- (ii) $+\infty$ -stable if $E_2 = 0$.
- (iii) $-\infty$ -stable if $E_1 = 0$.

Specifically, we consider the framed category $\mathcal{A}$ as a subcategory of $\text{Coh}(X)^{\Delta^1}$, whose objects are triples $E = (\mathcal{E}_0 \in \langle \mathcal{O}_X \rangle, \mathcal{E}_1, s_E)$, where $\langle \mathcal{O}_X \rangle$ is subcategory of $\text{Coh}(X)$ generated by extensions of $\mathcal{O}_X$. The degree of framing on $\mathcal{A}$ is given by $\nu(E) := \text{rk}(\mathcal{E}_0)$. We see $\mathcal{A}_0$ can be identified with $\text{Coh}(X)$ and $\mathcal{A}_1$ of framed objects are triples $(\mathcal{O}_X, \mathcal{E}_1, s_E)$, $\mathcal{E}_1 \in \text{Coh}(X)$, $s_E \in H^0(X, \mathcal{E}_1)$. The maximal unframed subobject of $E \in \mathcal{A}$ is $(0, \mathcal{E}_1, 0)$ and the maximal unframed quotient is $(0, \text{Coker}(s_E), 0)$. And the torsion pair in $\mathcal{A}_0$ can be chosen as the usual torsion-free decomposition of coherent sheaves. (Note the torsion pair is not uniquely chosen!)

Definition 3.2.21. Choose $\tau \in \mathbb{R}$, a framed object $E \in \mathcal{A}_1$ is called τ -**semistable** (resp. **stable**) if

- (i) For any monomorphism $F \hookrightarrow E$ with unframed F we have $\mu(F) \leq \tau$ (resp. $< \tau$).
- (ii) For any epimorphism $E \twoheadrightarrow F$ with unframed F we have $\mu(F) \geq \tau$ (resp. $> \tau$).

A framed object $E \in \mathcal{A}_1$ is called τ_+ -stable (resp. τ_- -stable) if E is $(\tau + \varepsilon)$ -semistable (resp. $(\tau - \varepsilon)$ -semistable) for $0 < \varepsilon \ll 1$. This is equivalent to say:

- (i) For any monomorphism $F \hookrightarrow E$ with $F \in \mathcal{A}_0$, we have $\mu(F) \leq \tau$ (resp. $< \tau$).
- (ii) For any epimorphism $E \twoheadrightarrow F$ with $F \in \mathcal{A}_0$, we have $\mu(F) > \tau$ (resp. $\geq \tau$).

Remark. Comparing the definition of τ -stability of pair $(\mathcal{E}, s)$ and τ -stability of triples, we found $(\mathcal{E}, s)$ is τ -(semi)stable if and only if the triple $(\mathcal{O}_X, \mathcal{E}, s)$ is τ -(semi)stable. Thus the moduli stack $\text{Bun}_{\alpha}^{\tau, \text{ss}}$ is identified with moduli stack of τ -semistable triples. To imitate the motivic integration for stable pairs, one need to extend the variables.

For a pair $E = (\mathcal{E}_0, \mathcal{E}_1, s_E)$, define its class to be $\text{cl}(E) = (\text{rk}(\mathcal{E}_1), \deg(\mathcal{E}_1), \text{rk}(\mathcal{E}_0)) \in \mathbb{Z}^3$. And for E, F two triples in $\mathcal{A}$, define $\chi(E, F) = \sum_{k=0}^2 (-1)^k \dim \text{Ext}^k(E, F)$, from the resolution sequence 3.12, we know $\chi(E, F) = \chi(\mathcal{E}_0, \mathcal{F}_0) + \chi(\mathcal{E}_1, \mathcal{F}_1) - \chi(\mathcal{E}_0, \mathcal{F}_1)$. Set also $\langle E, F \rangle = \chi(E, F) - \chi(F, E)$.

Let the quantum affine space $\mathbb{A}$ to be the completion of $R[x_1, x_2^{\pm 1}, x_3]$, recall $R = \widehat{K}_0(\text{Var}_{\mathbf{k}})[\mathbb{L}^{\pm \frac{1}{2}}]$. The multiplication $\circ$ on $\mathbb{A}$ is given by $x^{\hat{\alpha}} \circ x^{\hat{\beta}} = -\mathbb{L}^{\frac{1}{2} \langle \hat{\alpha}, \hat{\beta} \rangle} x^{\hat{\alpha} + \hat{\beta}}$. Let $\mathbf{H}(\mathcal{A})$ be the Hall algebra of framed abelian category $\mathcal{A}$ of triples, write the integration map to be:

$$I : \mathbf{H}(\mathcal{A}) \rightarrow \mathbb{A}, \quad [E = (\mathcal{E}_0, \mathcal{E}_1, s_E)] \mapsto -\mathbb{L}^{\frac{1}{2} \chi(E, E)} \frac{x^{\text{cl}(E)}}{[\text{Aut}(E)]}.$$

However, this integration map is not an algebra homomorphism. It compatible with algebra structure, namely $I([E] * [F]) = I([E]) \circ I([F])$ when $\text{Ext}^2(F, E) = 0$.

To obtain a motivic interpretation of higher rank zeta functions, we need to now the motivic class for semistable pairs. Define the invariants for a class $\alpha = (r, d)$

$$u(\alpha) = -\mathbb{L}^{\frac{1}{2} \chi(\alpha)} \beta_{\alpha}, \quad f_{\tau}(\alpha) := (\mathbb{L} - 1) \mathbb{L}^{(1-g)(r(r-1))} [\text{Bun}_{\alpha}^{\tau, \text{ss}}]$$

With the associated $u^h(\alpha) \in H(\mathcal{A}_0) = \widehat{\mathbf{H}}_X$ and $f_{\tau}^h(\alpha) \in \widehat{\mathbf{H}}(\mathcal{A})$ elements in Hall algebras so that

$$u(\alpha) x^{\alpha} = \left(-\mathbb{L}^{\frac{d}{2}} \right) \widehat{I}(u^h(\alpha)), \quad f_{\tau}(\alpha) x^{(\alpha, 1)} = (\mathbb{L} - 1) \left(-\mathbb{L}^{\frac{g_X - 1 + d}{2}} \right) I(f_{\tau}^h(\alpha))$$

Set also the generating series

$$u_{\tau} = 1 + \sum_{\mu(\alpha) = \tau} u(\alpha) x^{\alpha} \in \mathbb{A}_0, \quad f_{\tau} = \sum_{\alpha} f_{\tau}(\alpha) x^{(\alpha, 1)}, \quad u_{\geq \tau} = \prod_{\tau' \geq \tau} u_{\tau'}$$

where in the formulation of $u_{\geq \tau}$, we take the order of the product regards to the decreasing slope, notice also it is well-defined since $u_{\tau} \neq 1$ unless $\tau \in \mathbb{Q}$, and the product is thus countable. Also $u_{> \tau}$ is similarly well defined product.

Example 3.2.22. Take $\tau = +\infty$, then a framed ∞ -semistable object is $(\mathcal{O}_X, \mathcal{E}_1, s_E)$ with the maximal unframed quotient $E_2 = \text{Coker}(s_E)$ being torsion, assume $\mathcal{E}_1$ is of class (r, d) , we know $\mathcal{E}_1$ has rank $r = 1$ and $\text{Coker}(s_E)$ is torsion with length d . The endomorphism of E is $\mathbf{k}$. The moduli space $M_{(1, d)}^{\infty, \text{ss}}$ is the Hilbert scheme of points $S^d X$, and $\text{Bun}_{(1, d)}^{\infty, \text{ss}}$ is a $\mathbb{G}_m$ -gerbe over it. Thus:

$$f_{\infty} = \sum_{(1, d), d \geq 0} f_{\infty}((1, d)) x^{(1, d, 1)} = x_1 x_3 \sum_{d \geq 0} [S^d X] x_2^d = x_1 x_3 Z_{X, \text{mot}}(x_2).$$

Theorem 3.2.23. For any $\tau \in \mathbb{R}$ we have

$$\mathfrak{f}_\tau = (\mathfrak{u}_{>\tau}^{-1} \circ \mathfrak{f}_\infty \circ \mathfrak{u}_{\geq\tau})|_{\mu \leq \tau}.$$

where for a series $F = \sum_\alpha F_\alpha x^{(\alpha,1)}$, write $F|_{\mu < \tau} := \sum_{\mu(\alpha) < \tau} F_\alpha x^{(\alpha,1)}$. In particular

$$\mathfrak{f}_{\tau-} = (\mathfrak{u}_{\geq\tau}^{-1} \circ \mathfrak{f}_\infty \circ \mathfrak{u}_{\geq\tau})|_{\mu < \tau}, \quad \mathfrak{f}_{\tau+} = (\mathfrak{u}_{>\tau}^{-1} \circ \mathfrak{f}_\infty \circ \mathfrak{u}_{>\tau})|_{\mu \leq \tau}$$

where $\mathfrak{f}_{\tau\pm}$ are defined for moduli stack of $\tau_\pm$ -semistable triples.

Remark. Investigating the coefficient of $x^{(\alpha,1)}$ in $\mathfrak{f}_\tau$, we can extract the motivic classes as component of higher rank zeta functions.

To understand the formula and to prove this theorem, we need more analysis on stability.

Proposition 3.2.24. Fix $(\mathbb{T}, \mathbb{F})$ a torsion pair. Any $E \in \mathcal{A}_1$ framed object has a unique filtration $E' \subset E'' \subset E$ so that $E' \in \mathbb{T}$, $E''/E' \in \mathcal{A}_1$ is a $(\mathbb{T}, \mathbb{F})$ -stable framed object and $E/E'' \in \mathbb{F}$. Such a filtration is called a **canonical filtration**.

Proof. The construction can be concluded in a diagram
$$\begin{array}{ccccc} E' & \hookrightarrow & E_1 & & \\ & & \downarrow & & \\ E'' & \longrightarrow & E & \twoheadrightarrow & E_2 \\ & & \searrow & & \downarrow \\ & & & & E_2^{\text{cotor}} \end{array}$$
, where by hypothesis E_1 and E_2 are maximal unframed subobject or quotient respectively. We choose E' to be the torsion part of E_1 and $E'' := \text{Ker}(E \rightarrow E_2^{\text{cotor}})$. $\square$

Define for any $\sigma \in \mathbb{R}$ a slope μ_σ of triple E to be

$$\mu_\sigma(E) = \frac{\deg(\mathcal{E}_1) + \deg(\mathcal{E}_0) + \sigma \text{rk}(\mathcal{E}_0)}{\text{rk}(\mathcal{E}_1) + \text{rk}(\mathcal{E}_0)}$$

and the (semi)-stability of E with respect to μ_σ requires $\mu_\sigma(F) < \mu_\sigma(E)$ (resp. $\leq$) for F proper subobject of E .

Remark. The slope-stability for any σ is well defined, i.e. for any exact sequence of triples $0 \rightarrow F' \rightarrow E \rightarrow F'' \rightarrow 0$, we have the two signs of difference of slopes match, that is $\text{sgn}(\mu_\sigma(F'') - \mu_\sigma(E)) = \text{sgn}(\mu_\sigma(E) - \mu_\sigma(F'))$.

Lemma 3.2.25. Let $E = (\mathcal{O}_X, \mathcal{E}_1, s_E) \in \mathcal{A}_1$ be framed, $\mathcal{E}_1 \neq 0$. Then E is τ -semistable if and only if it is μ_σ semistable, where $\sigma = (r+1)\tau - d$ and class of $\mathcal{E}_1$ is (r, d) .

Proof. In such case $\mu_\sigma(E) = \frac{d+\sigma}{r+1} = \tau$, then the equivalence is clear by definition. $\square$

For $\tau \in \mathbb{R}$, other possible choices of torsion pair on $\text{Coh}(X) = \mathcal{A}_0$ could be $(\text{Coh}^{>\tau}, \text{Coh}^{\leq\tau})$ or $(\text{Coh}^{\geq\tau}, \text{Coh}^{<\tau})$, where $\text{Coh}^{>\tau}$ contains those coherent sheaves whose HN-filtration has minimal slope $> \tau$ (similar for other categories). It is clear

Lemma 3.2.26. (i) E is τ -(semi)stable if and only if $E_1 \in \text{Coh}^{\leq\tau}$, $E_2 \in \text{Coh}^{\geq\tau}$ (resp. $E_1 \in \text{Coh}^{<\tau}$, $E_2 \in \text{Coh}^{>\tau}$)
(ii) E is τ_+ -(semi)stable if and only if $E_1 \in \text{Coh}^{\leq\tau}$, $E_2 \in \text{Coh}^{>\tau}$, if and only if E is $(\text{Coh}^{>\tau}, \text{Coh}^{\leq\tau})$ -stable.
(iii) E is τ_- -(semi)stable if and only if $E_1 \in \text{Coh}^{<\tau}$, $E_2 \in \text{Coh}^{\geq\tau}$, if and only if E is $(\text{Coh}^{\geq\tau}, \text{Coh}^{<\tau})$ -stable.

We call the torsion pair in condition (ii) of above lemma the τ_+ -torsion pair (resp. τ_- -torsion pair for condition (iii)).

Proposition 3.2.27. Let $E = (\mathcal{O}_X, \mathcal{E}_1, s_E \neq 0)$ be τ -semistable framed triple. Let $\mathcal{E}_1$ have class (r, d) and set $\sigma = (r+1)\tau - d$ as before, then:

(i) The canonical filtration of E with respect to τ_+ -torsion pair is $0 = E' \subset E'' \subset E$, and if $E'' \subsetneq E$, then $\mu(E/E'') = \mu_\sigma(E'') = \tau$. The section $s_{E''}$ is automatically non-zero as long as $s_E \neq 0$.
(ii) The canonical filtration of E with respect to τ_- -torsion pair is $E' \subset E'' = E$, and if $E' \neq 0$, then $\mu(E') = \mu(E/E') = \tau$. If the slope $\mu(\mathcal{E}_1) < \tau$, then $s_{E/E'} \neq 0$.

Conversely, let $E \in \mathcal{A}_1$ be any framed object and assume $A \subseteq E, B = E/A$, then:

(iii) If $A \in \mathcal{A}_1$ is τ_+ -stable and B is semistable of slope τ (Note $B \in \mathcal{A}_0$ since $a \in \mathcal{A}_1$), then E is τ -semistable.

(iv) If $B \in \mathcal{A}_1$ is τ_- -stable and A is semistable of slope τ , then E is τ -semistable.

Proof. We prove only for (i) and (iii), the proof of (ii) and (iv) is similar. For (i), by property of canonical filtration, we know $E' \in \text{Coh}^{>\tau}$ if non-zero. But since E is assumed to be τ -semistable, then $\mathcal{E}_1 \in \text{Coh}^{\leq \tau}$. Thus E' can only be zero. If $E/E'' \neq 0$ in $\text{Coh}(X)$, then $\mu(E/E'') \geq \tau$. Still by torsion pair, we know $E/E'' \in \text{Coh}^{\leq \tau}$, it can only have slope τ . Since $E/E'' \in \mathcal{A}_0$, $\mu_\sigma(E/E'') = \mu(E/E'')\tau$, and $\mu_\sigma(E'') = \tau$ by considering the exact sequence $0 \rightarrow E'' \rightarrow E \rightarrow E/E'' \rightarrow 0$.

For (iii), by assumption, A is τ_+ -stable, then it is τ -semistable, then μ_σ -stable (for proper σ), so does B as $B \in \mathcal{A}_0$, hence the extension E is μ_σ -semistable with slope τ . By Lemma 3.2.25, E is τ -semistable. $\square$

Proof of Theorem 3.2.23. Take the generating series in (completed) Hall algebras $u_\tau^h := 1 + \sum_{\mu(\alpha)=\tau} u^h(\alpha)$, $f_\tau^h := \sum_\alpha f_\tau^h(\alpha)$.

Notice if $\mu(\alpha) > \tau$, then $\text{Bun}_\alpha^{\tau, \text{ss}}$ is empty, we have $f_\tau^h = f_\tau^h|_{\mu \leq \tau}$, $f_{\tau_+}^h = f_{\tau_+}^h|_{\mu \leq \tau}$, $f_{\tau_-}^h = f_{\tau_-}^h|_{\mu < \tau}$.

From (i) and (iii) of Proposition 3.2.27, we know those τ -semistable triples are exactly the extension of semistable coherent sheaf with slope τ by a framed τ_+ -stable object. Also from (ii) and (iv), we can decompose τ -semistable triples (with $\mu(\mathcal{E}_1) < \tau$) as the extension of τ_- -stable object by a semistable unframed coherent sheaf with slope τ . Hence in $\widehat{\mathbf{H}}(\mathcal{A})$, we have

$$f_\tau^h = f_{\tau_+}^h \circ u_\tau^h, \quad f_\tau^h|_{\mu < \tau} = u_\tau^h \circ f_{\tau_-}^h$$

Here we regard u_τ^h as an element in $\widehat{\mathbf{H}}(\mathcal{A})$, due to the identification $\mathcal{A}_0 = \text{Coh}(X) \subseteq \mathcal{A}$. Apply integration map,

$$f_\tau = f_{\tau_+} \circ u_\tau, \quad f_\tau|_{\mu < \tau} = u_\tau \circ f_{\tau_-}$$

Note the remark that the integration map compatible with the multiplication if the second order extension group vanishes. In both of our cases of extensions, this condition holds. Thus in $\mathbb{A}$, $f_\tau = f_{\tau_+} \circ u_\tau$, $f_\tau|_{\mu < \tau} = u_\tau \circ f_{\tau_-}$. And we get

$$f_{\tau_-} = (u_\tau^{-1} \circ f_{\tau_+} \circ u_\tau)|_{\mu < \tau}$$

Further, take the product over all $\tau' (\in \mathbb{Q}) \geq \tau$, we get the expression of

$$f_{\tau_-} = \left(u_{\geq \tau}^{-1} \circ f_\infty \circ u_{\geq \tau} \right)|_{\mu < \tau}$$

as the Theorem. Note that the operation to take the product on $f_{\tau'}$ makes sense, due to

Lemma 3.2.28. If $f_\tau(r, d) \neq 0$, then $d \geq 0$ and $\frac{d}{r} \leq \tau < \frac{\tau}{d-1}$.

Proof of Lemma. It follows from a direct computation. Let E be τ -semistable framed with class $(1, r, d)$ Using Proposition 3.2.27, assume E'' is the τ_+ -stable subobject, of class $(1, r'', d'')$ and E/E'' unframed of class (r', d') . Then $\frac{d'}{r'} = \tau$, $\frac{d''}{r''} \leq \tau < \frac{d''}{r''-1}$, thus $\frac{d'+d''}{r'+r''} \leq \tau < \frac{d'+d''}{r'+r''-1}$ (this also implies $d \geq 0$). $\square$

The part of f_{τ_+} is similarly obtained. Combining the expression of $f_{\tau_\pm}$, we get the one for f_τ . $\square$

3.2.4 Inversion formula for stabilities

To be added.

3.3 Integration on truncated Eisenstein series

3.3.1 Formulation

In this section, we investigate on Hall algebra of curve X to give a formulation of higher rank zeta functions.

For an arbitrary vector bundle $\mathcal{E}$, $q^{h^0(\mathcal{E})} - 1$ counts the number of sections, i.e. morphisms from $\mathcal{O}_X$ to $\mathcal{E}$. Especially if $\mathcal{E}$ is semistable, by a slope argument, $h^0(X, \mathcal{E}) \neq 0$ only when $\deg(\mathcal{E}) \geq 0$. Each non-zero morphism $\mathcal{O}_X \rightarrow \mathcal{E}$ is injective, whose image is identified with a line bundle of degree 0 in $\mathcal{E}$, while its automorphism group has volume $q - 1$. Then we know $\frac{q^{h^0(\mathcal{E})} - 1}{q - 1} = \# \{ \mathcal{L} \subseteq \mathcal{E} \mid \mathcal{L} \simeq \mathcal{O}_X \}$. Moreover, $\frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|}$ is thus equals to

$$q^{\frac{(g_X - 1)(r - 1) + d}{2}} \cdot (q - 1) (\mathbf{1}_{[\mathcal{E}]} \mid \mathbf{1}_{(r-1, d)} * \mathbf{1}_{[\mathcal{O}_X]})$$

as by definition, $\mathbf{1}_{(r-1, d)} * \mathbf{1}_{[\mathcal{O}_X]} = q^{\frac{(1-g_X)(r-1)-d}{2}} \sum_{\mathcal{V} \in \text{Obj}(\text{Coh}_{r,d})/\sim} \# \{ \mathcal{L} \subseteq \mathcal{V} \mid \mathcal{L} \simeq \mathcal{O}_X \}$ in $\widehat{\mathbf{H}}_X$, and by taking Hopf pairing, we mod out $\text{Aut}(\mathcal{E})$. As a result, we can rewrite the terms occur in higher rank zeta function:

$$\begin{aligned} \sum_{\mathcal{E} \in \text{Bun}_{r,rm}^{\text{ss}}} \frac{q^{h^0(\mathcal{E})} - 1}{|\text{Aut}(\mathcal{E})|} &= q^{\frac{(g_X - 1)(r - 1) + rm}{2}} \cdot (q - 1) (\mathbf{1}_{r,rm}^{\text{ss}} \mid \mathbf{1}_{r-1,rm} * \mathbf{1}_{[\mathcal{O}_X]}) \\ &= q^{\frac{(g_X - 1)(r - 1) + rm}{2}} \cdot \frac{q - 1}{|\text{Pic}^0(X)|} (\mathbf{1}_{r,rm}^{\text{ss}} \mid \mathbf{1}_{r-1,rm} * \mathbf{1}_{(1,0)}^{\text{Bun}}) \\ \text{subsheaf of vector bundle is a vector bundle} &= q^{\frac{(g_X - 1)(r - 1) + rm}{2}} \cdot \frac{q - 1}{|\text{Pic}^0(X)|} (\mathbf{1}_{r,rm}^{\text{ss}} \mid \mathbf{1}_{r-1,rm} * \mathbf{1}_{(1,0)}) \end{aligned}$$

For $\mathcal{E} \in \text{Bun}_{r,rm}^{\text{ss}}$, tensor it with a line bundle $\mathcal{L}'$ of degree $-m$ (assume $\text{Pic}^1(X) \neq 0$), one get a semistable vector bundle $\tilde{\mathcal{E}}$ of class $(r, 0)$. For vector bundle of such class, it is semistable if and only if it has minimal slope ≥ 0 for its HN-filtration. Denote $\mathbf{1}_{r,d}^{\geq \nu} = \sum_{\mathcal{V} \in \text{Coh}_{r,d}^{\geq \nu}} \mathbf{1}_{[\mathcal{V}]}$ (resp. $\mathbf{1}_{r,d}^{\text{Bun}, \geq \nu}$), then we have

$$Z_{x,r}(t) = q^{\frac{(g_X - 1)(r - 1)}{2}} \cdot \frac{(q - 1)}{|\text{Pic}^0(x)|} \sum_{m \geq 0} (\mathbf{1}_{r,0}^{\geq 0} \mid \mathbf{1}_{(r-1),m} * \mathbf{1}_{1,-m}) (t \cdot q^{\frac{r}{2}})^m \quad (3.14)$$

Set $E_r^{\geq 0}(w) = \sum_{k \geq 0} \mathbf{1}_{r,k}^{\geq 0} w^k$ to be the generating series for vector bundles with minimal slope ≥ 0 , called **truncated Eisenstein series**. Then the rank r zeta function is as the coefficient of an integration of Eisenstein series:

$$Z_{x,r}(t) = q^{\frac{(g_X - 1)(r - 1)}{2}} \cdot \frac{(q - 1)}{|\text{Pic}^0(X)|} (E_r^{\geq 0}(w) \mid E_{r-1}(z_2) * E_1(z_1)) \Big|_{w^0, \frac{z_2}{z_1} = tq^{\frac{r}{2}}} \quad (3.15)$$

This product of Eisenstein series is computed in [Sch16] in a general form, we then give a demonstration.

Recall our notation in the proof of 2.2.31,

$$E_{r_s, \dots, r_1}(z_s, \dots, z_1) = E_{r_s}(z_s) \dots E_{r_1}(z_1), \quad E_{r_s, \dots, r_1}^{\text{Bun}}(z_s, \dots, z_1) = E_{r_s}^{\text{Bun}}(z_s) \dots E_{r_1}^{\text{Bun}}(z_1)$$

and a theorem of harder 2.2.31. Want to compute (in $\mathbb{C}[[\frac{z_s}{z_{s-1}}, \dots, \frac{z_2}{z_1}, z_1; w]]$)

$$G_{r_s, \dots, r_1}^{\geq 0}(z_s, \dots, z_1; w) := (E_{r_s, \dots, r_1}(z_s, \dots, z_1) \mid E_r^{\geq 0}(w)), r = r_1 + \dots + r_s; \text{ and} \quad (3.16)$$

$$Y_{r_s, \dots, r_1}^{\geq 0}(z_s, \dots, z_1; w) := (E_{r_s, \dots, r_1}^{\text{Bun}}(z_s, \dots, z_1) \mid E_r^{\geq 0}(w)) \quad (3.17)$$

Remark. Note that we can expand them in the region $\mathbb{C}[[\frac{z_s}{z_{s-1}}, \dots, \frac{z_2}{z_1}, z_1; w]]$ because at first $G_r^{\geq 0} \in \mathbb{C}[[z_i^{\pm 1}; w]]$, and for any sheaf $\mathcal{F} \in \text{Coh}^{\geq 0}$, one filtration (not necessarily HN-type) of $\mathcal{F}$, $\mathcal{F}_1 \subsetneq \mathcal{F}_2 \subsetneq \dots \subsetneq \mathcal{F}_s = \mathcal{F}$ satisfies $\deg(\mathcal{F}/\mathcal{F}_{i-1}) = \sum_{l=s}^{i+1} \deg(\mathcal{F}/\mathcal{F}_{l-1}) \geq 0$.

Lemma 3.3.1. *We have the relations of truncated Eisenstein series easily deduced from HN-filtration:*

$$\begin{aligned} E_r(z) &= \sum_{\substack{s+t=r \\ s,t \geq 0}} q^{\frac{1}{2}(g_X-1)st} E_s^{<0}(q^{\frac{t}{2}} z) * E_t^{\geq 0}(q^{-\frac{s}{2}} z) \\ E_r^{\text{Bun}}(z) &= \sum_{\substack{s+t=r \\ s,t \geq 0}} q^{\frac{1}{2}(g_X-1)st} E_s^{<0}(q^{\frac{t}{2}} z) * E_t^{\text{Bun}, \geq 0}(q^{-\frac{s}{2}} z) \\ E_r^{\geq 0}(z) &= E_r^{\text{Bun}, \geq 0}(z) * E_0(q^{-\frac{r}{2}} z) \end{aligned}$$

Proposition 3.3.2 (Functional equation). *The functional equation holds for higher rank zeta functions, i.e. $\widehat{Z}_{X,r}(T) = \widehat{Z}_{X,r}(\frac{1}{QT})$.*

Proof. The proof only established on the functional equation of $Z_X(t)$. To change $t \mapsto \frac{1}{qt}$, we need to swap z_1 and z_2 in the product $G_{r-1,1}^{\geq 0}$.

First we observe that

$$E_{r-1}^{\text{Bun}}(z_2) E_1^{\text{Bun}}(z_1) = E_1^{\text{Bun}}(z_1^{-1}) E_{r-1}^{\text{Bun}}(z_2^{-1})$$

since we can consider the extension sequence $0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{E}'' \rightarrow 0$, take their dual sheaves we get an extension $0 \rightarrow \mathcal{E}''^\vee \rightarrow \mathcal{E}^\vee \rightarrow \mathcal{E}'^\vee \rightarrow 0$ that reverse the degree (corresponding to the power of z_i) and notice $\langle \mathcal{E}'', \mathcal{E}' \rangle = \langle \mathcal{E}'^\vee, \mathcal{E}''^\vee \rangle$.

We have the general commuting relation of essential eisenstein series:

$$\begin{aligned} E_1^{\text{Bun}}(z) E_r^{\text{Bun}}(q^{\frac{1-r}{2}} w) &= C_r \text{Res}_w (E_1^{\text{Bun}}(z) E_1^{\text{Bun}}(w_r) \dots E_1^{\text{Bun}}(w_1)) \\ &= \prod_{i=1}^r \frac{Z(\frac{z}{w_i})(\frac{z}{w_i})^{2(1-g_X)}}{Z(\frac{w_i}{z})} C_r \text{Res}_w (E_1^{\text{Bun}}(w_r) \dots E_1^{\text{Bun}}(w_1) E_1^{\text{Bun}}(z)), \quad w_i = q^{1-i} w \\ &= E_r^{\text{Bun}}(q^{\frac{1-r}{2}} w) E_1^{\text{Bun}}(z) \prod_{i=1}^r \frac{Z(q^{i-1} \frac{z}{w})(q^{i-1} \frac{z}{w})^{2(1-g_X)}}{Z(q^{1-i} \frac{w}{z})} \end{aligned}$$

That is to say

$$\begin{aligned} Z_{X,r}(T) &= Z_{X,r}(t^r) = (E_r^{\geq 0}(w) | E_1^{\text{Bun}}(z_1^{-1}) E_{r-1}^{\text{Bun}}(z_2^{-1})) Z_X(q^{-\frac{r}{2}} \frac{z_2}{z_1}) \Big|_{w^0, \frac{z_2}{z_1} = t^r q^{\frac{r}{2}}} \\ &= (E_r^{\geq 0}(w) | E_1^{\text{Bun}}(a) E_{r-1}^{\text{Bun}}(b)) \Big|_{w^0, \frac{b}{a} = t^{-r} q^{-\frac{r}{2}}} G'_r(t^r) Z_X(t^r) \\ &= Z_{X,r}(\frac{1}{QT}) \frac{G'_r(t^r) Z_X(t^r)}{Z_X((\frac{1}{qt})^r)}. \end{aligned}$$

where $G'_r(t) := \prod_{i=1}^{r-1} \frac{Z_X(q^i t) \cdot (q^i t)^{2(1-g_X)}}{Z_X(q^{-i} t^{-1})}$.

Using the functional equation of $Z_X(t)$, we can further simplify that

$$G'_r(t^r) \cdot \frac{Z_X(t^r)}{Z_X((qt)^{-r})} = \frac{Z_X(q^{r-1} t^r)}{Z_X((qt)^{-r})} q^{(r-1)(1-g_X)} = (q^{1-2r} t^{-2r})^{(1-g_X)} \cdot q^{(r-1)(1-g_X)} = q^{r(g_X-1)} t^{2r(g_X-1)}$$

Hence $\widehat{Z}_r(T) = T^{1-g_X} \frac{\widehat{Z}_r(\frac{1}{QT})}{(QT)^{g_X-1}} (QT^2)^{g_X-1} = \widehat{Z}_r(\frac{1}{QT})$, the functional equation is proved. $\square$

Proposition 3.3.3.

$$G_{r_s, \dots, r_1}^{\geq 0}(z_s, \dots, z_1; w) = X_{r_s, \dots, r_1}(z_s, \dots, z_1; w) \cdot Y_{r_s, \dots, r_1}^{\geq 0}(z_s, \dots, z_1; w)$$

for some $X_{r_s, \dots, r_1}(z_s, \dots, z_1; w) \in \mathbb{C}[[\frac{z_s}{z_{s-1}}, \dots, \frac{z_2}{z_1}, z_1; w]]$

Proof. This uses the commuting relation of eisenstein series.

$$\begin{aligned}
G_{r_s, \dots, r_1}^{\geq 0}(z_s, \dots, z_1; w) &= (E_{r_s}(z_s) * \dots * E_{r_1}(z_1) \mid E_r^{\text{Bun}, \geq 0}(w) E_0(q^{-\frac{r}{2}} w)) \\
&= (\Delta_{r,0}(E_{r_s}(z_s) \dots E_{r_1}(z_1)) \mid E_r^{\text{Bun}, \geq 0}(w) \otimes E_0(q^{-\frac{r}{2}} w)) \text{ (by Hopf property)} \\
&= (\Delta_{r_s,0}(E_{r_s}(z_s)) \dots \Delta_{r_1,0}(E_{r_1}(z_1)) \mid E_r^{\text{Bun}, \geq 0}(w) \otimes E_0(q^{-\frac{r}{2}} w)) \\
&= \left((E_{r_s}(z_s) \mathbf{k}_0 \otimes E_0(q^{-\frac{r_s}{2}} z_s)) \dots (E_{r_s}(z_s) \mathbf{k}_0 \otimes E_0(q^{-\frac{r_s}{2}} z_s)) \mid E_r^{\text{Bun}, \geq 0}(w) \otimes E_0(q^{-\frac{r}{2}} w) \right)
\end{aligned}$$

where $\delta_{r,0}$ is the restriction of the image of coproduct to $(\text{rk} = r, \text{rk} = 0)$ -part. Next,

$$G_{r_s, \dots, r_1}^{\geq 0}(z_s, \dots, z_1; w) = \underbrace{(E_{\underline{r}}(\underline{z}) \mid E_r^{\text{Bun}, \geq 0}(w))}_{\textcircled{1}} \cdot \prod_{i=1}^s \underbrace{(E_0(q^{-\frac{r_i}{2}} z_i) \mid E_0(q^{-\frac{r}{2}} w))}_{\textcircled{2}}$$

for $\textcircled{1}$,

$$\begin{aligned}
(E_{\underline{r}}(\underline{z}) \mid E_r^{\text{Bun}, \geq 0}(w)) &= (E_{r_s}^{\text{Bun}}(z_s) E_0(q^{-\frac{r_s}{2}} z_s) E_{r_1}^{\text{Bun}}(z_1) E_0(q^{-\frac{r_1}{2}} z_1) \mid E_r^{\text{Bun}, \geq 0}(w)) \\
&= \prod_{i=1}^s \prod_{j=i+1}^s \prod_{k=0}^{r_i-1} Z_X(q^{-\frac{r_i+r_j}{2}+k} \frac{z_j}{z_i}) \left(E_{\underline{r}}^{\text{Bun}}(\underline{z}) \prod_{i=1}^s E_0(q^{-\frac{r_i}{2}} z_i) \mid E_r^{\text{Bun}, \geq 0}(w) \right) \\
&\quad (\text{each } E_{r_i}^{\text{Bun}}(z_i) \text{ need to permute with } E_0(q^{-\frac{r_s}{2}} z_s) \dots E_0(q^{-\frac{r_{i+1}}{2}} z_{i+1})) \\
&= \text{Exp} \left(\frac{|X(\mathbb{F}_q)|}{q-1} \left[\sum_{i>j} \frac{z_i}{z_j} (q^{\frac{r_j}{2}} - q^{-\frac{r_j}{2}}) q^{-\frac{r_i}{2}} \right] \right) \cdot (E_{\underline{r}}^{\text{Bun}}(\underline{z}) \mid E_r^{\text{Bun}, \geq 0}(w))
\end{aligned}$$

and $\textcircled{2}$ contributes to $\text{Exp} \left(\frac{|X(\mathbb{F}_q)|}{q-1} q^{-\frac{r+r_i}{2}} z_i w \right)$.

We conclude that $X_{r_s, \dots, r_1}(z_s, \dots, z_1; w) = \text{Exp} \left(\frac{|x(\mathbb{F}_q)|}{q-1} \left[\sum_i q^{-\frac{1}{2}(r+r_i)} z_i w + \sum_{i>j} \frac{z_i}{z_j} \left(q^{\frac{r_j}{2}} - q^{-\frac{r_j}{2}} \right) q^{-\frac{r_i}{2}} \right] \right)$ and the Proposition holds. $\square$

Further, to study on truncated $Y_{\underline{r}}^{\geq 0}(\underline{z})$, define also the completed version and the complementary truncation to be

$$Y_{\underline{r}}^*(\underline{z}; w) := (E_{\underline{r}}^{\text{Bun}}(\underline{z}) \mid E_r^*(w)), \quad * \in \{\emptyset, \geq 0, < 0\}$$

Using the same argument as in previous remark, we see $Y_{\underline{r}}^{< 0}(\underline{z}; w) \in z_1^{-1} \mathbb{C}[[z_s^{-1}, \frac{z_s}{z_{s-1}}, \dots, \frac{z_2}{z_1}; w^{-1}]]$.

The main consequence of this section is

Proposition 3.3.4. *Let $Y_{\underline{1}}^{\geq 0}(\underline{z}; w) := Y_{1, \dots, 1}^{\geq 0}(z_s, \dots, z_1; w)$, then for any $s \geq 1$,*

$$Y_{\underline{1}}^{\geq 0}(\underline{z}; w) = \frac{q^{\frac{1}{4}(g_X-1)s(s-1)} \text{Vol}_1^s}{\prod_{i<j} \widehat{Z}_X\left(\frac{z_i}{z_j}\right)} \cdot \sum_{\sigma \in \mathfrak{S}_s} \sigma \left[\prod_{i<j} \widehat{Z}\left(\frac{z_i}{z_j}\right) \cdot \frac{1}{\prod_{i<s} (1 - q^{\frac{z_{i+1}}{z_i}})} \cdot \frac{1}{1 - q^{\frac{1-s}{2}} z_1 w} \right] \quad (3.18)$$

$$Y_{\underline{1}}^{< 0}(\underline{z}; w) = (-1)^s \frac{q^{\frac{1}{4}(g_X-1)s(s-1)} \text{Vol}_1^s}{\prod_{i<j} \widehat{Z}_X\left(\frac{z_i}{z_j}\right)} \cdot \sum_{\sigma \in \mathfrak{S}_s} \sigma \left[\prod_{i<j} \widehat{Z}_X\left(\frac{z_i}{z_j}\right) \cdot \frac{1}{\prod_{i<s} (1 - q^{-1} \frac{z_i}{z_{i+1}})} \cdot \frac{1}{1 - q^{\frac{s-1}{2}} z_s w} \right] \quad (3.19)$$

where the expansion of $Y_{\underline{1}}^*(\underline{z}; w)$ is as previous remark.

Proof. • When $s = 1$, by definition we have

$$Y_1^{\geq 0}(z_1; w) = \sum_{d \geq 0} (\mathbf{1}_{1,d}^{\text{Bun}} \mid \mathbf{1}_{1,d}^{\text{Bun}}) (z_1 w)^d = \frac{\text{Vol}_1}{1 - z_1 w}, \quad w \ll 1$$

$$Y_1^{< 0}(z_1; w) = \sum_{d < 0} (\mathbf{1}_{1,d}^{\text{Bun}} \mid \mathbf{1}_{1,d}^{\text{Bun}}) (z_1 w)^d = -\frac{\text{Vol}_1}{1 - z_1 w}, \quad w \gg 1$$

We then use the induction for $s > 1$, notice using the relation $E_r^{\text{Bun}}(z) = \sum_{s+t=r} q^{\frac{1}{2}(gx-1)st} E_s^{<0}(q^{\frac{t}{2}}z) * E_t^{\text{Bun}, \geq 0}(q^{-\frac{s}{2}}z)$ we have

$$Y_{\underline{1}}(z_s, \dots, z_1; w) = \left(E_{\underline{1}}^{\text{Bun}}(\underline{z}) \mid E_s^{\geq 0}(w) + E_s^{<0}(w) + \sum_{\substack{u+t=s \\ u,t>0}} q^{\frac{1}{2}(gx-1)ut} E_s^{<0}(q^{\frac{t}{2}}z) * E_t^{\text{Bun}, \geq 0}(q^{-\frac{u}{2}}z) \right)$$

$$= Y_{\underline{1}}^{\geq 0}(\underline{z}; w) + Y_{\underline{1}}^{<0}(\underline{z}; w) + \sum_{\substack{u+t=s \\ u,t>0}} q^{\frac{1}{2}(gx-1)ut} \left(\Delta_{u,t} \left(E_{\underline{1}}^{\text{Bun}}(\underline{z}) \right) \mid E_u^{<0} \left(q^{\frac{t}{2}}w \right) \otimes E_t^{\geq 0} \left(q^{-\frac{u}{2}}w \right) \right) \quad (3.20)$$

Lemma 3.3.5. For $r_i \geq 1$, we can compute $Y_{\underline{r}}(\underline{z}; w)$ to be

$$Y_{\underline{r}}(\underline{z}; w) = q^{\frac{1}{2}(gx-1)} \sum_{i>j} r_i r_j \prod_i \text{Vol}_{r_i} \prod_i \left\{ \sum_{l \in \mathbb{Z}} z_i^l w^l q^{\frac{1}{2}l(r_{<i}-r_{>i})} \right\} \quad (3.21)$$

where $r_{<i} := \sum_{k<i} r_k$, $r_{>i} = \sum_{k>i} r_k$, $r_{[i,j]} := \sum_{k=1}^j r_k$.

Proof of Lemma. It follows again from the computation using Hopf property:

$$Y_{\underline{r}}(\underline{z}; w) = (E_{r_s}^{\text{Bun}}(z_s) \dots E_{r_1}^{\text{Bun}}(z_1) \mid E_r(w)) = (E_{r_s}^{\text{Bun}}(z_s) \otimes \dots \otimes E_{r_1}^{\text{Bun}}(z_1) \mid \Delta(E_r(w)))$$

where the coproduct Δ can be restrict to the part $(r_s, \dots, r_1)$. This restriction contributes to

$$\Delta_{r_s, \dots, r_1}(E_r(w)) = q^{\frac{1}{2}(gx-1) \sum_{i>j} r_i r_j} \bigotimes_{i=1}^s E_{r_1}(q^{\frac{1}{2}(r_{<i}-r_{>i})} w) \quad \square$$

To use the induction hypothesis, we decrease the rank of the coproduct in formula 3.20, and we further determine its expression. Consider:

$$\Delta(E_{\underline{1}}^{\text{Bun}}(z_s, \dots, z_1)) = \Delta(E_1^{\text{Bun}}(z_s)) \dots \Delta(E_1^{\text{Bun}}(z_1)) = \overrightarrow{\prod}_i \left(E_1^{\text{Bun}}(z_i) \otimes 1 + E_0(q^{\frac{1}{2}}z_i) E_0(q^{-\frac{1}{2}})^{-1} \mathbf{k}_1 \otimes E_1^{\text{Bun}}(z_i) \right)$$

$$=: \sum_{\sigma} \mathbf{X}_{\sigma}$$

where $\sigma : \{1, \dots, s\} \rightarrow 1, 2$ is a shuffle mapping, while for each σ , $\mathbf{X}_{\sigma}$ is defined as

$$\mathbf{X}_{\sigma} = \overrightarrow{\prod}_i C_{\sigma(i)}(z_i); \quad C_1(z) := E_1^{\text{Bun}}(z) \otimes 1, \quad C_2(z) = E_0(q^{\frac{1}{2}}z) E_0(q^{-\frac{1}{2}})^{-1} \mathbf{k}_1 \otimes E_1^{\text{Bun}}(z)$$

$$= \underbrace{\prod_{\substack{(i,j), i<j \\ \sigma(i)=1, \sigma(j)=2}} \frac{\widehat{Z}(\frac{z_j}{z_i})}{\widehat{Z}(\frac{z_i}{z_j})}}_{H_{\sigma}(\underline{z})} \cdot \left(\overrightarrow{\prod}_{i, \sigma(i)=1} E_1^{\text{Bun}}(z_i) \overrightarrow{\prod}_{j, \sigma(j)=2} E_0(q^{\frac{1}{2}}z_j) E_0(q^{-\frac{1}{2}})^{-1} \mathbf{k}_1 \right) \otimes \prod_{j, \sigma(j)=2} E_1^{\text{Bun}}(z_j)$$

where in the second line, σ is chosen as $\#\{\sigma^{-1}(1)\} = u, \#\{\sigma^{-1}(2) = t\}$, denote $\text{Sh}_{u,t}$ to be the set of such shuffles as in 2.2.4. The restriction of coproduct to certain rank is the same as the restriction of shuffle maps. Namely

$$Y_{\underline{1}}(z_s, \dots, z_1; w) = Y_{\underline{1}}^{\geq 0}(\underline{z}; w) + Y_{\underline{1}}^{< 0}(\underline{z}; w) + \sum_{\substack{u+t=s \\ u, t > 0}} \sum_{\sigma \in \text{Sh}_{u,t}} Y_{\sigma}(z_s, \dots, z_1; w) \quad (3.22)$$

$$Y_{\sigma}(z_s, \dots, z_1; w) = q^{\frac{1}{2}(g_X-1)ut} H_{\sigma}(\underline{z}) Y_{\underline{1}}^{< 0} \left(\underbrace{z_{i_u}, \dots, z_{i_1}}_{\text{decreasing order of } \sigma^{-1}(1)}; q^{\frac{t}{2}} w \right) Y_{\underline{1}}^{\geq 0}(z_{j_t}, \dots, z_{j_1}; q^{-\frac{u}{2}} w) \quad (3.23)$$

so it remains to check the equality using the induction hypothesis and the formula in Lemma 3.21. **To be added for further analysis.** $\square$

As a corollary of Proposition 3.3.4, we have

$$\begin{aligned} Y_{r_s, \dots, r_1}^{\geq 0}(q^{\frac{1-r_s}{2}} z_s, \dots, q^{\frac{1-r_1}{2}} z_1; w) &= \left(E_{r_s}^{\text{Bun}}(q^{\frac{1-r_s}{2}} z_s) \dots E_{r_1}^{\text{Bun}}(q^{\frac{1-r_1}{2}} z_1) \mid E_r^{\geq 0} w \right) \\ &= \prod_{i=1}^s C_{r_i} \left(\text{Res}_{z_s}(E_{1, \dots, 1}^{\text{Bun}}(z_s)) \dots \text{Res}_{z_1}(E_{1, \dots, 1}^{\text{Bun}}(z_1)) \mid E_r^{\geq 0}(w) \right) \\ &= \prod_{i=1}^s C_{r_i} \text{Res}_{\{z_s, \dots, z_1\}} Y_{1, \dots, 1}^{\geq 0}(z_s, \dots, z_1; w) \end{aligned} \quad (3.24)$$

Where z_j is the sequence $(z_j^{(r_j)}, z_j^{(r_j-1)}, \dots, z_j^{(1)})$ with $z_j^{(1)} = z_j$. And $\text{Res}_{\{z_s, \dots, z_1\}}$ is a s-Step iterated residue over $(z_s, \dots, z_1)$, taking as $\text{Res}_{\frac{z_s^{(r_s)}}{z_s^{(r_s-1)}} = \dots = \frac{z_s^{(2)}}{z_s^{(1)}} = q^{-1}} \circ \dots \circ \text{Res}_{\frac{z_1^{(r_1)}}{z_1^{(r_1-1)}} = \dots = \frac{z_1^{(2)}}{z_1^{(1)}} = q^{-1}}$

Example 3.3.6. We imposing the result formula 3.24 to the expression of higher rank 2, 3 zeta functions.

- When $r = 2$, we have (abbreviate Z_X as Z)

$$\begin{aligned} Y_{1,1}^{\geq 0}(z_2, z_1; w) \mid_{w^0, \frac{z_2}{z_1} = tq} &= \frac{q^{\frac{1}{2}(g_X-1)} \text{Vol}_1^2}{\widehat{Z}(\frac{z_1}{z_2})} \left[\widehat{Z}(\frac{z_1}{z_2}) \frac{1}{1 - q \frac{z_2}{z_1}} \cdot \frac{1}{1 - q^{-\frac{z_1 w}{2}}} + \widehat{Z}(\frac{z_2}{z_1}) \frac{1}{1 - q \frac{z_1}{z_2}} \cdot \frac{1}{1 - q^{-\frac{z_2 w}{2}}} \right] \mid_{w^0, \frac{z_2}{z_1} = tq} \\ &= q^{\frac{1}{2}(g_X-1)} \text{Vol}_1^2 \left(\frac{1}{1 - tq^2} + \frac{\widehat{Z}(tq)}{\widehat{Z}((tq)^{-1})} \cdot \frac{t}{t-1} \right) \\ \widehat{Z}_2(t) &= q^{(g_X-1)} \frac{(q-1) \text{Vol}_1^2}{|\text{Jac}_X(\mathbb{F}_q)|} \left(\frac{\widehat{Z}(t)}{1 - tq^2} + \frac{t \widehat{Z}(tq)}{t-1} \right) = q^{(g_X-1)} \text{Vol}_1 \left(\frac{\widehat{Z}(t)}{1 - tq^2} + \frac{t \widehat{Z}(tq)}{t-1} \right) \end{aligned} \quad (3.25)$$

- When $r = 3$, we have

$$\begin{aligned} Y_{1,1,1}^{\geq 0}(z_3, z_2, z_1; w) &= q^{\frac{3}{2}(g_X-1)} \text{Vol}_1^3 \cdot \frac{1}{\widehat{Z}(\frac{z_1}{z_2}) \widehat{Z}(\frac{z_1}{z_3}) \widehat{Z}(\frac{z_2}{z_3})} \left[\widehat{Z}(\frac{z_1}{z_2}) \widehat{Z}(\frac{z_1}{z_3}) \widehat{Z}(\frac{z_2}{z_3}) \frac{1}{(1 - q \frac{z_2}{z_1})(1 - q \frac{z_3}{z_2})} \cdot \frac{1}{1 - q^{-z_1 w}} \right. \\ &+ \widehat{Z}(\frac{z_2}{z_1}) \widehat{Z}(\frac{z_2}{z_3}) \widehat{Z}(\frac{z_1}{z_3}) \frac{1}{(1 - q \frac{z_1}{z_2})(1 - q \frac{z_3}{z_1})} \cdot \frac{1}{1 - q^{-z_2 w}} + \widehat{Z}(\frac{z_3}{z_2}) \widehat{Z}(\frac{z_3}{z_1}) \widehat{Z}(\frac{z_2}{z_1}) \frac{1}{(1 - q \frac{z_2}{z_3})(1 - q \frac{z_1}{z_2})} \cdot \frac{1}{1 - q^{-z_3 w}} \\ &+ \widehat{Z}(\frac{z_1}{z_3}) \widehat{Z}(\frac{z_1}{z_2}) \widehat{Z}(\frac{z_3}{z_2}) \frac{1}{(1 - q \frac{z_3}{z_1})(1 - q \frac{z_2}{z_3})} \cdot \frac{1}{1 - q^{-z_1 w}} + \widehat{Z}(\frac{z_2}{z_3}) \widehat{Z}(\frac{z_2}{z_1}) \widehat{Z}(\frac{z_3}{z_1}) \frac{1}{(1 - q \frac{z_2}{z_3})(1 - q \frac{z_1}{z_2})} \cdot \frac{1}{1 - q^{-z_2 w}} \\ &\left. + \widehat{Z}(\frac{z_3}{z_1}) \widehat{Z}(\frac{z_3}{z_2}) \widehat{Z}(\frac{z_1}{z_2}) \frac{1}{(1 - q \frac{z_3}{z_1})(1 - q \frac{z_2}{z_1})} \cdot \frac{1}{1 - q^{-z_3 w}} \right] \end{aligned}$$

Thus

$$\begin{aligned}
Y_{2,1}^{\geq 0}(z_2, z_1) \big|_{w^0, \frac{z_2}{z_1} = tq^{\frac{3}{2}}} &= C_2 \operatorname{Res}_{\frac{z'_2}{z'_1} = q^{-1}} \left(Y_{1,1,1}^{\geq 0}(z'_3, z'_2 = q^{\frac{1}{2}} z_2, z_1) \right) \big|_{w^0, \frac{z'_2}{z'_1} = q^2 t} \\
&= q^{(g_X-1)} \operatorname{Vol}_1 \operatorname{Vol}_2 \left[\frac{1}{1-q^3 t} + \frac{P(q^{-1}) q^{g_X-1} \widehat{Z}(qt) \widehat{Z}(q^2 t)}{(1-q^{-1}) \widehat{Z}(q^{-2} t^{-1}) \widehat{Z}(q)} \frac{1}{(1-q^2)(1-q^{-1} t^{-1})} \right. \\
&\quad \left. + \frac{P(q^{-1})}{\widehat{Z}(q)(1-q^{-1})} \frac{q^{g_X-1}}{(1-q^2 t)(1-q^2)} + \frac{\widehat{Z}(q^2 t) \widehat{Z}(qt)}{\widehat{Z}(q^{-2} t^{-1}) \widehat{Z}((qt)^{-1})} \frac{1}{1-t^{-1}} + \frac{\widehat{Z}(qt) P(q^{-1})}{\widehat{Z}((qt)^{-1}) \widehat{Z}(q)(1-q^{-1})} \frac{q^{g_X-1}}{(1-t^{-1})(1-q^3 t)} \right]
\end{aligned}$$

And we have

$$\begin{aligned}
\widehat{Z}_3(t) &= t^{1-g_X} \frac{q^{g_X-1}}{\operatorname{Vol}_1} \widehat{Z}(t) Y_{2,1}^{\geq 0}(z_2, z_1; w) \big|_{w^0, \frac{z_2}{z_1} = tq^{\frac{3}{2}}} \\
&= q^{2g_X-2} \operatorname{Vol}_2 \frac{\widehat{Z}(t)}{1-q^3 t} + q^{3g_X-3} \operatorname{Vol}_1^2 \frac{qt \widehat{Z}(q^2 t)}{(1-q^2)(qt-1)} + q^{3g_X-3} \operatorname{Vol}_1^2 \frac{\widehat{Z}(t)}{(1-q^2)(1-q^2 t)} \\
&\quad + q^{2g_X-2} \operatorname{Vol}_2 \frac{t \widehat{Z}(q^2 t)}{t-1} + q^{3g_X-3} \operatorname{Vol}_1^2 \frac{t \widehat{Z}(qt)}{(1-q^3 t)(t-1)}
\end{aligned} \tag{3.26}$$

It is exactly the formula as in [MR14]. Hence the uniformity conjecture in rank $r = 2, 3$ is obtained.

Appendix A

A

A.1 Recollection on stacks

A.1.1 Fibered categories

Let $\mathcal{C}$ be any category and $p_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{C}$ be a category over $\mathcal{C}$.

Definition A.1.1. An arrow $\phi : y \rightarrow x$ in $\mathcal{F}$ is said to be **(strongly) Cartesian** if for any arrow $\psi : z \rightarrow x$ in $\mathcal{F}$ and any $h : p_{\mathcal{F}}(z) \rightarrow p_{\mathcal{F}}(y)$ so that $p_{\mathcal{F}}(\phi) \circ h = p_{\mathcal{F}}(\psi)$, there exists a unique $\theta : z \rightarrow y$ so that $h = p_{\mathcal{F}}(\theta)$. Namely we have the following fiber product

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{F}}(z, y) & \xrightarrow{\phi \circ -} & \mathrm{Hom}_{\mathcal{F}}(y, x) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{\mathcal{C}}(p_{\mathcal{F}}(z), p_{\mathcal{F}}(y)) & \xrightarrow{p_{\mathcal{F}}(\phi) \circ -} & \mathrm{Hom}_{\mathcal{C}}(p_{\mathcal{F}}(x), p_{\mathcal{F}}(x)) \end{array}$$

We have the following properties, see [FGI05, Proposition 3.4].

Lemma A.1.2. (i) *The composition of strongly Cartesian arrows in $\mathcal{F}$ is strongly Cartesian.*

(ii) *The composition of strongly Cartesian arrows along functors is strongly Cartesian. That is, if $F : \mathcal{G} \rightarrow \mathcal{F}$ is category over $\mathcal{F}$ and $\phi : y \rightarrow x$ in $\mathcal{G}$ is strongly Cartesian over its image $F(\phi) : F(y) \rightarrow F(x)$ and $F(\phi)$ is strongly Cartesian over its image, then $\phi : y \rightarrow x$ is strongly Cartesian over $p_{\mathcal{F}} \circ F(y) \rightarrow p_{\mathcal{F}} \circ F(x)$.*

Definition A.1.3. For $\mathcal{F}$ over $\mathcal{C}$, we say that $\mathcal{F}$ is a **fibered category** over $\mathcal{C}$ if for any arrow $f : U \rightarrow V$ and $x \in \mathcal{F}$ with image V , there is a strongly Cartesian arrow $\phi : y \rightarrow x$ with $p_{\mathcal{F}}(\phi) = f$.

If $p_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{C}$ is fibered, for any object $U \in \mathrm{Obj}(\mathcal{C})$ we set $\mathcal{F}_U$ be its **fiber**, which is a category with $\mathrm{Obj}(\mathcal{F}_U) := \{x \in \mathcal{F} | p_{\mathcal{F}}(x) = U\}$ and arrows $\phi : x \rightarrow y$ in $\mathcal{F}$, $x, y \in \mathrm{Obj}(\mathcal{F}_U)$ so that $p_{\mathcal{F}}(\phi) = \mathrm{id}_U$.

The arrow between two fibered categories over $\mathcal{C}$ is natural, that is, a functor $F : \mathcal{F} \rightarrow \mathcal{G}$ so that $p_{\mathcal{G}} \circ F = p_{\mathcal{F}}$ and sending strongly Cartesian arrows to strongly Cartesian ones. These conditions of functor F allow us to restrict F to $\mathcal{F}_U$ for any $U \in \mathrm{Obj}(\mathcal{C})$.

Next Proposition tells us that fiberwise equivalence of fibered categories is an equivalence of fibered categories, the proof is straightforward, see for example [FGI05, Proposition 3.36].

Proposition A.1.4. *Let $F : \mathcal{F} \rightarrow \mathcal{G}$ be an arrow of fibered categories, then F is an equivalence if and only if the restriction $F_U : \mathcal{F}_U \rightarrow \mathcal{G}_U$ is an equivalence for any $U \in \mathrm{Obj}(\mathcal{C})$.*

We recall the definition of a groupoid is a category whose arrows are all invertible.

Definition A.1.5. For any category $\mathcal{C}$, we let $\mathrm{Gpd}_{\mathcal{C}}$ be the 2-category fibered in groupoids over $\mathcal{C}$ as the following

- Objects of $\mathrm{Gpd}_{\mathcal{C}}$ are those fibered categories $\mathcal{F}$ over $\mathcal{C}$, whose fiber $\mathcal{F}_U$ at any object $U \in \mathcal{C}$ is a groupoid.

- The 1-morphisms are as arrows of fibered categories over $\mathcal{C}$.
- The 2-morphisms are natural transforms between functors.

We are particularly interested in the case when the base category is taken as $\mathcal{C} = \text{Sch}/S$, where S is some scheme, call an object $\mathcal{X} \in \text{Gpd}_{\text{Sch}/S}$ a S -groupoid. Notice that if $X : (\text{Sch}/S)^{\text{op}} \rightarrow \text{SET}$ is an (fppf) presheaf, then it can be regarded as a S -groupoid by taking the tangent space of $U \in \text{Obj}(\text{Sch}/S)$ as $X(U) \in \text{SET}$, which is discrete (as category) and therefore a groupoid and the arrow $X(V) \rightarrow X(U)$ induced from $U \rightarrow V$ is the restriction map. For example, if T is a scheme over S , then it is also a S -groupoid.

In fact, for arbitrary category $\mathcal{C}$, we have the following inversion for those fibered in discrete categories, see [FGI05, Proposition 3.26].

Lemma A.1.6. *There is an equivalence between the category of categories fibered in sets over $\mathcal{C}$ and the category of functors $\mathcal{C}^{\text{op}} \rightarrow \text{SET}$.*

We see fibered categories as presheaves over $\mathcal{C}$ (take $\mathcal{C}$ as a site), while the so called stacks over $\mathcal{C}$ are those sheaves of categories over $\mathcal{C}$. We introduce some descent theory to give an explanation for this.

A.1.2 Some descent theory

Let $\mathcal{C}$ be a site and $\mathcal{F}$ be a category over $\mathcal{C}$.

Definition A.1.7. Let $\mathcal{U} = \{U_i \rightarrow U\}$ be a covering in $\mathcal{C}$, denote by $\text{pr}_{1,2}$ the projection of $U_i \times_U U_j$ into each factor. Define the **category of descent data** of $\mathcal{F}$ associate with $\mathcal{U}$ is a category $\text{DD } \mathcal{F}(\mathcal{U})$ with object (descent datum) (ξ_i, ϕ_{ij}) where $\xi_i \in \mathcal{F}(U_i)$ is an object in the fiber and $\phi_{ij} : \text{pr}_1^* \xi_i \xrightarrow{\sim} \text{pr}_2^* \xi_j$ is an isomorphism in $\mathcal{F}(U_i \times_U U_j)$ such that the following cocycle condition is satisfied

$$\text{pr}_{13}^* \phi_{ik} = \text{pr}_{12}^* \phi_{ij} \circ \text{pr}_{23}^* \phi_{jk} : \text{pr}_{13}^* \xi_i \longrightarrow \text{pr}_{13}^* \xi_k$$

where pr_{ab} , $a, b = 1, 2, 3$ is the projection of $U_i \times_U U_j \times_U U_k$ onto corresponding factors.

For (ξ_i, ϕ_{ij}) and (η_i, ψ_{ij}) two descent data, then an arrow $(\xi_i, \phi_{ij}) \rightarrow (\eta_i, \psi_{ij})$ is the collections of $\alpha_i : \xi_i \rightarrow \eta_i$ in the fiber $\mathcal{F}(U_i)$ such that the following diagram commutes

$$\begin{array}{ccc} \text{pr}_2^* \xi_j & \xrightarrow{\text{pr}_2^* \alpha_j} & \text{pr}_2^* \eta_j \\ \phi_{ij} \downarrow & & \downarrow \psi_{ij} \\ \text{pr}_1^* \xi_i & \xrightarrow{\text{pr}_1^* \alpha_i} & \text{pr}_1^* \eta_i \end{array}$$

The descent data is used to describe the "descent" problems, namely when the data attached to the covering is induced from the data from the base. For instance, consider if $\{U_i \rightarrow U\}_i$ is a covering in category SET , and if $V_i \rightarrow U_i$ are U_i -sets, we wonder if there exists $V \rightarrow U$ being a U -set, such that $V_i \simeq U_i \times_U V$ for all i .

Definition A.1.8. Fix a covering $\mathcal{U} = \{\sigma_i : U_i \rightarrow U\}$ in $\mathcal{C}$, can construct a functor $\mathcal{F}(U) \rightarrow \text{DD}(\mathcal{U})$ by sending $\xi \mapsto \sigma_i^* \xi =: \xi_i$ and thus the resulting isomorphisms are $\phi_{ij} := \text{pr}_1^* \sigma_i \xi \rightarrow \text{pr}_2^* \sigma_j \xi$.

For any object in $\text{DD } \mathcal{F}(\mathcal{U})$, we say it is an **effective descent datum** if it is the essential image of $\mathcal{F}(U)$ with respect to above functor.

Example A.1.9 (Descending of sets). As we just mentioned, for some set $U \in \text{SET}$ and $\mathcal{U} = U_i \rightarrow U$ a covering (we are in the case of fibered category $\text{SET} \rightarrow \text{SET}$ with discrete topology), can consider $\text{DD}(\mathcal{U})$ with objects $(V_i \rightarrow U_i, \phi_{ij} : V_i \times_U U_j \rightarrow V_j \times_U U_j)$.

In such case, we have the fact that $\text{SET}_U \rightarrow \text{DD}(\mathcal{U})$ is an equivalence of categories, i.e. all descent data is effective.

Example A.1.10 (Descending of modules). Let $A \rightarrow B$ be a map in the category CommRing of commutative rings, and let Mod_A be the category of A -modules (we are in the case of $\text{Ab} \rightarrow \text{CommRing}^{\text{op}}$). Define the category $\text{DDMod}_{A \rightarrow B}$ with objects being

the pair (N, ψ) , where N is a B -module and $\psi : N \otimes_A B \simeq B \otimes_A N$ is an isomorphism of $B^{\otimes 2}$ -modules such that

$$\psi_3 = \psi_1 \circ \psi_2 : N \otimes_A B \otimes_A B \longrightarrow B \otimes_A B \otimes_A N.$$

where $\psi_a, a = 1, 2, 3$ are associated maps

$$\begin{aligned}\psi_1 : B \otimes_A N \otimes_A B &\rightarrow B \otimes_A B \otimes_A N = \text{id}_B \otimes \psi, \\ \psi_2 : N \otimes_A B \otimes_A B &\rightarrow B \otimes_A N \otimes_A B = \psi \otimes \text{id}_B, \\ \psi_3 : N \otimes_A B \otimes_A B &\rightarrow B \otimes_A B \otimes N\end{aligned}$$

An arrow $\beta : (N, \psi) \rightarrow (N', \psi')$ is a homomorphism of B -modules $\beta : N \rightarrow N'$, making the following diagram commute

$$\begin{array}{ccc} N \otimes_A B & \xrightarrow{\psi} & B \otimes_A N \\ \downarrow \beta \otimes \text{id}_B & & \downarrow \text{id}_B \otimes \beta \\ N' \otimes_A B & \xrightarrow{\psi'} & B \otimes_A N' \end{array}$$

There is an obvious functor $\text{Mod}_A \longrightarrow \text{DMod}_{A \rightarrow B}$ by sending $M \in \text{Mod}_A \mapsto B \otimes_A M$. A useful criterion for (effective) descent data associated with $A \rightarrow B$ is the following [FGI05, Theorem 4.21]

Lemma A.1.11. *If B is faithfully flat over A , then there is an equivalence of categories $\text{Mod}_A \xrightarrow{\simeq} \text{DMod}_{A \rightarrow B}$.*

Example A.1.12 (Descending of quasi-coherent sheaves). Let S be a scheme, consider the fibered category $\text{QCoh}/S \rightarrow \text{Sch}/S$ whose fiber at some scheme U over S is the abelian category of quasi-coherent sheaves on U .

For $\mathcal{U} = \{U_i \rightarrow U\}_i$ open covering (can take the site associated with the requiring topology)

$$\text{Sch}/S = (\text{Sch}/S)_{\text{Zar}}, (\text{Sch}/S)_{\text{fpqc}}, (\text{Sch}/S)_{\text{fppf}}, \dots$$

Define the descent data $\text{DDQCoh}(\mathcal{U})$ to be the category with objects $(\mathcal{F}_i, \phi_{ij})$, where $\mathcal{F}_i$ is a quasi-coherent sheaf over U_i and $\phi_{ij} : \text{pr}_1^* \mathcal{F}_i \xrightarrow{\simeq} \text{pr}_2^* \mathcal{F}_j$ satisfies cocycle condition.

It is just a tautology that under Zariski topology, there is an equivalence of category $\text{QCoh}(U) \simeq \text{DDQCoh}(\mathcal{U})$. we would like to see if quasi-coherent sheaves can be effectively descended for fpqc topology.

Theorem A.1.13. [Stacks, Tag 023T] *Under the fpqc topology, the functor $\text{QCoh}(U) \rightarrow \text{DDQCoh}(\mathcal{U})$ is an equivalence of categories.*

The proof of the above theorem for affine case is deduced from the previous example of descent for modules and the following lemma.

Lemma A.1.14. [FGI05, Lemma 4.25] *Let $\mathcal{F} : (\text{Sch}/S)^{op} \rightarrow \text{SET}$ be a functor, assume if*

- (i) $\mathcal{F}$ is a sheaf with respect to Zariski topology and,
 - (ii) For arbitrary $V \rightarrow U$ faithfully flat morphism of affine schemes, we have the exact sequence $\mathcal{F}(U) \rightarrow \mathcal{F}(V) \rightrightarrows \mathcal{F}(V \times_U V)$.
- Then $\mathcal{F}$ is a sheaf with respect to the fpqc topology.*

For the proof of general case in gluing the affine data, see [Stacks, Tag 023T].

Now we can give the definition of stacks:

Definition A.1.15. Let $\mathcal{F} \rightarrow \mathcal{C}$ be a fibered category on a site $\mathcal{C}$. (i) Say $\mathcal{F}$ is a **prestack** over $\mathcal{C}$ if for each covering $\mathcal{U} = \{U_i \rightarrow U\}$ in $\mathcal{C}$, the functor $\mathcal{F}(U) \rightarrow \text{DD}\mathcal{F}(\mathcal{U})$ is fully faithful.

(ii) Say $\mathcal{F}$ is a **stack** over $\mathcal{C}$ if for each covering $\mathcal{U} = \{U_i \rightarrow U\}$ in $\mathcal{C}$, the functor $\mathcal{F}(U) \rightarrow \text{DD}\mathcal{F}(\mathcal{U})$ is an equivalence.

In particular, we see from the Example A.1.12, the fibered category $\mathrm{QCoh}/S \rightarrow (\mathrm{Sch}/S)_{fpgc}$ is a stack (for a proof of fully-faithfulness, see [FGI05, Lemma 2.60]).

Remark. We shall note here that the fibered category $\mathrm{QCoh}/S \rightarrow (\mathrm{Sch}/S)_{fppf}$ is no longer a stack, neither a prestack.

For a scheme S and the fppf-topology on Sch/S , with which we would like to associate the S -stacks, as mentioned in previous section, the stacks fibered in groupoids.

Definition A.1.16. Let S be a base scheme, by a S -**stack** we mean a stack $\mathcal{X} \rightarrow (\mathrm{Sch}/S)_{fppf}$ fibered in groupoids.

Lemma A.1.17. (2-Yoneda Lemma) Let $\mathcal{X}$ be a S -groupoid and T be an S -scheme, considered as category Sch/T fibered over Sch/S , then there is an equivalence of categories

$$\mathrm{Hom}_{\mathrm{Gpd}/S}(T, \mathcal{X}) \simeq \mathcal{X}_T$$

where a 1-morphism $F : T \rightarrow \mathcal{X}$ is sent to $F(\mathrm{id}_T) \in \mathcal{X}_T$ and a 2-morphism $\eta : F \rightarrow G$ is sent to $F(\mathrm{id}_T) \rightarrow G(\mathrm{id}_T)$.

A.1.3 Algebraic spaces and algebraic stacks

Along those abstract constructions of stacks, we introduce algebraic spaces and algebraic stacks that are often used to represent the 'geometric objects' associated with moduli problem, and geometrically can be viewed as the generalization of schemes.

Definition A.1.18. A **space** is an fppf-sheaf: $(\mathrm{Sch}/S)^{\mathrm{op}} \rightarrow \mathrm{SET}$ (Notice a scheme is tautologically a space). Spaces form a category with arrows are morphisms as sheaves. We say a space is representable if it is the essential image of the natural embedding functor $\mathrm{Sch} \rightarrow \mathrm{Space}$.

Let $X \rightarrow X'$ be a morphism of spaces, then it is **representable** if for any scheme Y with $Y \rightarrow X'$, the space $X \times_{X'} Y$ is representable. Further, a representable map of spaces can be endowed with a property (P) (stable under base change and fppf-local) of morphisms of schemes if the morphism of schemes $\mathrm{Spec} A \times_{X'} X \rightarrow \mathrm{Spec} A$ has property (P) for any ring A .

A space X is an **algebraic space** if the following two are satisfied:

- (i) The diagonal map $\Delta : X \rightarrow X \times X$ is representable and,
- (ii) There exist some scheme Y (called a presentation of X) with $Y \rightarrow X$ is a surjective and étale map of spaces.

Remark. The presentation of X might not be unique. In fact, there is an another equivalent definition of algebraic space by using 'equivalence relations', see [Stacks, Tag 02WW] for a description.

Back to the stacky case we concentrate on, the definition of algebraic stack is tautology as defining spaces who are algebraic.

Definition A.1.19. [Stacks, Tag 026O] also [LM00, Définition 4.1] with separated, quasi-compact conditions.

Let S be a scheme, a 1-morphism $F : \mathcal{X} \rightarrow \mathcal{X}'$ is representable by an algebraic space (resp. scheme) if $\forall U \in \mathrm{Obj}(\mathrm{Sch}/S)$ with map $U \rightarrow \mathcal{X}'$, the fiber product $\mathcal{X} \times_{\mathcal{X}'} U$ is an algebraic space (resp. scheme), the property (P) of a morphism of S -stacks is similarly defined as in definition A.1.18.

Assume $F : \mathcal{X} \rightarrow \mathcal{X}'$ is a 1-morphism of S -stacks, then say F is surjective if $\forall U \in \mathrm{Obj}(\mathrm{Sch}/S), x \in \mathcal{X}'_U$, there is a covering $\sigma_i : U_i \rightarrow U$ and associated objects $y_i \in \mathrm{Obj}(\mathcal{X}_{U_i})$ such that $F(y_i) \simeq \sigma_i^* x$.

Definition A.1.20. An **algebraic stack** (or **Artin stack**) $\mathcal{X}$ over S is an S -stack with the following properties:

- (i) The diagonal map $\Delta : \mathcal{X} \rightarrow \mathcal{X} \times_S \mathcal{X}$ is representable by algebraic spaces.
- (ii) There exists $U \in \mathrm{Obj}(\mathrm{Sch}/S)$ so that there is a 1-morphism $U \rightarrow \mathcal{X}$ being smooth and surjective.

Example A.1.21 (Quotient stacks). For a site $\mathcal{C}$, let $G : \mathcal{C} \rightarrow \mathrm{Gp}$ be a sheaf in group, and on some topological space X , define a G -**torsor** on X is a sheaf $\mathcal{G}$ on X endowed with a G -action ($G(U) \times \mathcal{G}(U) \rightarrow \mathcal{G}(U)$ a group action on set, $\forall U \rightarrow X$) satisfies:

- (i) For any open U with $\mathcal{G}(U) \neq \emptyset$, the $G(U)$ -action on $\mathcal{G}(U)$ is simple, transitive.
- (ii) There exists a covering $U_i \rightarrow X$ such that $\mathcal{G}(U_i) \neq \emptyset, \forall i$.

Let X be an S -scheme and G be an affine smooth S -scheme acting on X (sheaf in group on $(\text{Sch}/S)_{\text{fppf}}$), then we can consider the fibered category $[X/G] \rightarrow \text{Sch}/S$ defined as follows.

The objects of $[X/G]$ are diagrams $U \xleftarrow{p} E \xrightarrow{\phi} X$, where $U \rightarrow S$ is an S -scheme, $p : E \rightarrow U$ is a U -scheme and a $G \times_S U$ -torsor for fppf topology, $\phi : E \rightarrow X$ is G -equivariant. An arrow in $[X/G]$ from (U, E, p, ϕ) to (U', E', p', ϕ') is diagram

$$\begin{array}{ccccc} U & \xleftarrow{p} & E & \xrightarrow{\phi} & X \\ f \downarrow & & g \downarrow & \nearrow \phi' & \\ U' & \xleftarrow{p'} & E' & & \end{array}$$

so that both the left square and the right triangle commute.

First, we notice that $[X/G]$ is an S -stack. By definition of a G -torsor, it is an (fppf) sheaf so that satisfies the fully-faithfulness of descent functor. For the essentially-surjective of descent data, we use the fact that G is affine, then any descent datum is effective for fppf topology, see [Stacks, Tag 0245].

Moreover, this quotient stack is an algebraic one. To prove this, we shall see if the diagonal map is representable by algebraic spaces, this follows from

Lemma A.1.22. *Let S be a scheme, and $\mathcal{X}$ fibered in groupoid over $(\text{Sch}/S)_{\text{fppf}}$, then the diagonal $\Delta : \mathcal{X} \rightarrow \mathcal{X} \times \mathcal{X}$ is representable by algebraic space if and only if for any $U \in \text{Obj}(\text{Sch}/S)$, $\forall x, y \in \mathcal{X}_U$, the presheaf $\text{Isom}(x, y) : V \mapsto \text{Isom}(x|_V, y|_V)$ is an algebraic space (over $(\text{Sch}/U)_{\text{fppf}}$).*

Proof. For any $U \in \text{Obj}(\text{Sch}/S)$, $x, y \in \mathcal{X}_U$, consider the fiber product $\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}, (x, y)} U \rightarrow U$ is the category of objects $(\sigma : V \rightarrow U, z \in \mathcal{X}_V, \alpha : z \xrightarrow{\sim} a^*x, \beta : z \xrightarrow{\sim} a^*y)$, and its fiber $(\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}, (x, y)} U)_V$ at V is identified with $\text{Isom}(x|_V, y|_V)$ by taking $\beta \circ \alpha^{-1}$. $\square$

Lemma A.1.23. *Let G be an affine group scheme over S , X a scheme over S , then $\forall U \in \text{Obj}(\text{Sch}/S)$, $x, y \in \text{Obj}([X/G]_U)$, $\text{Isom}(x, y) \rightarrow (\text{Sch}/U)_{\text{fppf}}$ is an algebraic space.*

Proof. We can use the 'Bootstrap' to reduce to some fppf covering of U , and then a set theoretical argument concludes the proof, for details, see [Stacks, Tag 04TB]. $\square$

To show that there is an smooth and surjective presentation onto $[X/G]$, we consider the functor $X \rightarrow [X/G], (U \xrightarrow{f} X) \mapsto$

$$(U/S, E = G \times U, \phi : G \times U \rightarrow X), \text{ where } \phi \text{ is the map of fiber product along the diagram } \begin{array}{ccc} G \times U & \longrightarrow & U \\ \downarrow & & \downarrow f \\ G \times X & \xrightarrow{a} & X \end{array} \text{ . Notice that this}$$

gives a surjective morphism of stacks by definition of G -torsors and definition of surjective map of stacks, it is smooth since G is a smooth scheme and $X \times_{[X/G]} Y$ is representable by $G \times_S Y$ for $Y \in \text{Obj}(\text{Sch}/S)$.

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